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الجامعة المصرية اليابانية للعلوم والتكنولوجيا
EGYPT-JAPAN UNIVERSITY OF SCIENCE AND TECHNOLOGY
エジプト日本科学技術大学

Introduction to **Energy**, Environmental and Chemical Engineering

CPE211

Dr. Ahmed Elwardany
Energy Resources Engineering (ERE) Department

Course Team – Grading system

Members	Part
Dr. Ahmed Elwardany	Energy Resources (5 weeks)
Dr. Mahmoud Nasr	Environmental Engineering (5 weeks)
Prof. Ahmed Elshazly	Chemical Engineering (5 weeks)

Grading system: 100 Mark

Item	Weight
Mid-term Exam (9th week)	**30%**
Course work	**30%**
Final Exam	**40%**

Detailed grading system

Item	Dr. Elwardany	Dr. Mahmoud	Prof. Elshazly	Total/item
Mid Term	20	10	0	**30**
Class Work	10	10	10	**30**
Final Exam	0	15	25	**40**
Total/Prof	**30**	**35**	**35**	**100**

Energy: Outlines

- ✓ Introduction to energy resources
- ✓ Renewable energy
- ✓ Applications of renewable energy

References

[1] Capareda, Sergio C. *Introduction to Renewable Energy Conversions*. CRC Press, 2019.

[2] Goswami, Y. and Kreith, F., (2017), Energy Conversion, second edition, CRC press, Taylor and Francis.

[2] Outlook, W.E., (2016). International energy agency. *Organization for Economic Co-operation and Development, Paris*.

Applications



Power stations



Oil and gas





Refineries

Solar and Wind Energies



Marine and Aviation Engines

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Introduction

Renewable Energy

Renewable energy is defined as energy from resources that are replenished at the same rate that they are used

- ✓ A resource will cease to qualify as renewable if it is used in an unsustainable manner.
- ✓ **Solar energy** is a prime example of a renewable energy resource. The energy received by the earth from the sun each day is simply reradiated back into the atmosphere.
- ✓ Solar energy is utilized by simply delaying the return of this energy and converting it into some useful temporal applications.
- ✓ Most other renewable energies, such as biomass, wind, hydropower, and wave energy, would directly and indirectly be coming from or influenced by solar energy.

Renewable Energy

- ✓ By the above definition, geothermal energy ***does not*** fall into the category of renewable energy.
- ✓ This is because the process of harnessing geothermal energy removes heat from the earth's core and releases the energy to the atmosphere.
- ✓ There is no way the energy may be fully returned to the earth's core.
- ✓ However, for practical purposes, since the amount of energy currently removed from geothermal resources is still a small fraction of the overall available energy from earth's heat at its core, we can consider geothermal energy as a renewable resource.

Biomass → photosynthetic **Hydropower → Hydrologic cycle**

Wind → Temperature gradient

Advantages and disadvantages of RE

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Advantages

1. Ability to sustain itself when used properly.
2. Available in any part of the world.
3. Renewable energy could be extracted with limited resources.
4. Environment-friendly (Low CO2 emissions).
5. It provides diversity, balancing the harmful effects of using just a single energy or power source. e. Wide ranges of products can now be derived from renewables.
6. Some renewable energy technologies do not require sophisticated components, parts, and materials (simple installation).

Advantages and disadvantages of RE

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Disadvantages

1. Renewables usually require large tracks of land.

2. Most renewable energy resources are not consistently available, and timeliness of production and use may be a challenge.

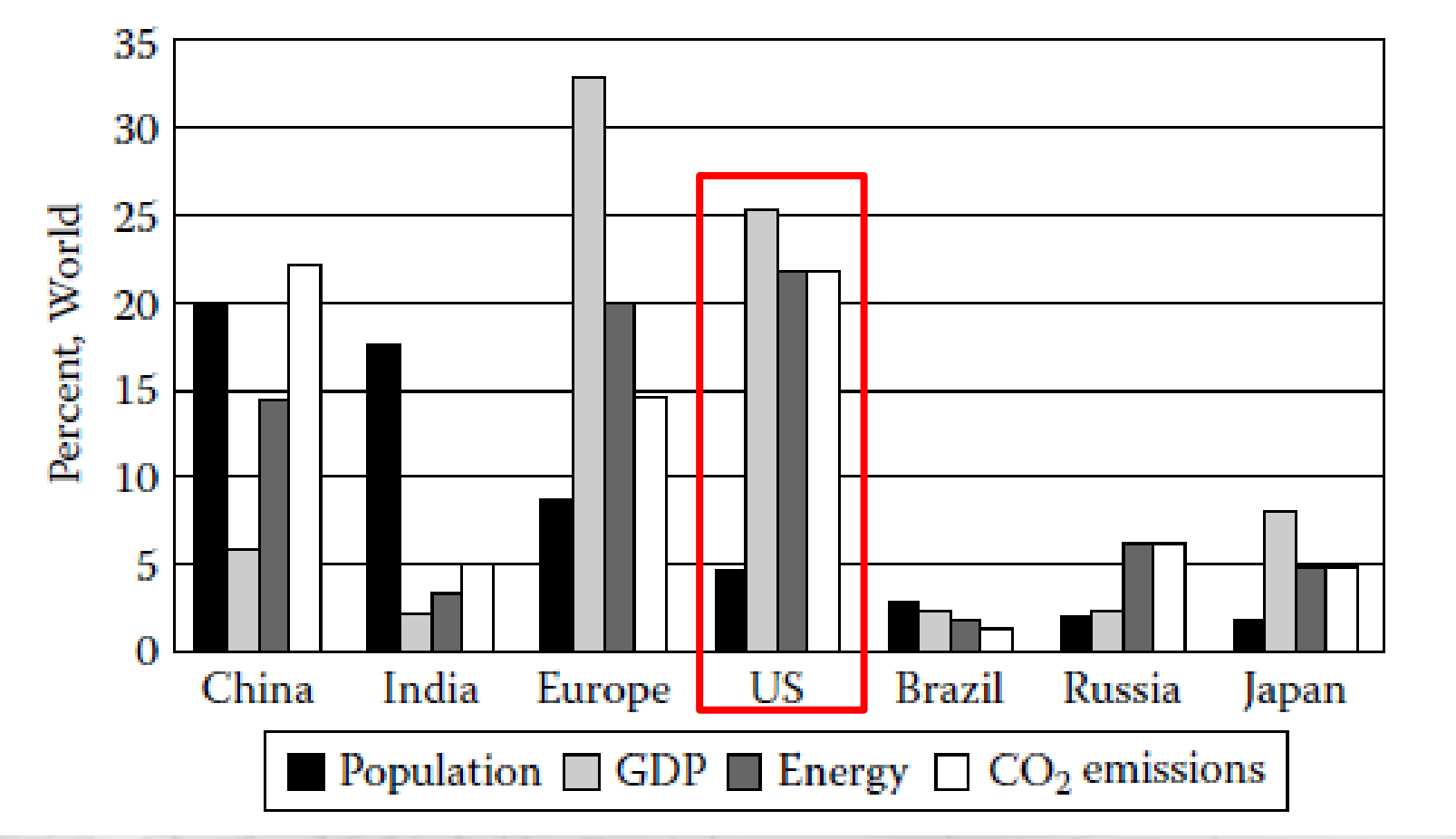
3. Storage of energy and power will also be a challenge for most renewables—some are available in large amounts for short periods, and this necessitates expenses on energy storage facilities.

4. Some renewables require special, costly materials (e.g., solar photovoltaic [PV] cells) with substantial carbon footprints during manufacturing.

5. Economics—for example, while the price of the solar PV cell has gone down below $1/watt peak (Wp) [$746/hp], the balance-of-systems costs (i.e., wirings, transmission, battery banks, etc.) are still relatively high.

Energy and Society

Industrialized societies run on energy

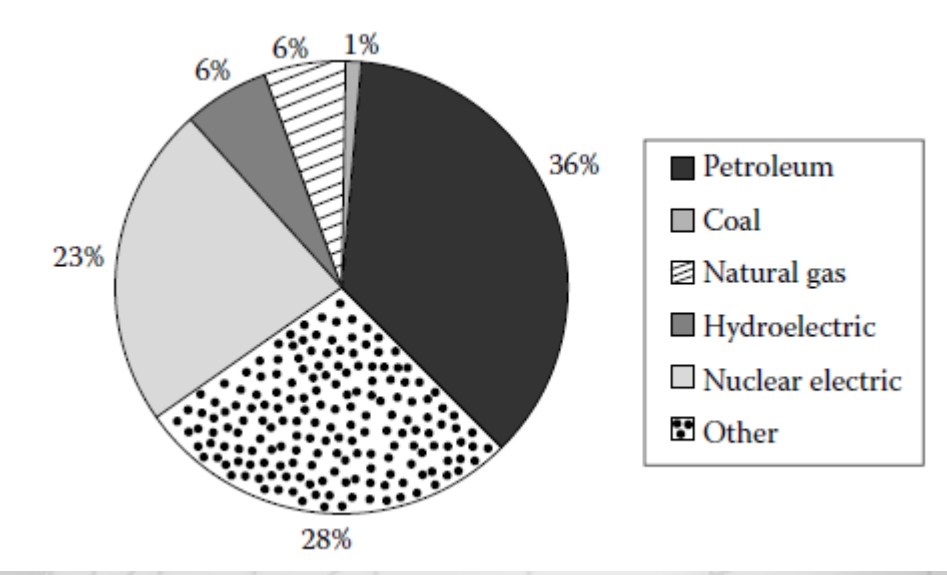


GDP: gross domestic product

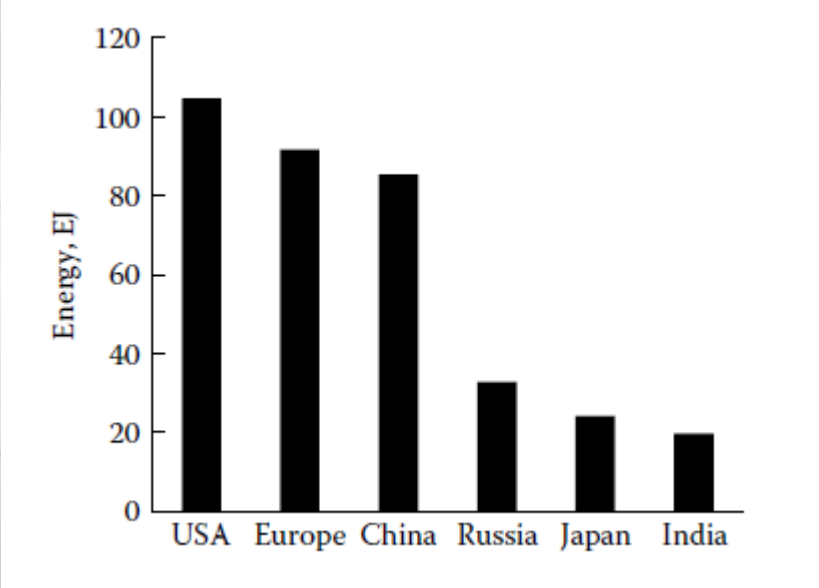
Energy sectors

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Sectoral shares in world primary energy

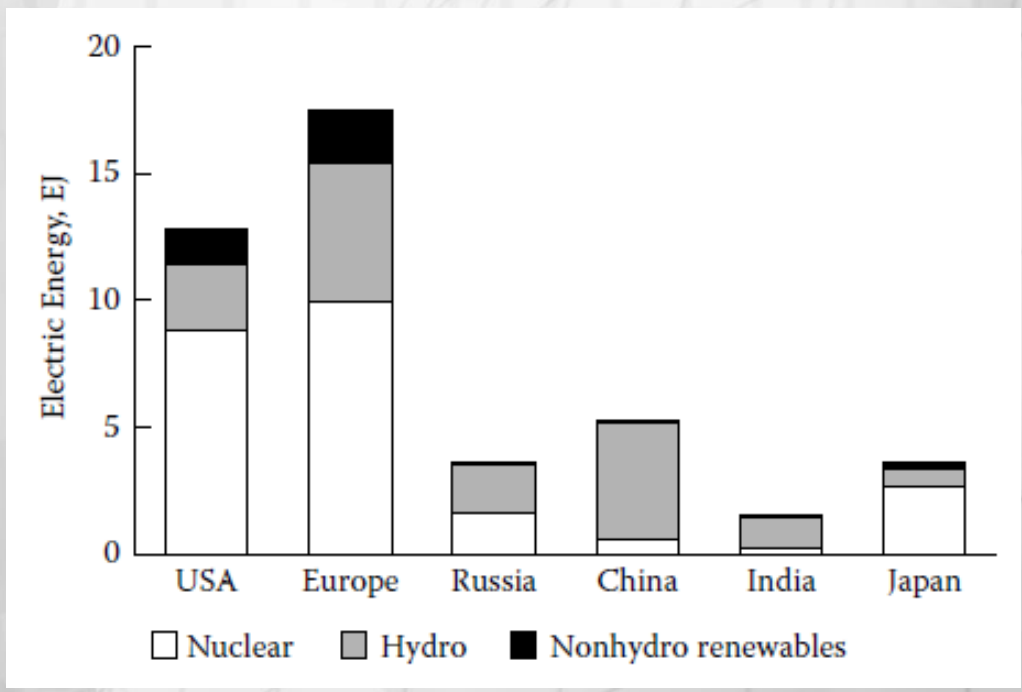


Consumption of world energy

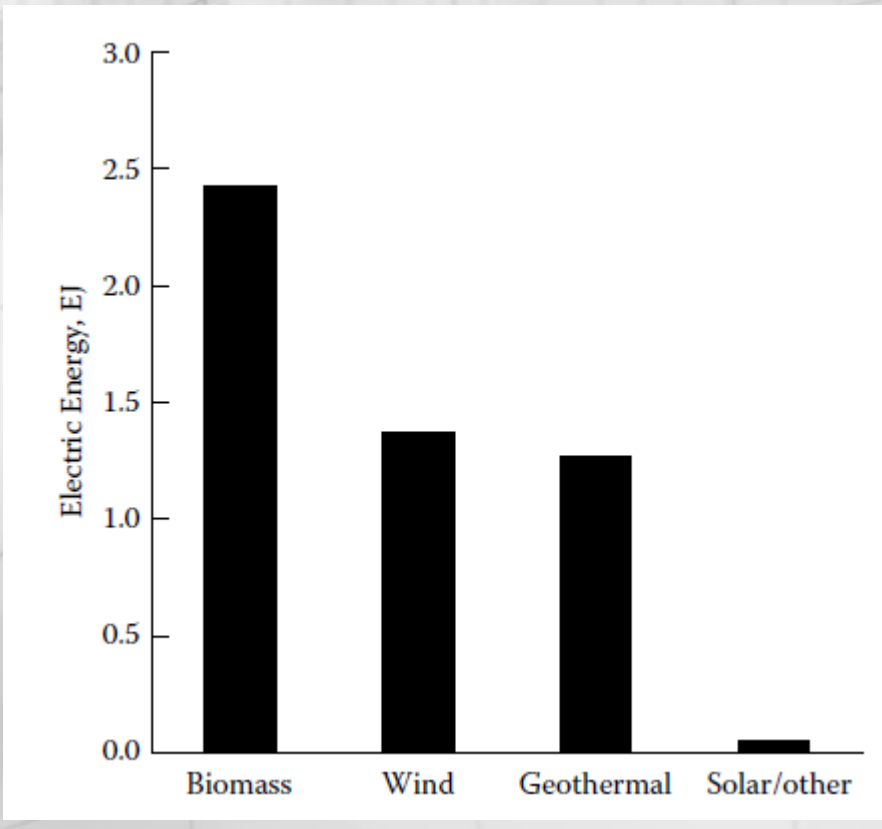


Consumption of energy, 2007

Energy sectors



Electric energy from nonfossil fuels



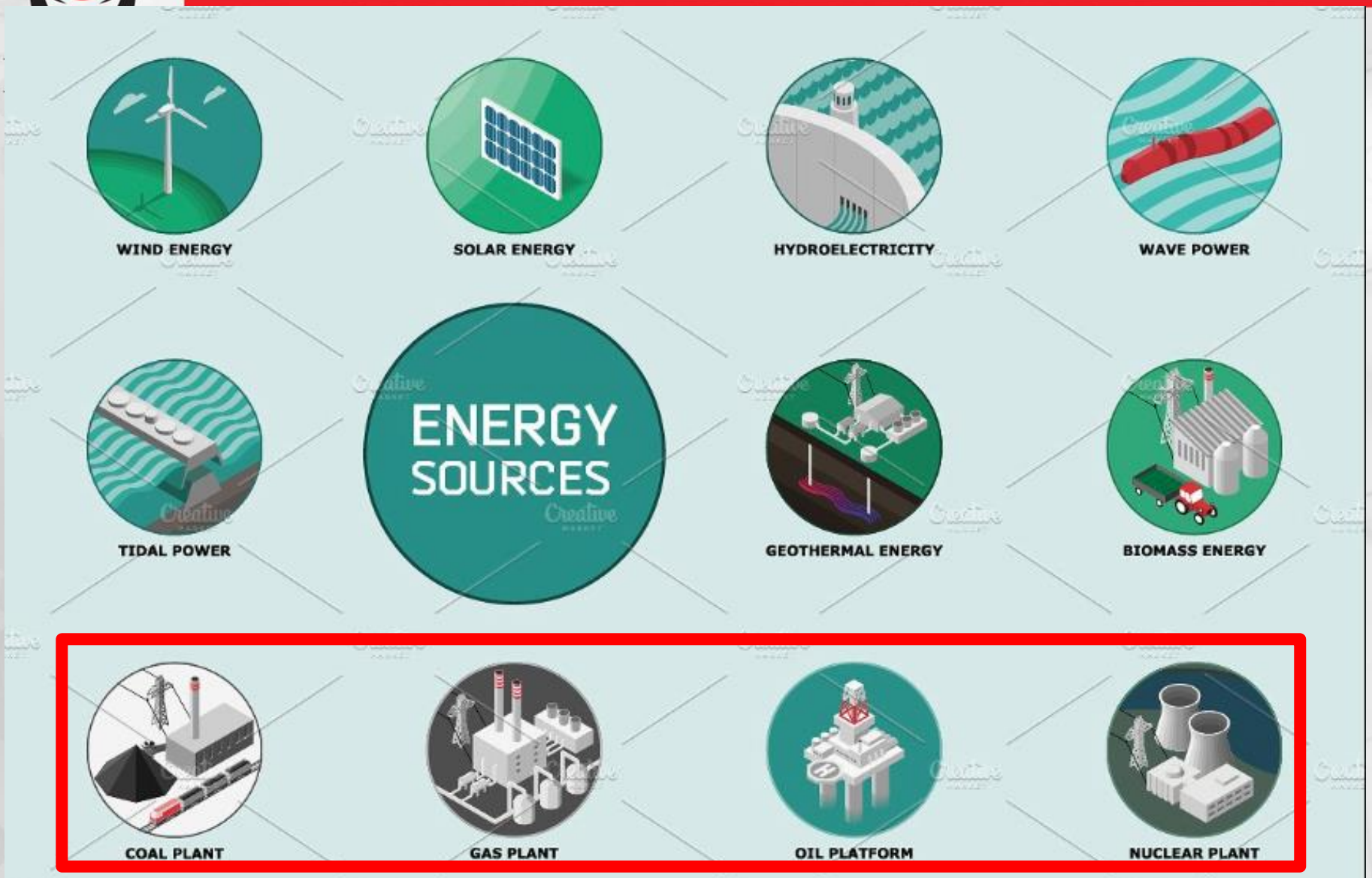
World electric energy from nonhydro renewables

Global Energy Demand

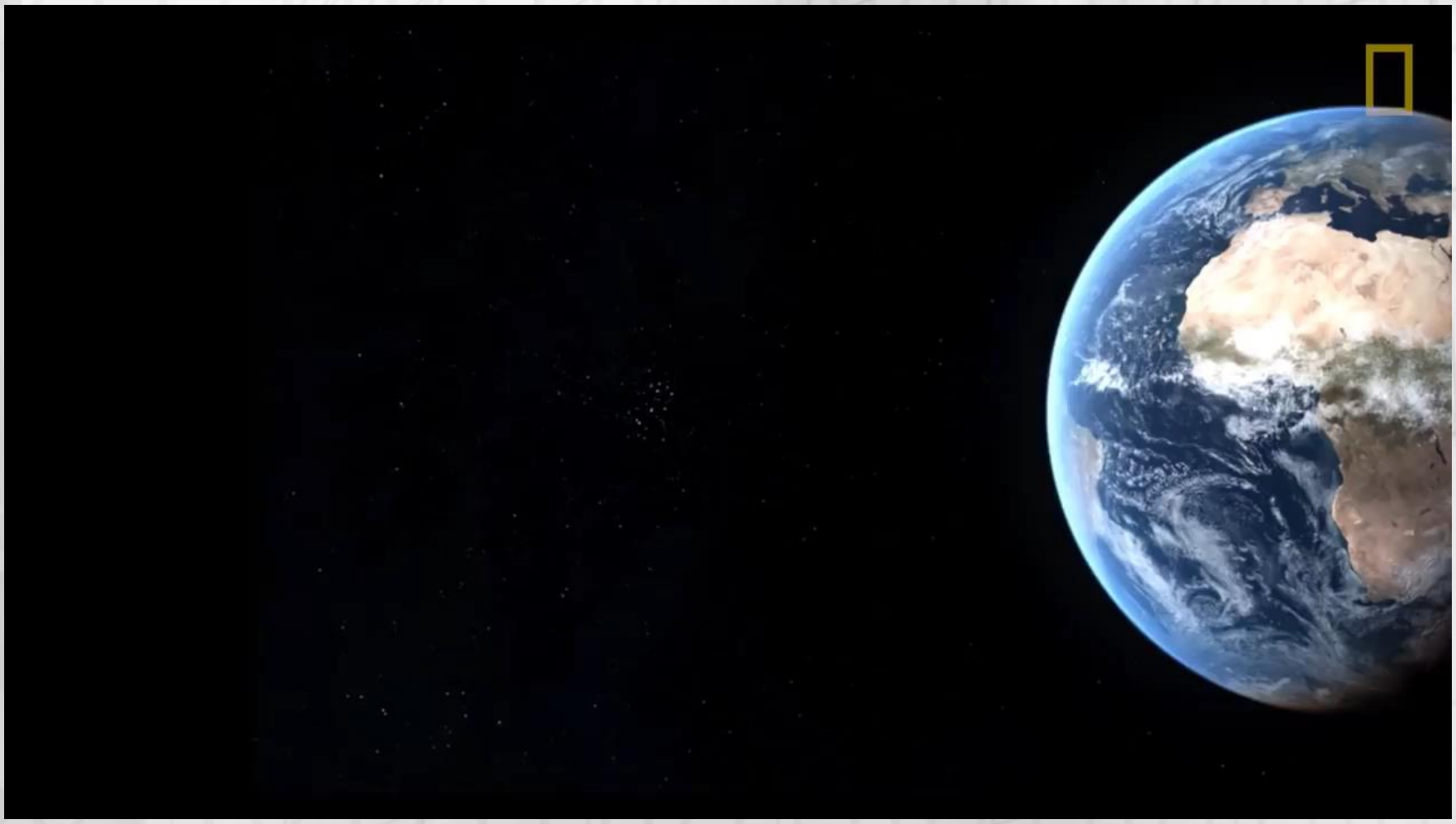
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Energy Resources

Energy Resources



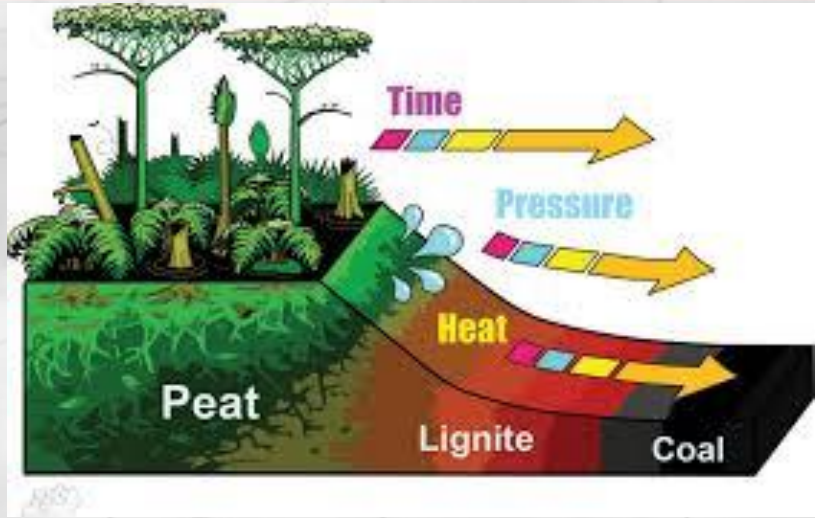
Energy Resources



Energy Resources

Conventional (Primary)	Renewable
✓Fossil fuels (oil, coal, natural gas, LPG)	✓Solar energy
✓Nuclear energy	✓Wind energy
	✓Hydropower
	✓Geothermal
	✓Biomass and municipal waste

Fossil Fuels: Coal



Coal is a sedimentary rock formed by the accumulation and decay of organic substances, derived from *plant tissues and exudates*, which have been buried over periods of time.



- ✓ Coal is classified by **type** (plant sources) and **rank** (the age of the coal, **coalification**).

Fossil Fuels: Coal

- ✓ The study of coal by type is known as coal petrography.
- ✓ Coal type is determined from the examination of polished sections of a coal sample using a reflected-light microscope.
- ✓ The degree of reflectance and the color of a sample are identified with specific residues of the original plant tissues.

Coal rank

- ✓ Coal rank is the most important property of coal because rank initiates the classification of coal for use.
- ✓ Coalification describes the process that the buried organic matter undergoes to become coal.
- ✓ When first buried, the organic matter has a certain elemental composition and organic structure.
- ✓ However, as the material becomes subjected to heat and pressure, the composition and structure slowly change.
- ✓ Certain structures are broken down, and others are formed.
- ✓ Some elements are *lost* through volatilization, while others are *concentrated* through a number of processes, including exposure to underground flows.
- ✓ Thus, coal rank could be determined through the measurement of one or more of the changing properties.

Fossil Fuels: Coal

- ✓ The rank classification scheme defined by the American Society of Testing and Materials (ASTM) has become the standard.
- ✓ In this scheme, the properties of **gross calorific value** and **fixed carbon** or **volatile matter** content are used to classify a coal by rank.
- ✓ Gross calorific value is a measure of the energy content of the coal and is usually expressed in units of energy per unit mass.
- ✓ Calorific value increases as the coal proceeds through coalification.
- ✓ Fixed carbon content is a measure of the mass remaining after heating a dry coal sample under conditions specified by the ASTM.
- ✓ The rank of a coal proceeds from lignite, the "youngest" coal, through subbituminous, bituminous, and semibituminous, to anthracite, the "oldest" coal.

Coal analysis and properties

Proximate analysis

- ✓ Volatile matter
- ✓ Moisture content
- ✓ Fixed carbon
- ✓ Ash content

- ✓ ASTM D 3174, 3173

Ultimate (elemental) analysis

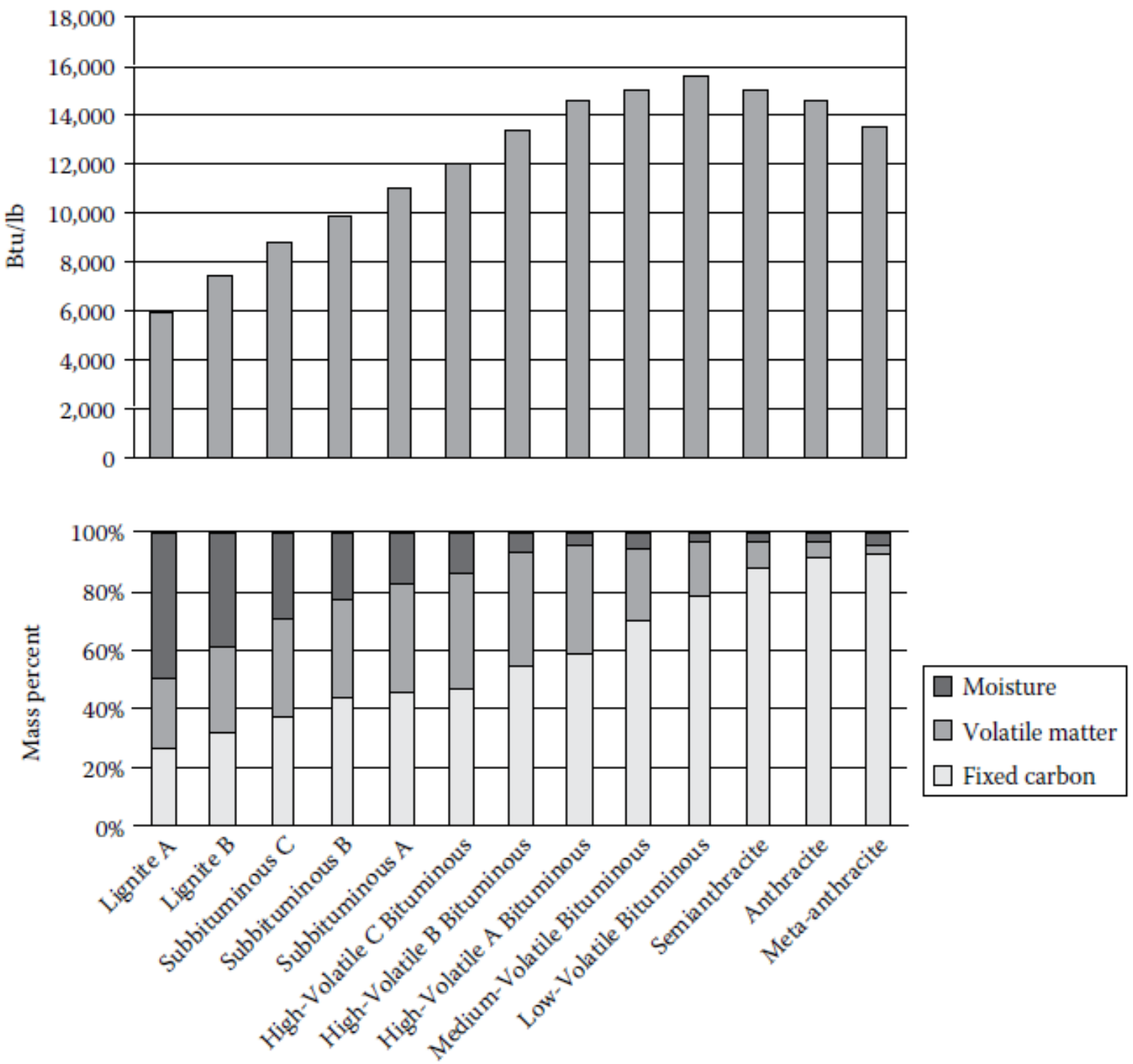
- ✓ Carbon
- ✓ Hydrogen
- ✓ Nitrogen
- ✓ Sulfur
- ✓ Oxygen

ASTM D 3176

Proximate and ultimate analysis

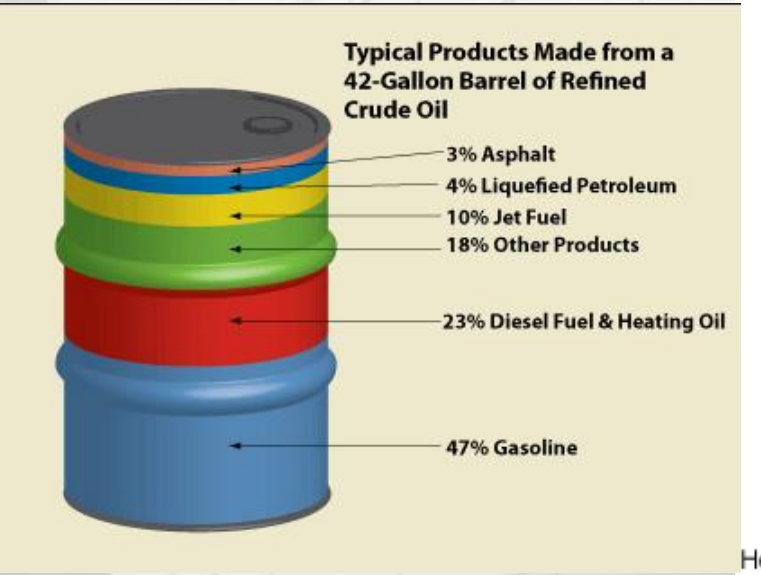
- ✓ Proximate analysis is measured by heating coal under controlled temperature and heating rate conditions.
- ✓ The total weight loss from holding the coal temperature at 100°C represents its moisture content.
- ✓ Volatile matter is the fraction of coal that decomposes into gases at moderate temperatures of about 400°C in an inert environment.
- ✓ The remaining fraction is a mixture of solid carbon (fixed carbon) and mineral matter (ash).
- ✓ The ash content can be determined by introducing oxygen and burning the remaining carbon material.

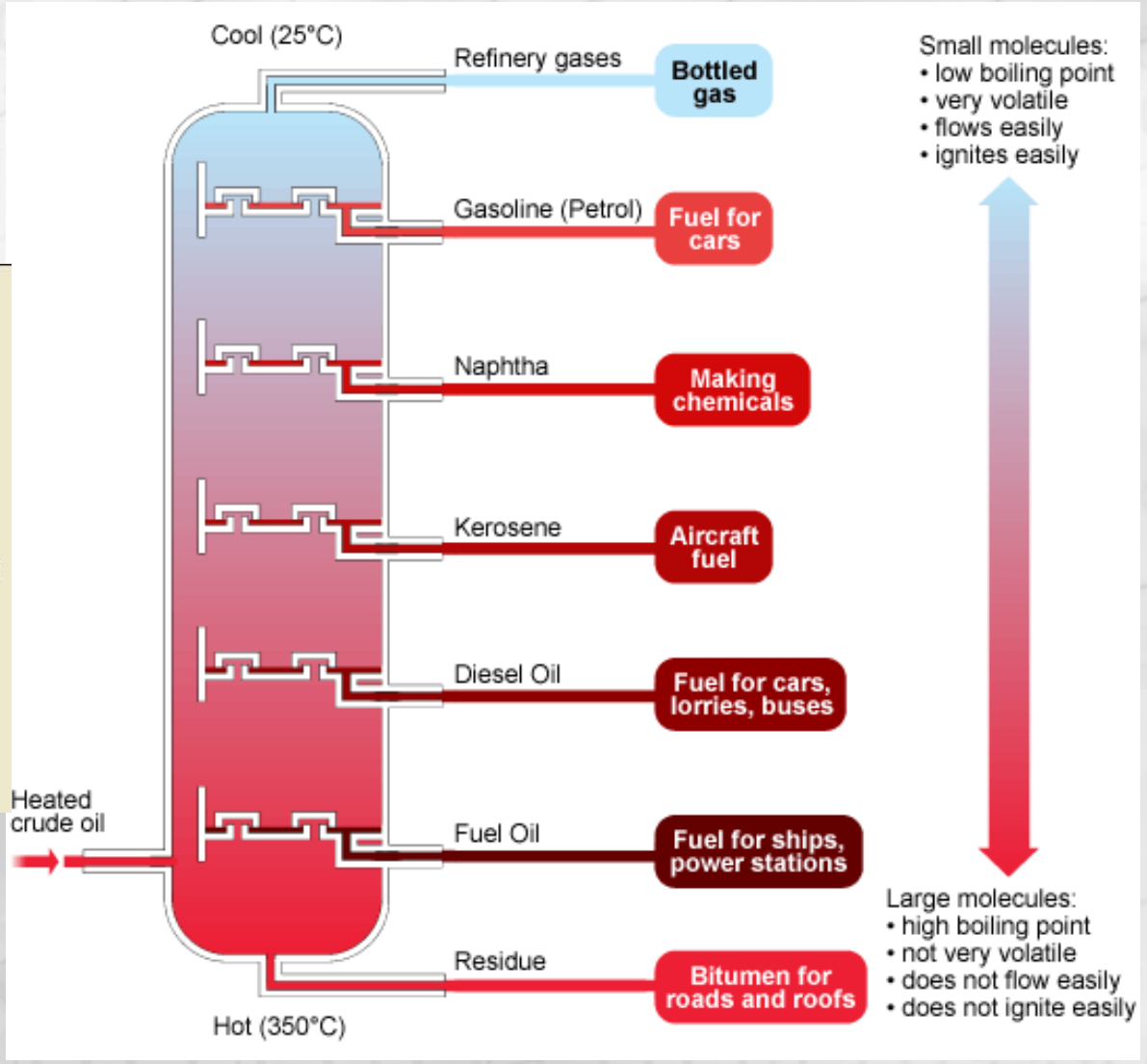
Ultimate analysis is often reported on a dry, ash-free (daf) basis and often used to estimate thermal biomass properties.


18,000
16,000
14,000
12,000
10,000
8,000
6,000
4,000
2,000
0
Btu/lb
100%
80%
60%
40%
20%
0%
Mass percent
Moisture
Volatile matter
Fixed carbon
Lignite A
Lignite B
Subbituminous C
Subbituminous B
Subbituminous A
High-Volatile C Bituminous
High-Volatile B Bituminous
High-Volatile A Bituminous
Medium-Volatile Bituminous
Low-Volatile Bituminous
Semianthracite
Anthracite
Meta-anthracite

Oil

Distillation tower



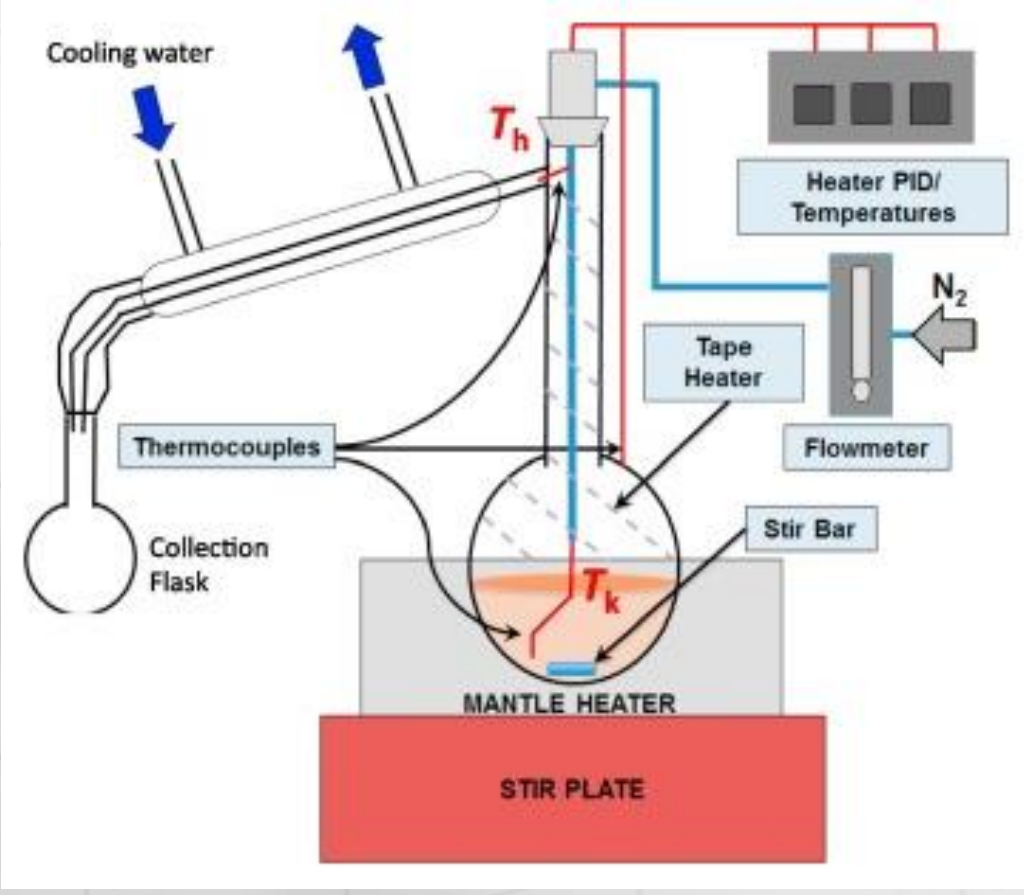


Oil

Liquid hydrocarbon fuels basically are mixtures of specific compounds from the following inorganic families:

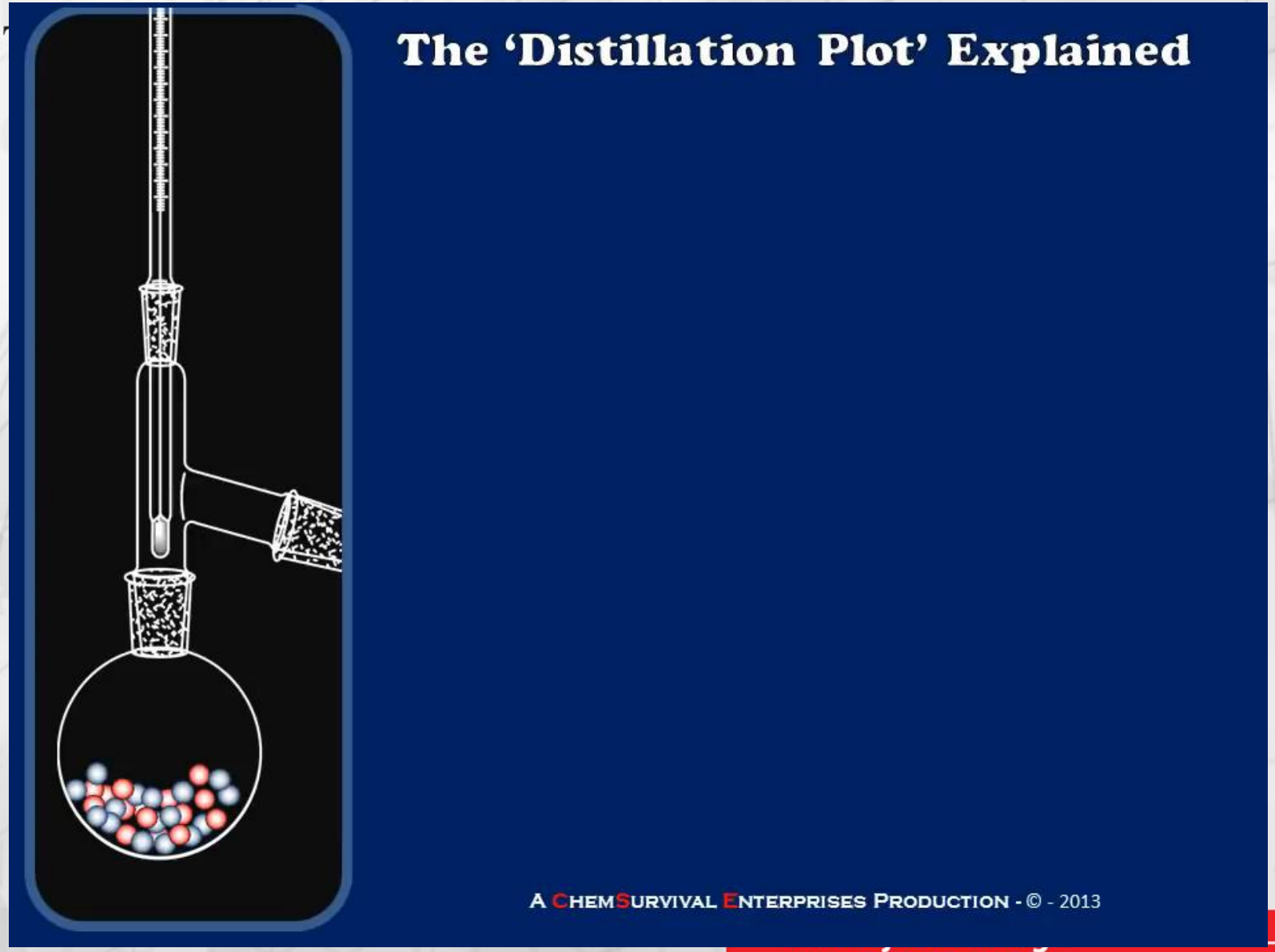
- ✓ Paraffins
- ✓ Olefins
- ✓ Naphthenes
- ✓ Aromatics

Fractional distillation separates high-boiling-point hydrocarbons from those with lower boiling points.

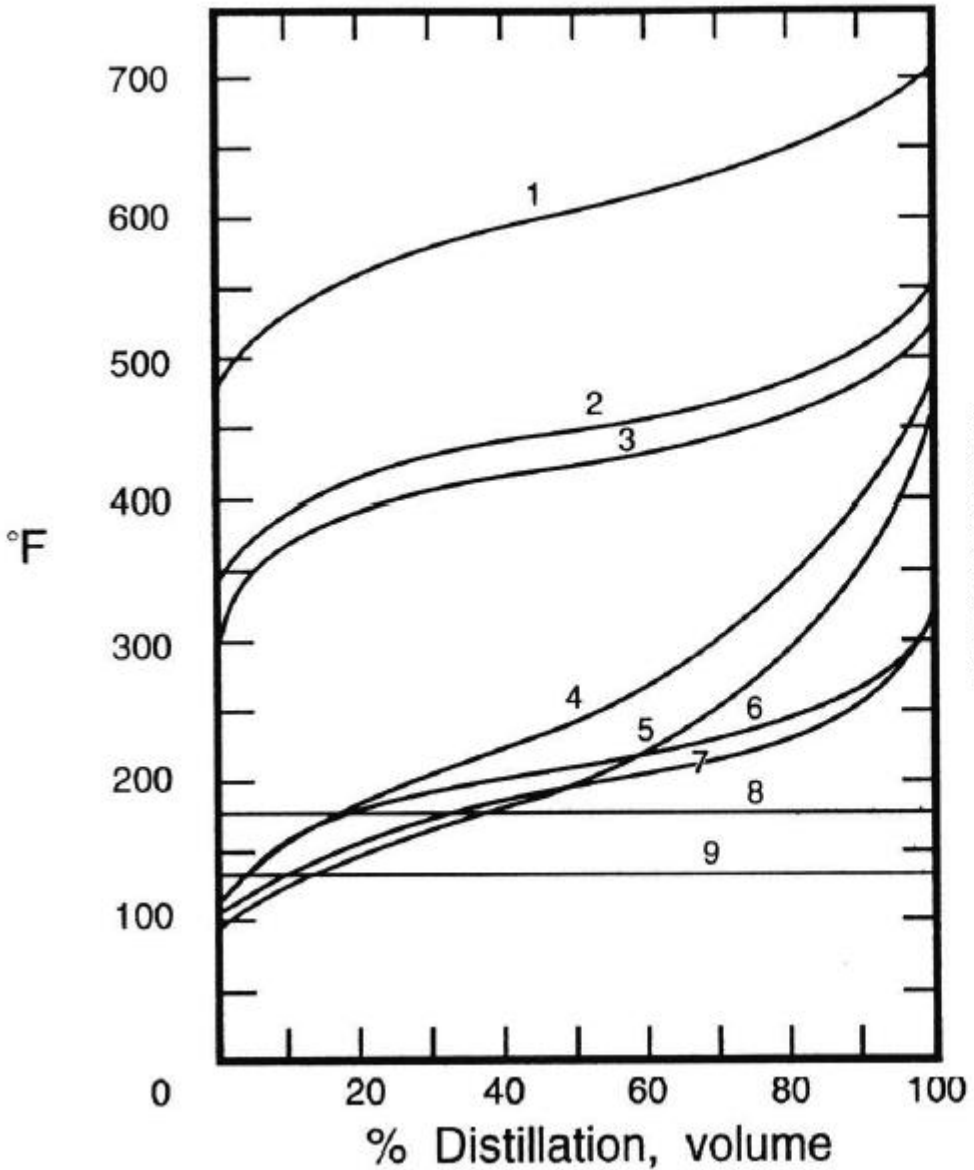


Distillation apparatus

Oil


The 'Distillation Plot' Explained
A ChemSurvival Enterprises Production - © - 2013

Oil



1. heavy-diesel oil
2. distillate
3. kerosene
4. summer gasoline
5. winter gasoline
6. aviation gasoline
7. low exhaust emission gasoline
8. ethyl alcohol
9. benzene

Oil: properties

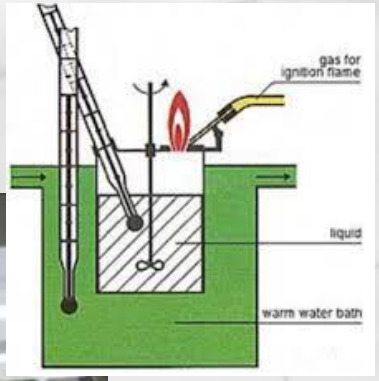
Specific gravity

Liquid fuel densities can be expressed in dimensionless form in terms of their *specific gravity,* that is, density of the reference liquid liquid at a specified temperature and pressure to a (usually water).

Flash point

Flash point is the lowest temperature at which a fuel will liberate vapor at a sufficient rate such that the vapor will form a mixture with ambient air that will ignite in the presence of an ignition source.





Oil: properties

Pour point





It is defined as the lowest temperature (in F or C) at which a liquid will flow easily (meaning it still behaves as a fluid). Hence, pour point is a rough indication of the lowest temperature at which oil is readily pumped.

Cloud point

- ✓ It is the temperature at which wax crystals begin to form in a petroleum product as it is cooled. Wax crystals depend on nucleation sites to initiate growth.
- ✓ The difference in the cloud points between two samples can sometimes be explained by the fact that any fuel additive will increase the number of nucleation sites, which initiates clouding.



Natural gas

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- ✓ Natural gas occurs deep beneath the earth's surface.
- ✓ Natural gas consists mainly of methane, a compound with one carbon atom and four hydrogen atoms.
- ✓ Natural gas has been called the environmentally friendly fossil fuel because it releases fewer harmful contaminants.
- ✓ World production of dry natural gas was 73.7 trillion ft^3 and accounted for over 20% of world energy production

- ✓ **CNG? LNG? LPG?**

World Natural Gas Reserves by Country as of January 1, 2003

Country	Reserves (trillion ft^3)	Percent of World Total
World	5501	100.0
Top 20 countries	4879	88.7
Russia	1680	30.5
Iran	812	14.8
Qatar	509	9.2
Saudi Arabia	224	4.1
United Arab Emirates	212	3.9
U.S.	183	3.3
Algeria	160	2.9
Venezuela	148	2.7
Nigeria	124	2.3
Iraq	110	2.0
Indonesia	93	1.7
Australia	90	1.6
Norway	77	1.4
Malaysia	75	1.4
Turkmenistan	71	1.3
Uzbekistan	66	1.2
Kazakhstan	65	1.2
Netherlands	62	1.1
Canada	60	1.1
Egypt	59	1.1
Rest of World	622	11.3

Africa emerges as a new gas province thanks to large offshore developments in Mozambique and Tanzania. Egypt makes a comeback with the start of production from the major Zohr field, as does Argentina with the development of its promising shale gas resource in the

Ex 1

Determine the amount of carbon dioxide needed (in tonnes) to produce a tonne of glucose using

$$6CO_2 + 6H_2O + Solar\ Energy \rightarrow C_6H_{12}O_6 + 6O_2$$

$$\frac{6CO_2}{C_6H_{12}O_6} = \left[\frac{6\times(12+2\times(16))}{1\times[(6\times12)+(12\times1)+(6\times16)]}\right] = \frac{264\ g}{180\ g} = 1.5\frac{tonne\ CO_2}{tonne\ C_6H_{12}O_6}$$

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Conventional (Primary)

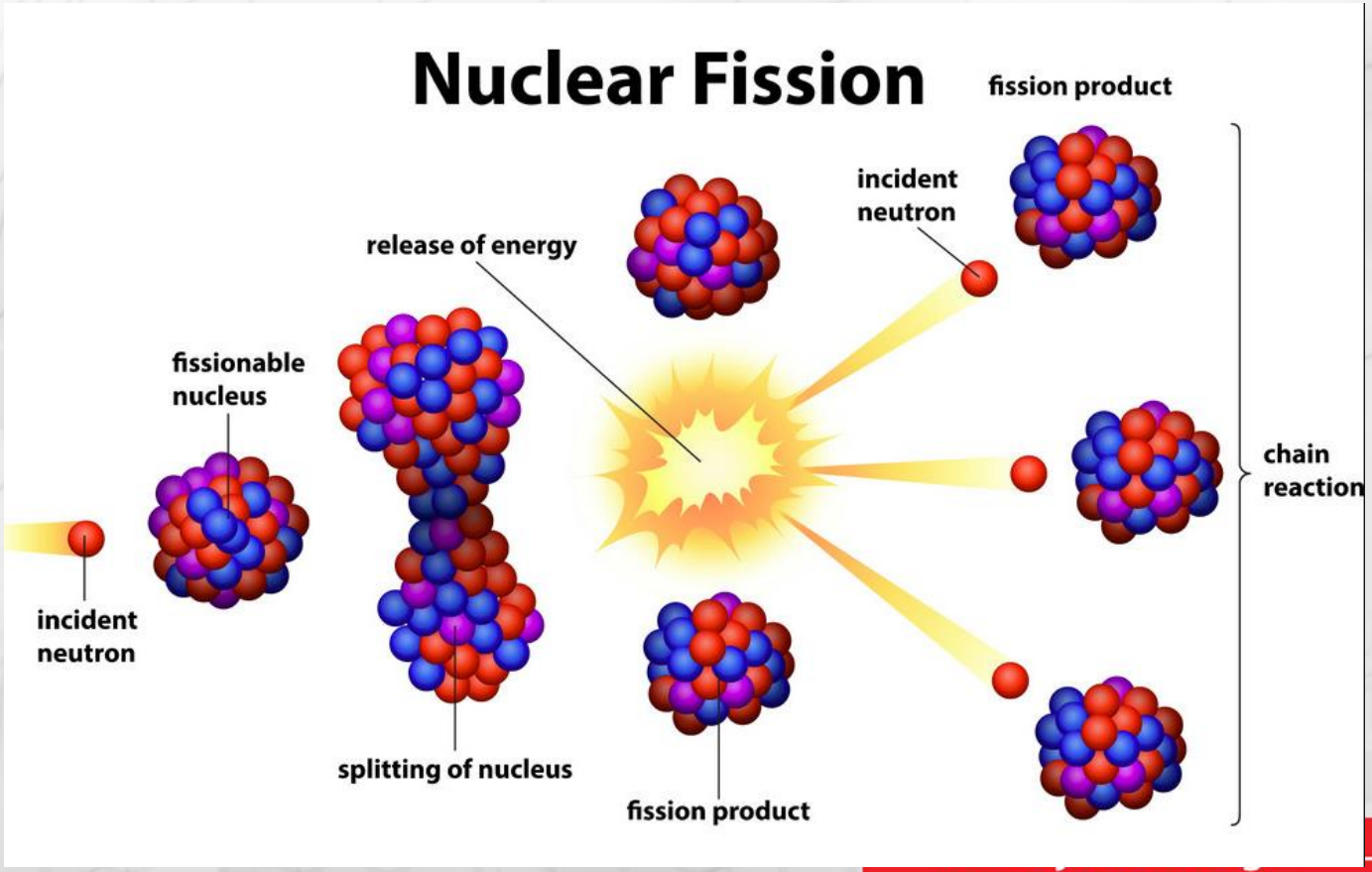
- Fossil fuels (oil, coal, natural gas, LPG)
- Nuclear energy

Nuclear power is derived from the fission of heavy element nuclei or the fusion of light element nuclei

Nuclear energy

Why is it conventional?

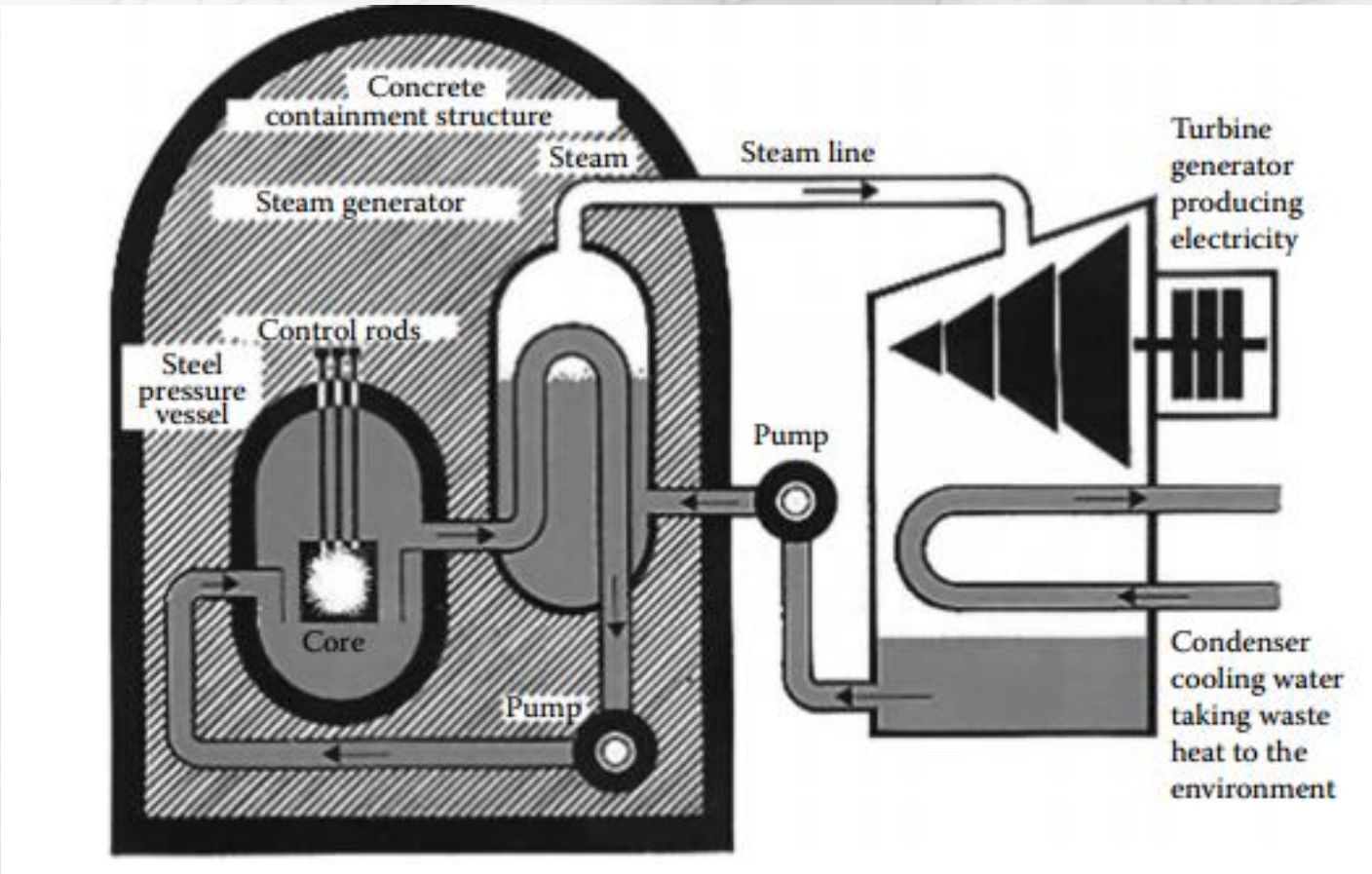
- ✓ Nuclear power can use two naturally occurring elements, uranium, and thorium, as the sources of its fissioning energy.



Nuclear energy



Pressurized water reactor power plant



Renewable energy

Renewable

- Solar energy
- Wind energy
- Hydropower
- Geothermal
- Biomass and municipal waste

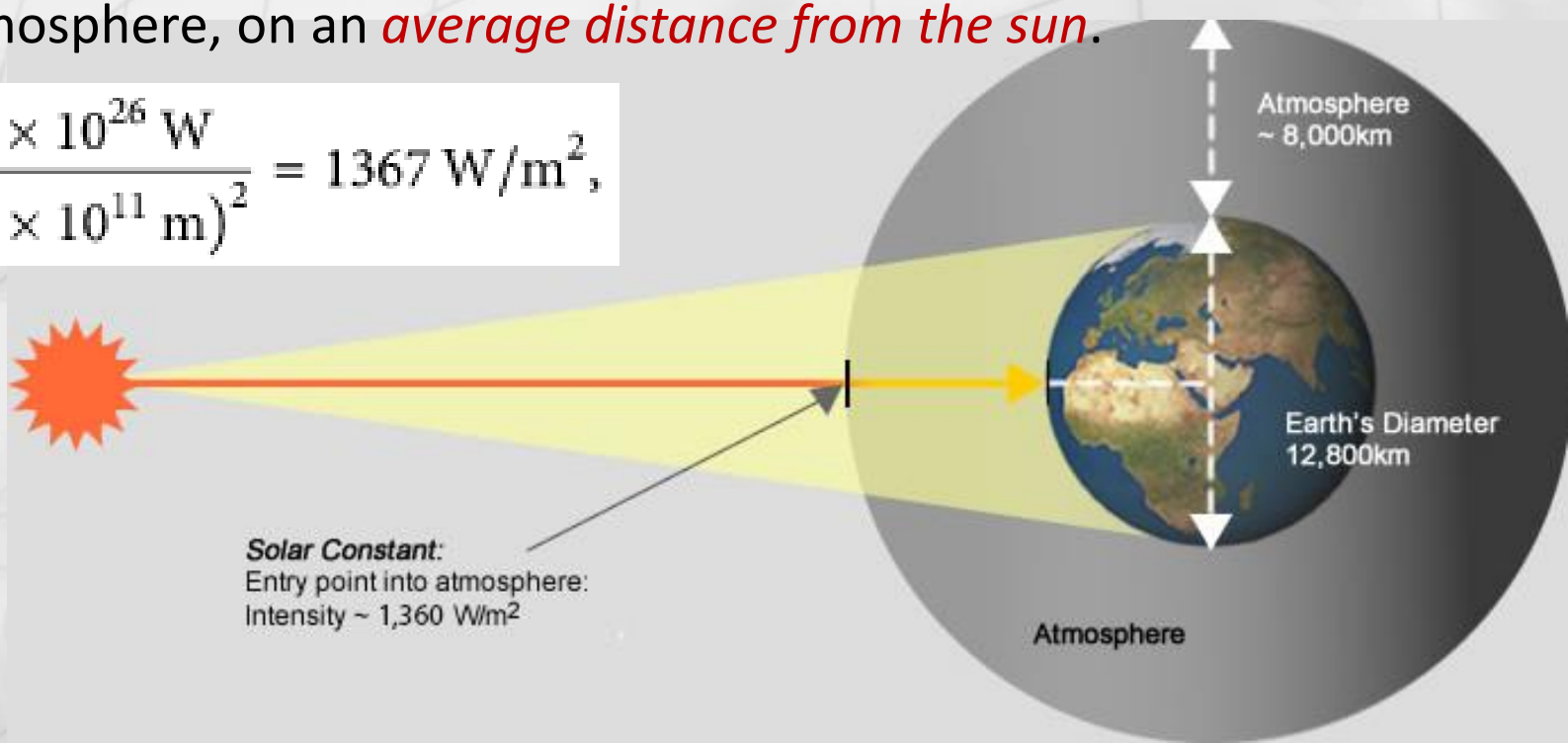


Solar Constant

- ✓ Solar energy is the world's most abundant permanent source of energy.
- ✓ The amount of solar energy intercepted by the planet Earth is a tiny fraction of the solar radiation emitted by the sun.

Solar constant (G_{sc}): The amount of energy received (solar irradiation) on a unit area of a surface perpendicular to the ray of the sun, outside of the earth's atmosphere, on an *average distance from the sun*.

$$\frac{3.865 \times 10^{26}\ \mathrm{W}}{4\pi\left(1.5 \times 10^{11}\ \mathrm{m}\right)^2} = 1367\ \mathrm{W/m^2},$$



Extraterrestrial solar irradiance

Extraterrestrial solar radiation: the amount of solar energy received in the earth's atmosphere, that varies at any given season, time of day, and time of year.

Given

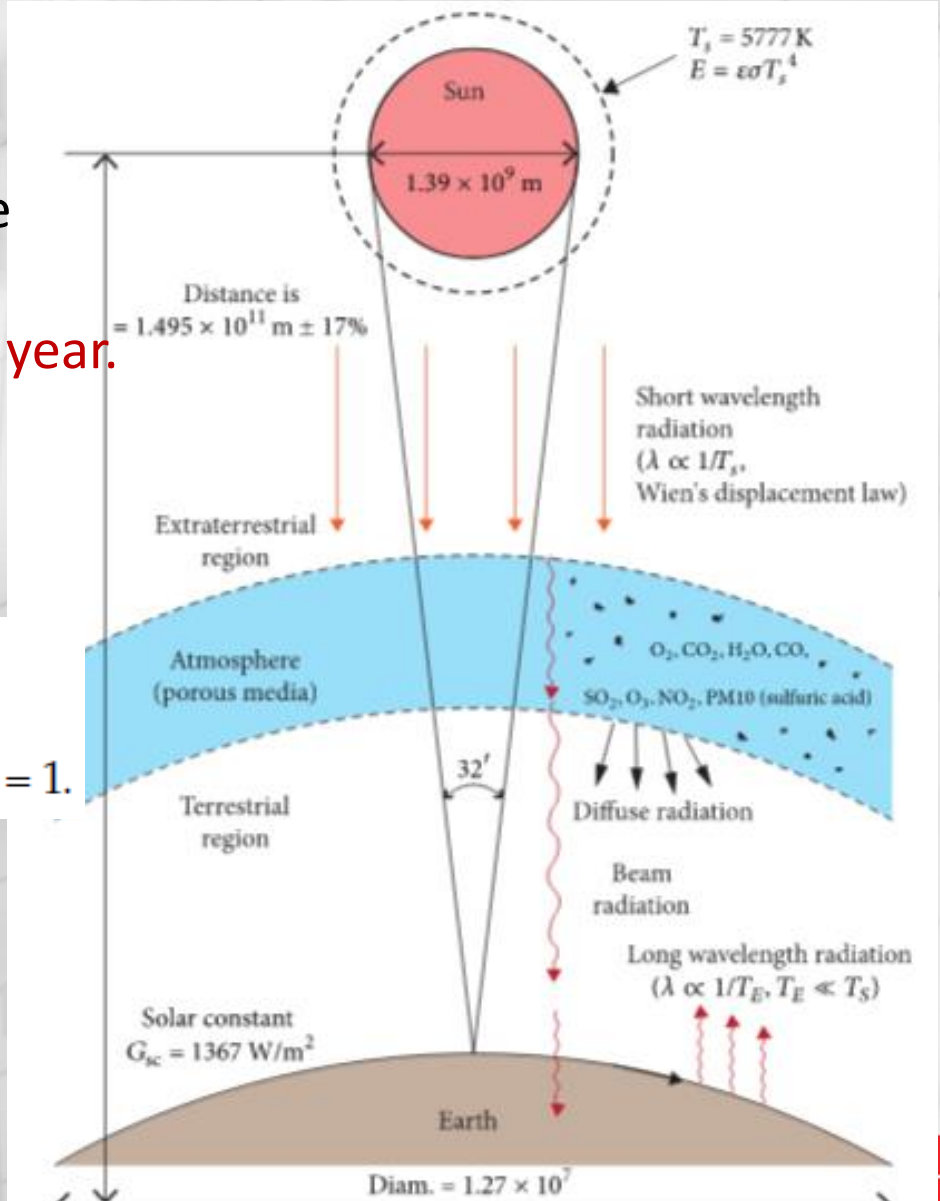
$$G_{on} = G_{sc} \times \left(1 + 0.033 \times cos\frac{360n}{365}\right)$$

where

G_{sc} = the solar constant, 1,367 W/m²

n = day of the year, with January 1 equal to $n = 1$.

You must convert degree angles into radians by multiplying the degree angle with **π/180.**



Extraterrestrial solar irradiance: Example

Determine the extraterrestrial solar radiation received on earth in January 1. Plot the variation of extraterrestrial solar radiation received with the time of year on a normal plane. Use $G_{sc} = 1{,}367\ W/m^2$.

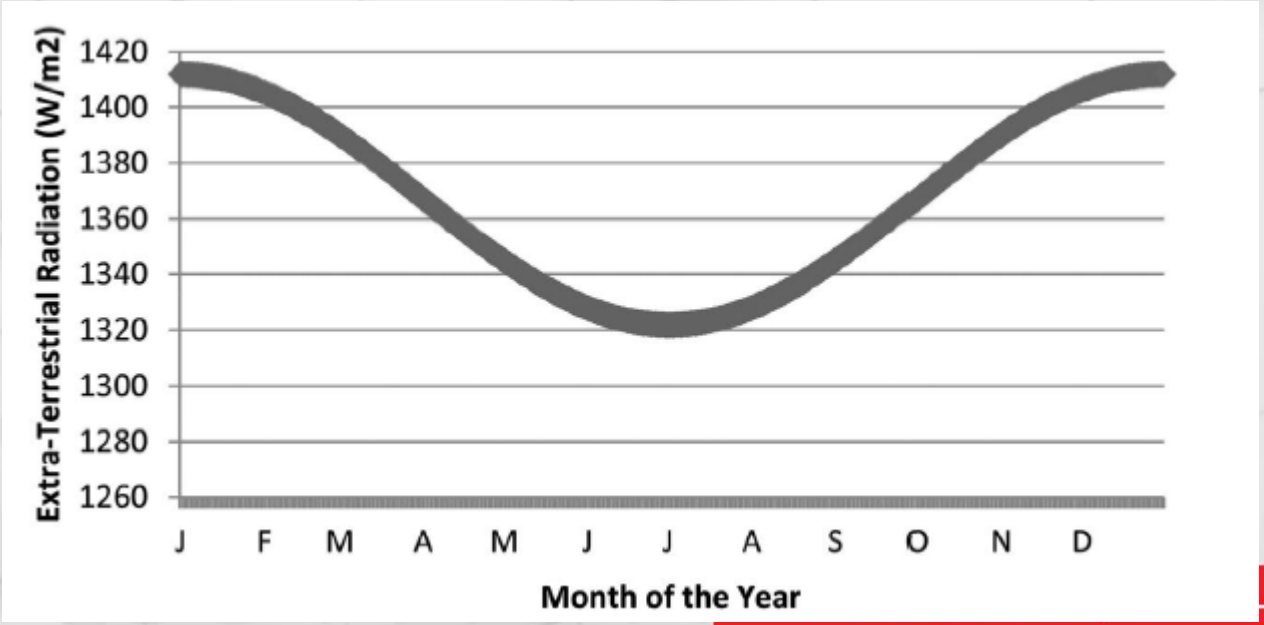
$$G_{on} = G_{sc} \times \left(1 + 0.033 \times cos\frac{360n}{365}\right)$$

$$G_{on} = 1{,}367 \times \left(1 + 0.033 \times cos\frac{360 \times 1}{365}\right) = 1{,}412\frac{W}{m^2}$$

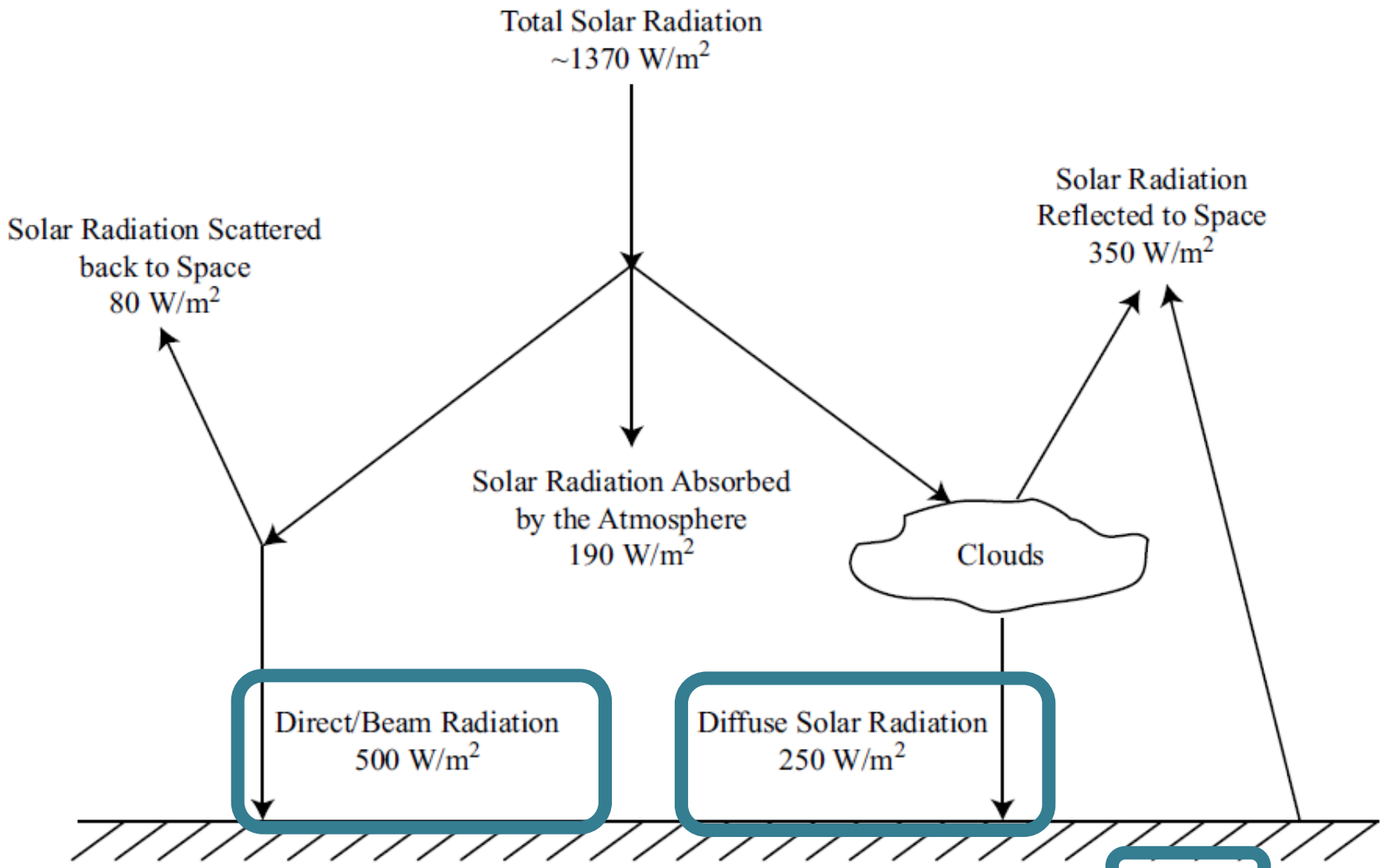
Table to Easily Estimate the Value of n for a Given (Non–Leap-Year) Month

Month	n for ith Day of Month	Month	n for ith Day of Month
January	i	July	$181 + i$
February	$31 + i$	August	$212 + i$
March	$59 + i$	September	$243 + i$
April	$90 + i$	October	$273 + i$
May	$120 + i$	November	$304 + i$
June	$151 + i$	December	$334 + i$

Given



Estimation of Terrestrial Solar Radiation

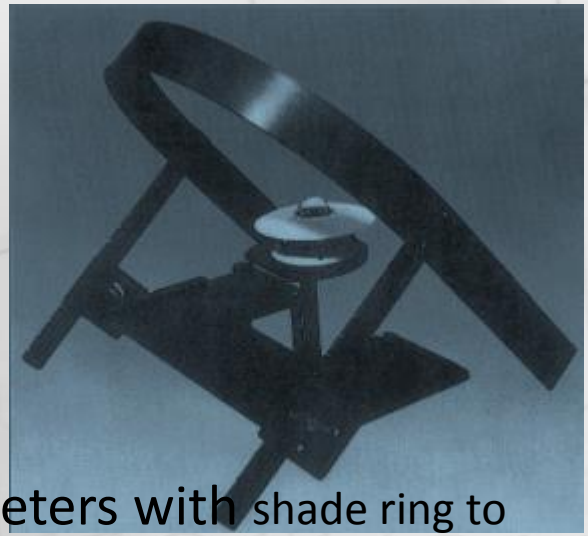


Instruments for Measuring Solar Radiation

- There are two basic types of instruments used to measure solar radiation, *pyranometer* and *pyrheliometer*.
- A **pyranometer** has a hemispherical view of the surroundings and therefore is used to measure total, direct and diffuse, solar radiation on a surface.
- A pyrheliometer, on the other hand, has a restricted view (about 5°) and is, therefore, often used to measure the direct or beam solar radiation by pointing it toward the sun.



Pyranometers with thermal detector

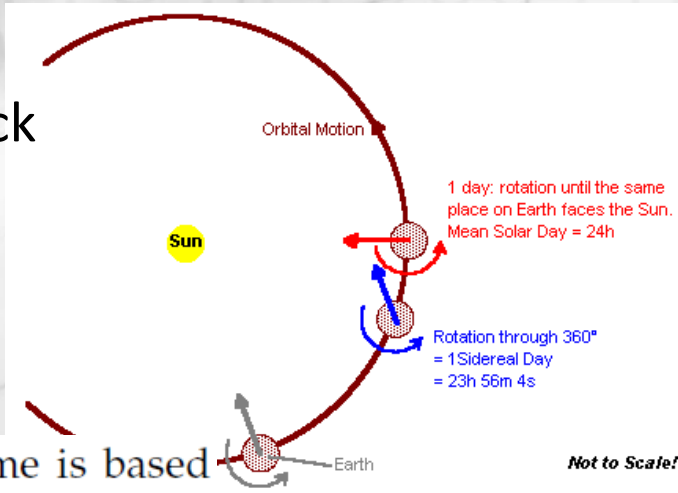


Pyranometers with shade ring to measure sky diffuse radiation

Solar Time

Solar time: the time based on the apparent angular motion of the sun across the sky with solar noon, the time the sun crosses the meridian of the observer.

Solar time is not the same as our standard or clock time.



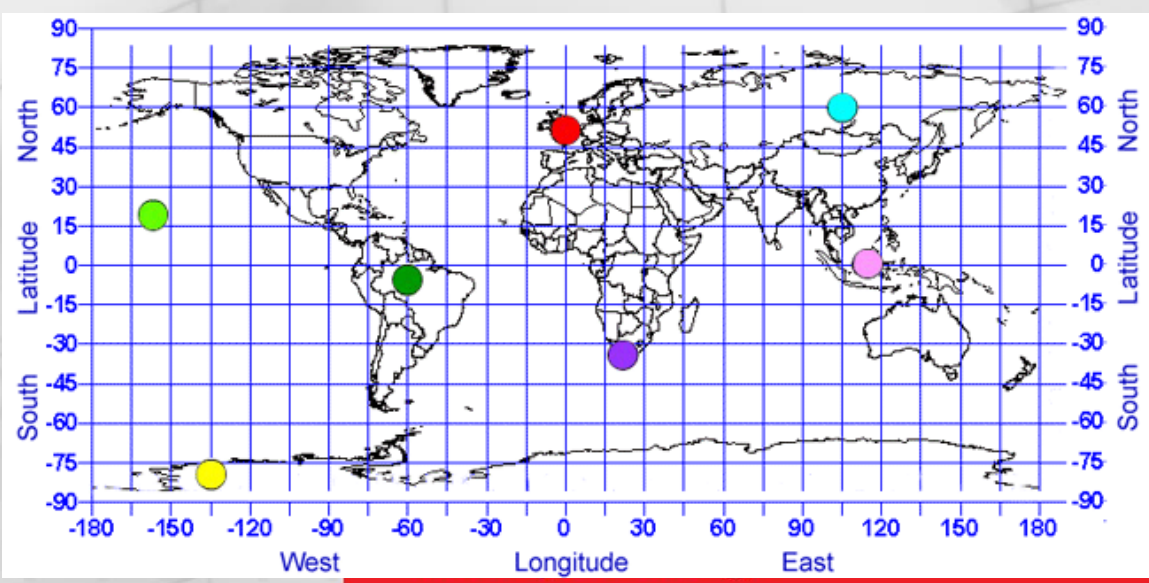
$$\text{Solar Time} = \text{Standard Time} + 4 \times (L_{std} - L_{loc}) + E$$

standard time = our clock time

L_{std} = meridian or standard longitude on which the local time is based

L_{loc} = meridian of the observer

E = the equation of time, in minutes,



Egypt Meridian: 30, 30

Solar Time

$$\text{Solar Time} = \text{Standard Time} + 4 \times (L_{std} - L_{loc}) + E$$

Longitude correction

- The standard clock time is reckoned from a selected meridian near the center of a time zone or from the **standard meridian, the Greenwich**, which is at longitude of 0.
- Since the sun takes 4 min to transverse 1 of longitude, a *longitude correction term of 4* (Standard longitude [SL], Local longitude [LL]) should be either added or subtracted to the standard clock time of the locality.

- This correction is constant for a particular longitude, and the following rule must be followed with respect to sign convention.
- If the location is **east** of the standard meridian (**Greenwich**) , the correction is **added** to the clock time.
- If the location is west, it is subtracted.
- The general equation for calculating the AST is:

DS: Day light saving (60 min)

Given

$$\text{AST} = \text{LST} + \text{ET} \pm 4(\text{SL} - \text{LL}) - \text{DS}$$

Solar Time

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E = the equation of time, in minutes

$$E = 229.2 \times (0.000075 + 0.00186 \cos B - 0.032077 \sin B - 0.014615 \cos 2B - 0.04089 \sin 2B)$$

$$B = (n-1) \times \frac{360}{365}, \ in\ degrees$$

City	Coordinates
Cairo	30.06263, 31.24967
Alexandria	31.20176, 29.91582
Giza	30.00944, 31.20861
Port Said	31.26531, 32.3019
Suez	29.97371, 32.52627
Luxor	25.69893, 32.6421

Example: Find the equation of AST for the city of Nicosia, Cyprus.

For the locality of Cyprus, the SL is 30°E. The city of Nicosia is at a LL of 33.33° east of Greenwich. Therefore, the longitude correction is $-4 \times (30-33.33) = +13.32$ min. Thus, Eq (2.3) can be written as:

$$\text{AST} = \text{LST} + \text{ET} + 13.32 \ (\text{min})$$

Solar Time: Example

Determine the solar time corresponding to 10:30 a.m. Central Standard Time on October 25, 2014, in College Station, Texas. The longitude of College Station, Texas, is 96.32 West, and this town's time was based on a 90° meridian.

$$\text{AST} = \text{LST} + \text{ET} \pm 4(\text{SL} - \text{LL}) - \text{DS}$$

$$E = 229.2 \times (0.000075 + 0.00186 \cos B - 0.032077 \sin B - 0.014615 \cos 2B - 0.04089 \sin 2B)$$

$$B = (n-1) \times \frac{360}{365}, \text{ in degrees}$$

a. For October 25, 2014, $n = 298$

b. B is then calculated using

$$B = (n-1) \times \frac{360}{365} = (298-1) \times \frac{360}{365} = 292.9°$$

The equation of time E is then calculated using

$$E = 229.2 \times (0.000075 + 0.00186 \cos B - 0.032077 \sin B - 0.014615 \cos 2B - 0.04089 \sin 2B)$$

$$E = 229.2 \times \left(0.000075 + 0.00186 \cos(292.9) - 0.032077 \sin(292.9) - 0.014615 \cos\left(2 \times (292.9)\right) - 0.04089 \sin\left(2 \times (292.9)\right)\right) = 16.01$$

Solar Time: Example

Determine the solar time corresponding to 10:30 a.m. Central Standard Time on October 25, 2014, in College Station, Texas. The longitude of College Station, Texas, is 96.32 West, and this town's time was based on a 90° meridian.

d. The longitude correction is then calculated:

$$\text{Longitude Correction} = 4 \times (90 - 96.32) = -25.28 \text{ minutes}$$

$$\text{AST} = \text{LST} + \text{ET} \pm 4(\text{SL} - \text{LL}) - \text{DS}$$

e. The solar time is calculated as follows:

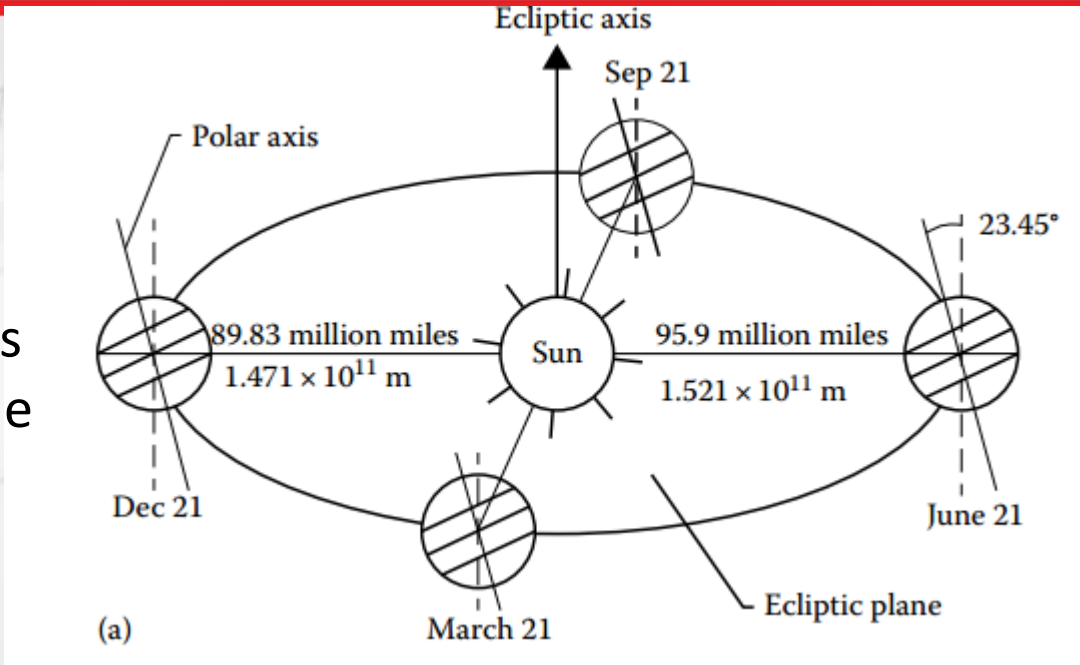
$$\text{Solar Time} = 10.50 + (-25.28/60) + (16.01/60) = 10.35$$

f. Note that the longitude correction and the equation of time, *E* is converted into units of hours to be consistent with the calculation of solar time. The fraction may be brought back into units of minutes by multiplying this by 60 minutes to an hour.

g. So, the solar time is actually 10:21 a.m.

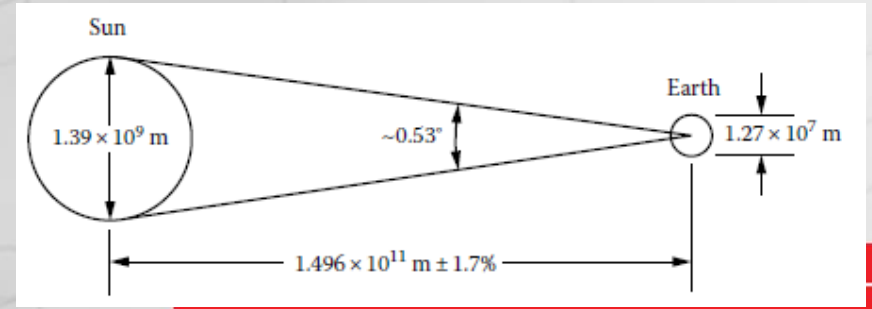
Sun-Earth Geometric Relationship

- ✓ The axis of the earth's daily rotation around itself is at an angle of 23.45° to the axis of its ecliptic orbital plane around the sun.

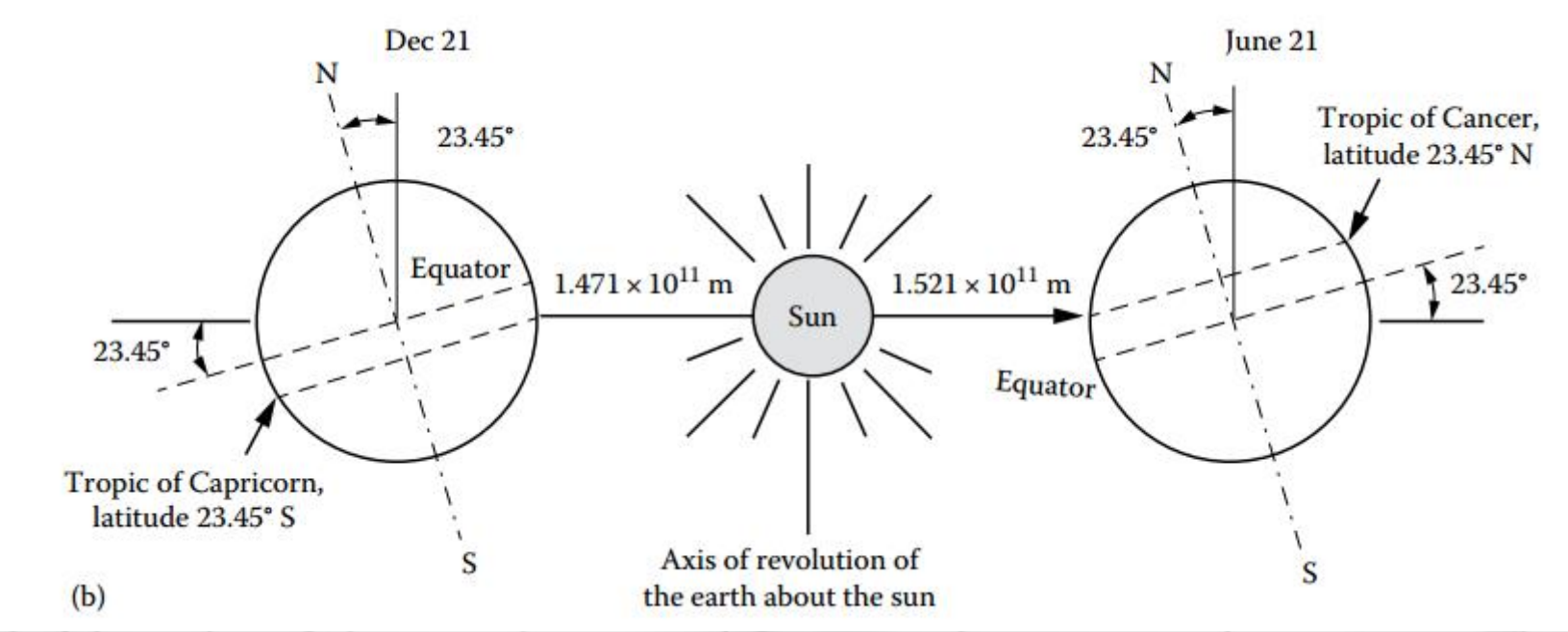


The annual orbit of the earth around the sun.

- ✓ This tilt is the major cause of the seasonal variation of the solar radiation available at any location on the earth.



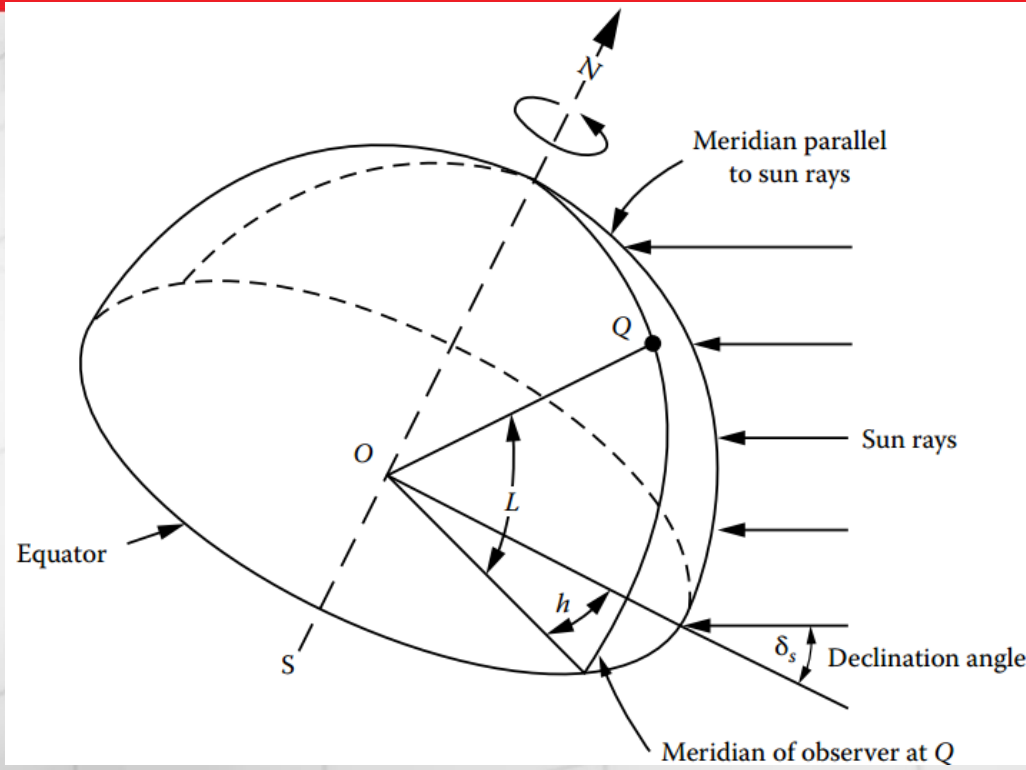
Sun-Earth Geometric Relationship



- ✓ The angle between the earth–sun line (through their centers) and the plane through the equator is called the *solar declination*, δ_s.
- ✓ The declination varies between -23.45° on December 21 and +23.45° on June 21.
- ✓ The declination has the same numerical value as the latitude at which the sun is directly overhead at solar noon on a given day.
- ✓ The tropics of Cancer (23.45° N) and Capricorn (23.45° S) are at the extreme latitudes where the sun is overhead at least once a year

Sun-Earth Geometric Relationship

- The latitude angle *L* (*φ*)is the angle between the line from the center of the earth to the *site* and the *equatorial plane*.
- N +ve, S -ve



Sun-Earth Geometric Relationship

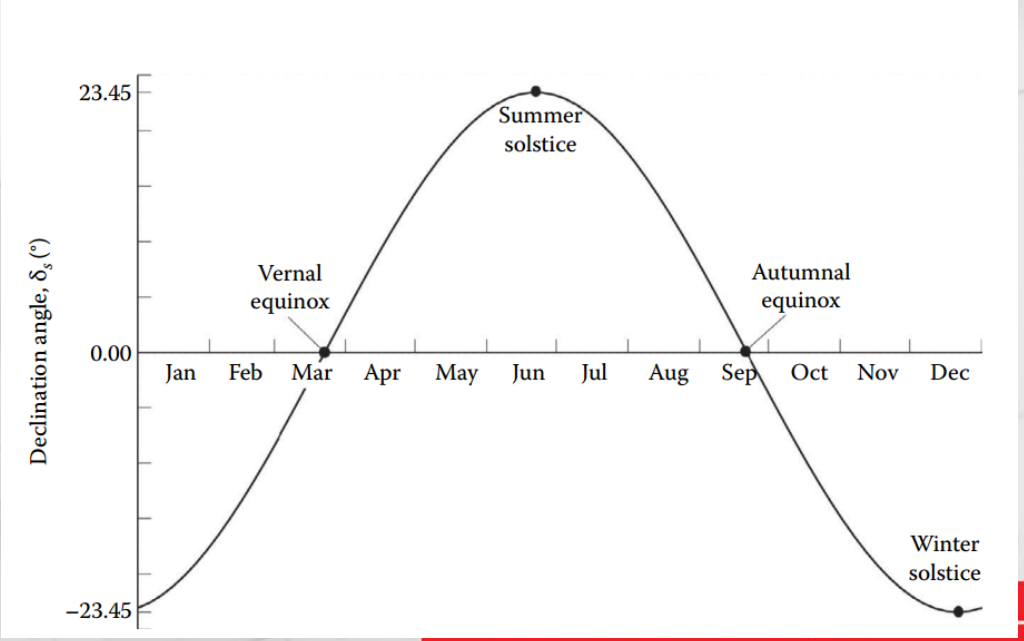
E-JUST

- ✓ Declinations north of the equator (summer in the northern hemisphere) are positive; those south, negative.
- ✓ The solar declination may be estimated by the relation

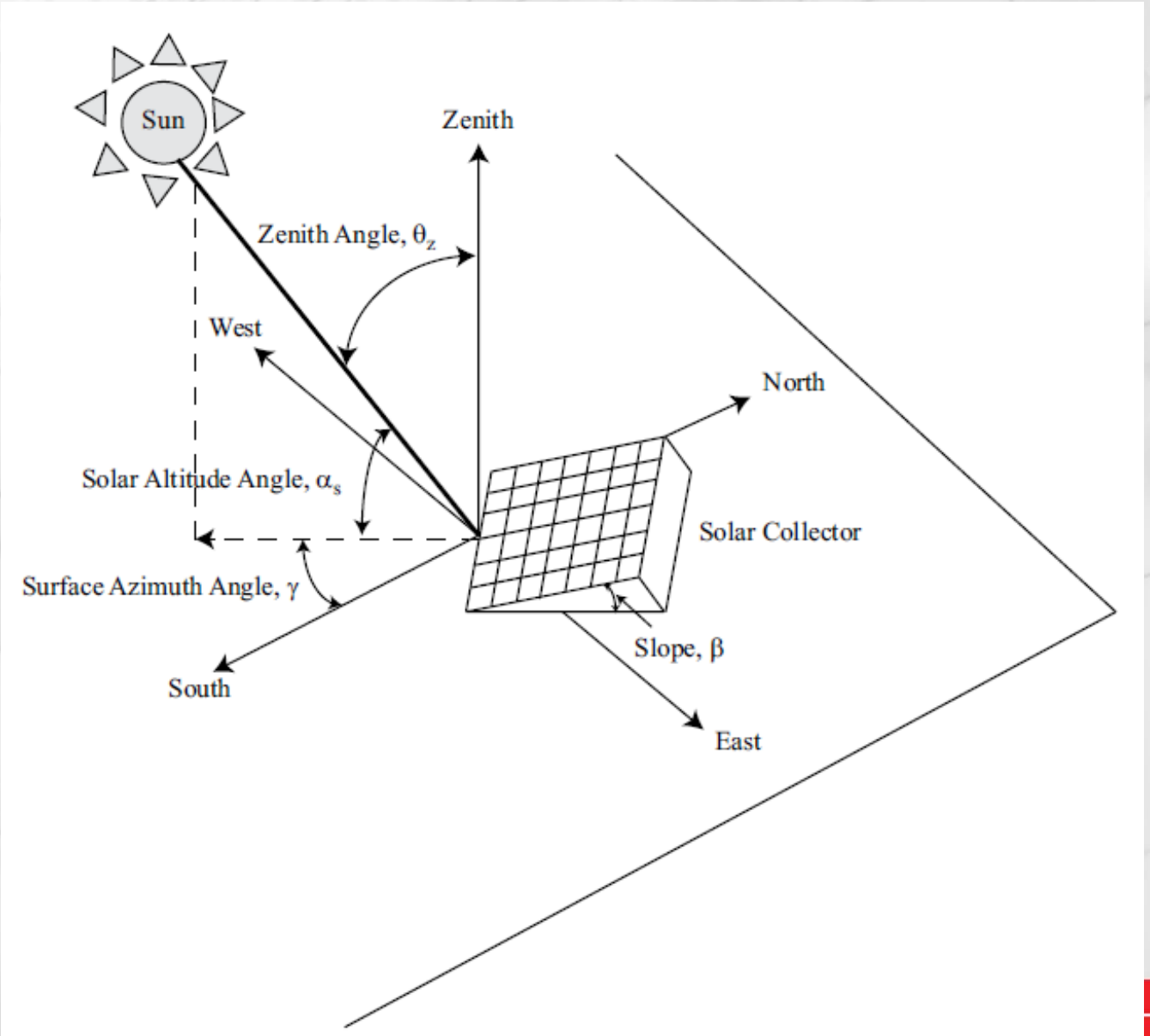
$$\delta = 23.45 \times sin\left(360 \times \frac{284 + n}{365}\right)$$

Given

where *n* is the day number during a year with January 1 being *n* = 1.



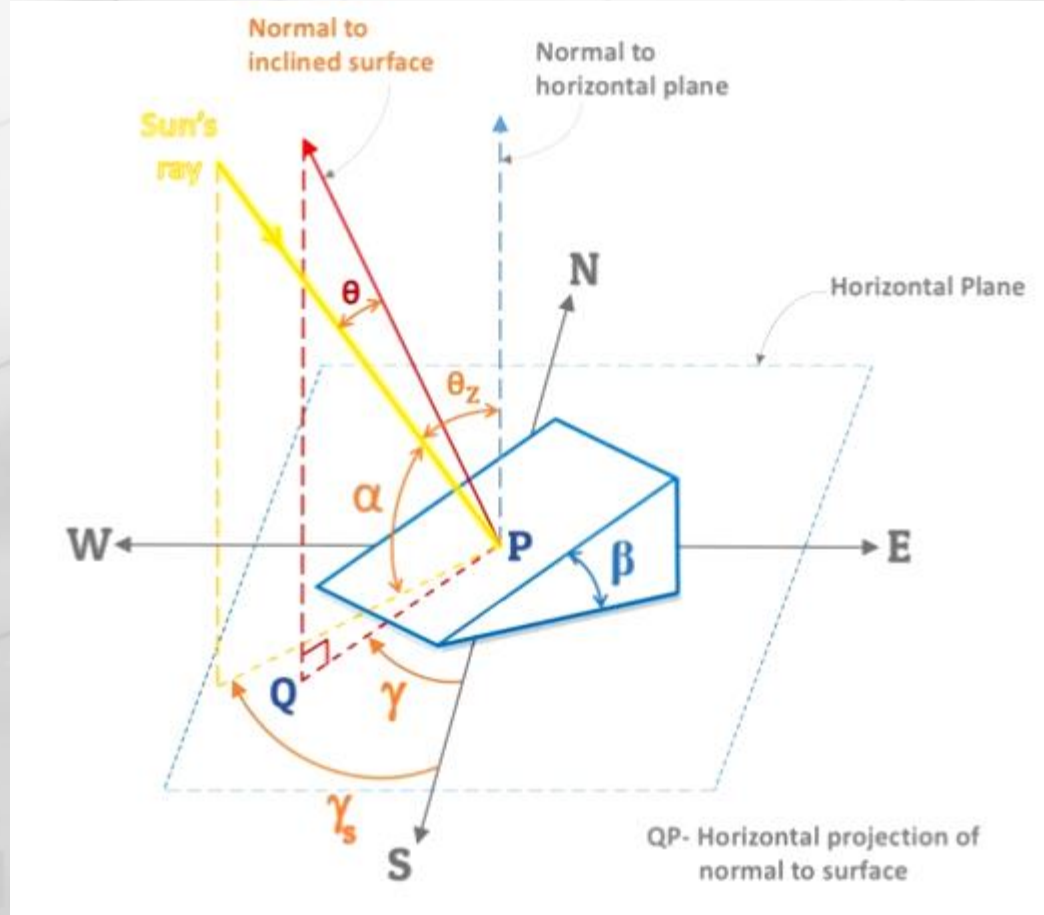
Sun-Earth Geometric Relationship



Sun-Earth Geometric Relationship

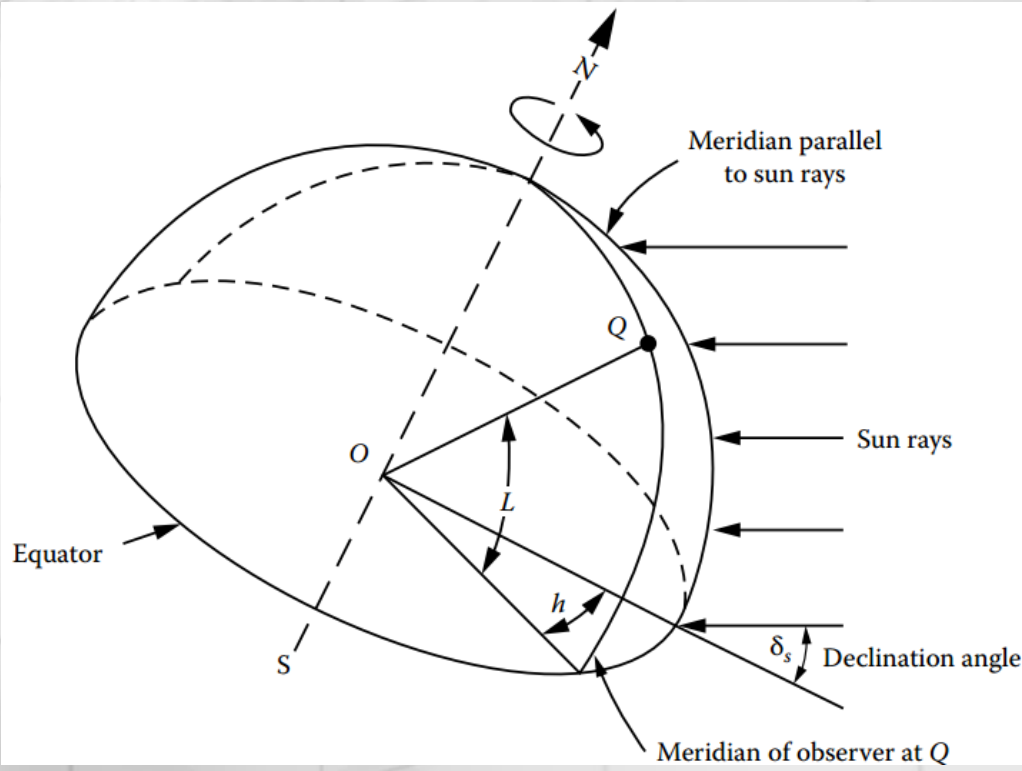
- ✓ The position of the sun can be described at any time by two angles, the **altitude and azimuth angles**.
- ✓ The *solar altitude angle*, α, is the angle between a line collinear with the sun's rays and the horizontal plane.

- ✓ The *solar azimuth angle*, γ_s, is the angle between a due south line and the *projection* of the site to sun line on the horizontal plane.
- ✓ The sign convention used for the azimuth angle is positive west of south and negative east of south.
- ✓ The *solar zenith angle*, θz, is the angle between the site to sun line and the vertical at the site



Sun-Earth Geometric Relationship

- ✓ Hour angle, h (ω): The angular displacement of the sun east or west of the local meridian due to the rotation of the earth on its axis at 15° per hour.
- ✓ The convention to be followed is negative for the morning and positive for the afternoon, with zero being solar noon.



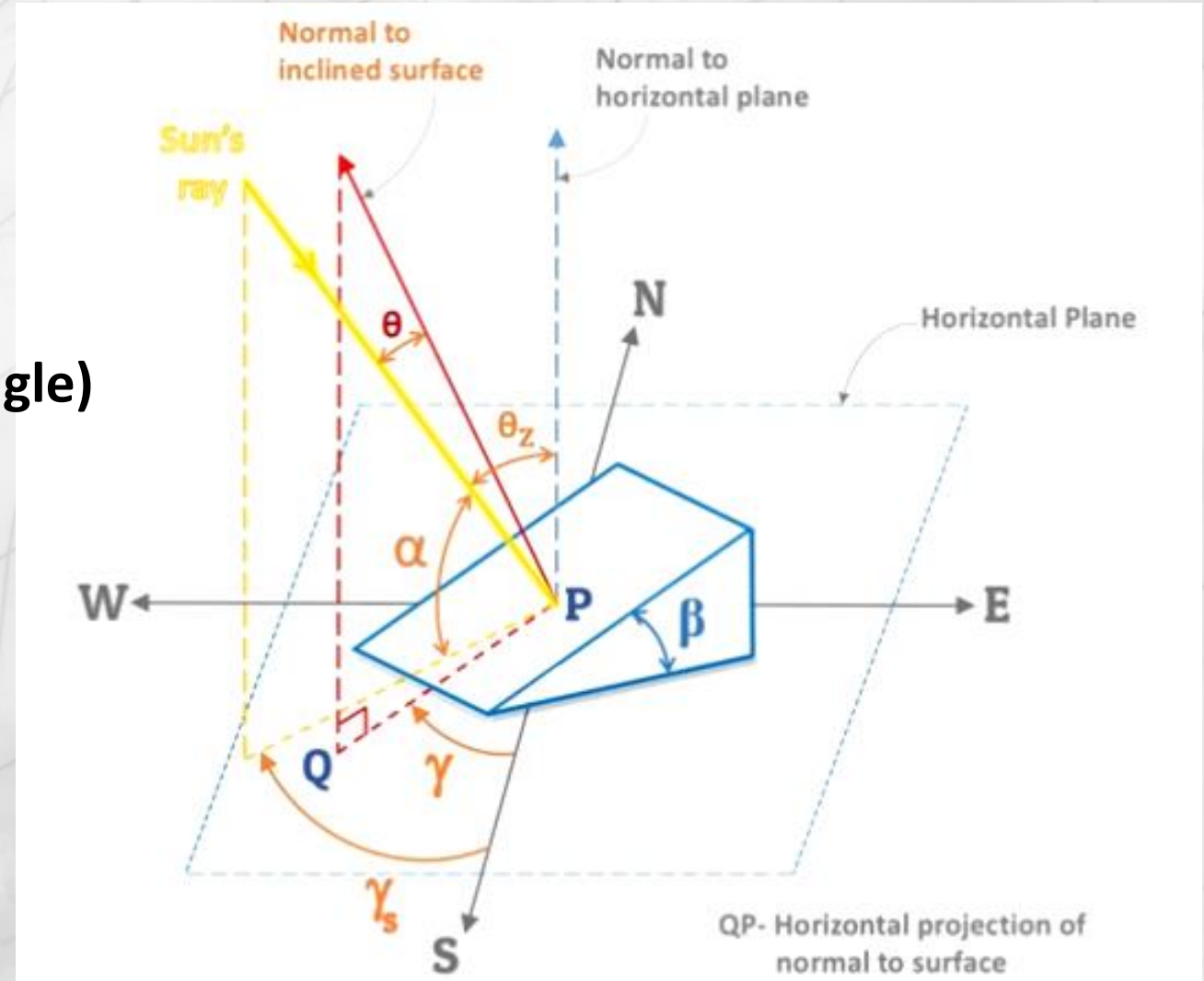
$$h_s = (15°/\text{h}) \cdot (\text{Hours from local solar noon}) = \frac{\text{Minutes from local solar noon}}{4 \text{ min/degree}}.$$

$$h = (\text{AST} - 12)15$$

Sun-Earth Geometric Relationship

Θ: Incidence angle

B: Tilt angle (plate angle)



Given

$$Cos\theta = sin\delta sin\phi cos\beta - sin\delta cos\phi sin\beta cos\gamma + cos\delta cos\phi cos\beta cos\omega + cos\delta sin\phi sin\beta cos\gamma cos\omega + cos\delta sin\beta sin\gamma sin\omega$$

Example

Calculate the angle of incidence of beam radiation on a solar collector surface located in College Station, Texas, at 10:30 a.m. (solar time) on October 25, 2014, if the surface is tilted at an angle of 30° and pointed 15° west of south. The latitude of College Station, Texas, is 30.61° North.

First calculate the value of $n = 298$.

$$h = (\text{AST} - 12)15 \quad \omega = -22.5°, \quad \gamma = 15°, \beta = 30° \text{ and } \varphi = 30.61°.$$

$$\begin{aligned}\cos\theta = {} & \sin(-13.12)\sin(30.61)\cos(30) - \sin(-13.12)\cos(30.61)\sin(30)\cos(15) \\ & + \cos(-13.12)\cos(30.61)\cos(30)\cos(-22.5) \\ & + \cos(-13.12)\sin(30.61)\sin(30)\cos(15)\cos(-22.5) \\ & + \cos(-13.12)\sin(30)\sin(15)\sin(-22.5)\end{aligned}$$

$$\begin{aligned}\cos\theta = {} & [(-0.227)\times(0.509)\times(0.866)] - [(-0.227)\times(0.861)\times(0.50)\times(0.966)] \\ & + [(0.974)\times(0.861)\times(0.866)\times(0.924)] \\ & + [(0.974)\times(0.509)\times(0.50)\times(0.966)\times(0.924)] \\ & + [(0.974)\times(0.50)\times(0.259)\times(-0.383)] = 0.838\end{aligned}$$

e. $\cos\theta = 0.838$ and $\theta = 33.04°$.

f. If the magnitude of solar energy received on the surface is known, about 83.8% of this is actually absorbed by the collector at this time of the day.

Sunset and Sunrise Hour Angle

$$Cos\theta = sin\delta sin\phi cos\beta - sin\delta cos\phi sin\beta cos\gamma + cos\delta cos\phi cos\beta cos\omega + cos\delta sin\phi sin\beta cos\gamma cos\omega + cos\delta sin\beta sin\gamma sin\omega$$

The equation for horizontal surfaces leads to the calculation of the sunset (or sunrise) hour angle or the estimate of the sunrise and sunset for a given location and day of the year.

$$Cos\theta_z = cos\delta cos\phi cos\omega + sin\delta sin\phi$$

The angle of incidence becomes the zenith angle

It could be rearranged to calculate for the value of ω, which we now call the sunset (or sunrise) hour angle (ω*s*),

when θ_z is equal to 90°:

$$cos\omega_s = -\frac{sin\delta sin\phi}{cos\delta cos\phi} = -tan\delta tan\phi$$

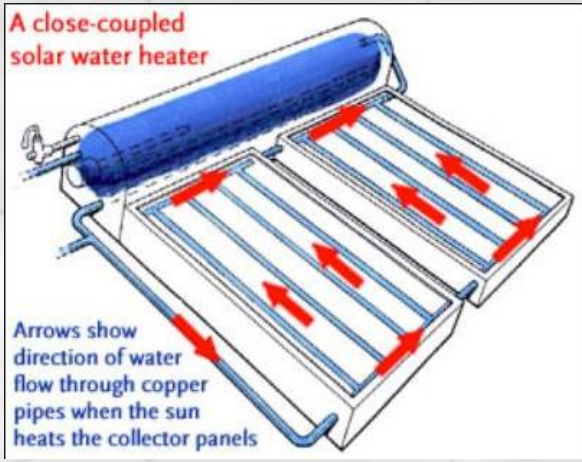
E-JUST

Energy Resources Applications

Solar-Thermal energy Conversion

Active solar heating systems

Solar Collectors



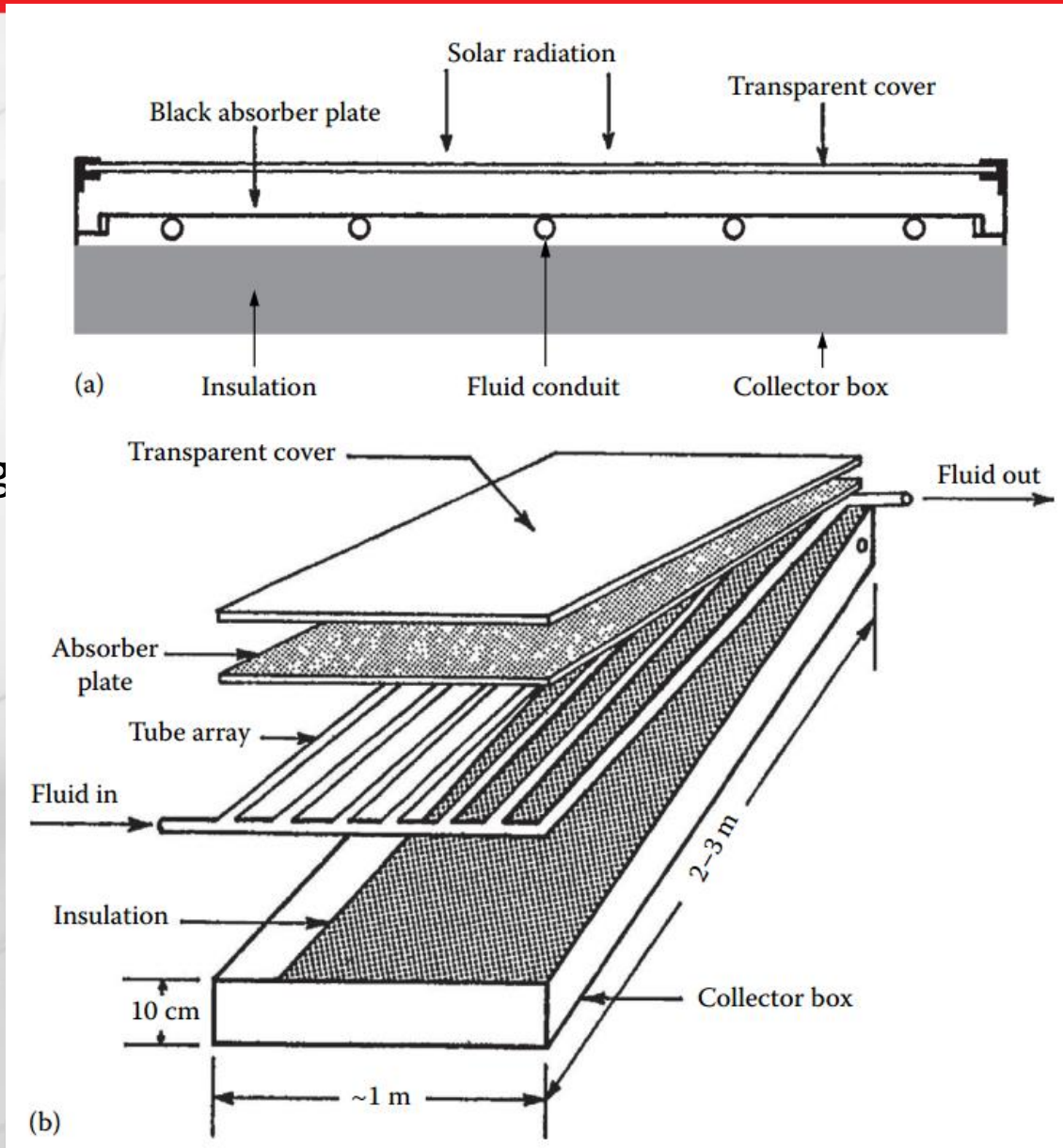
- ✓ A solar thermal collector is a heat exchanger that converts radiant solar energy into heat.
- ✓ In essence this consists of a receiver that absorbs the solar radiation and then transfers the thermal energy to a working fluid.
- ✓ The analysis and design of solar collectors present unique and unconventional problems in *heat transfer, optics, and material science*.
- ✓ The classification of solar collectors can be made according to the type of working fluid (water, air, or oils) or the type of solar receiver used (non-tracking or tracking).

Collector types

- ✓ Most commonly used *working fluids* are water (*glycol* being added for *freeze protection*) and air.
- ✓ Because of the poorer heat transfer characteristics of air (k_{air} = 0.02 W/m.K) with solar absorber, the air collector may operate at a higher temperature than a liquid-filled collector (k_{water} = 0.6 W/m.K), resulting in greater thermal losses and, consequently, a lower efficiency.
- ✓ The choice of the working fluid is usually dictated by the application.
- ✓ Air collectors are suitable for space heating and convective drying applications, while liquid collectors are the obvious choice for domestic and industrial hot-water applications.
- ✓ In certain high-temperature applications, special types of oils are used that provide better heat transfer characteristics.

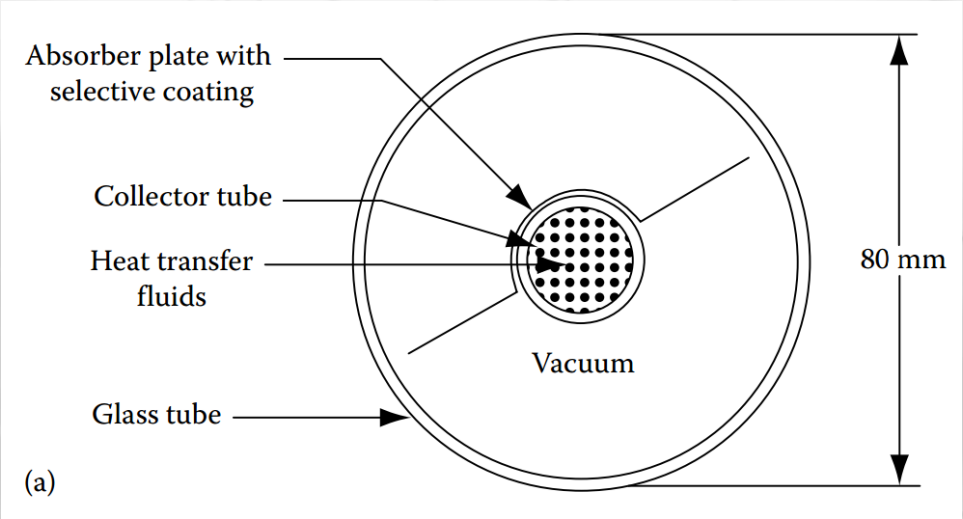
Flat-plate collectors

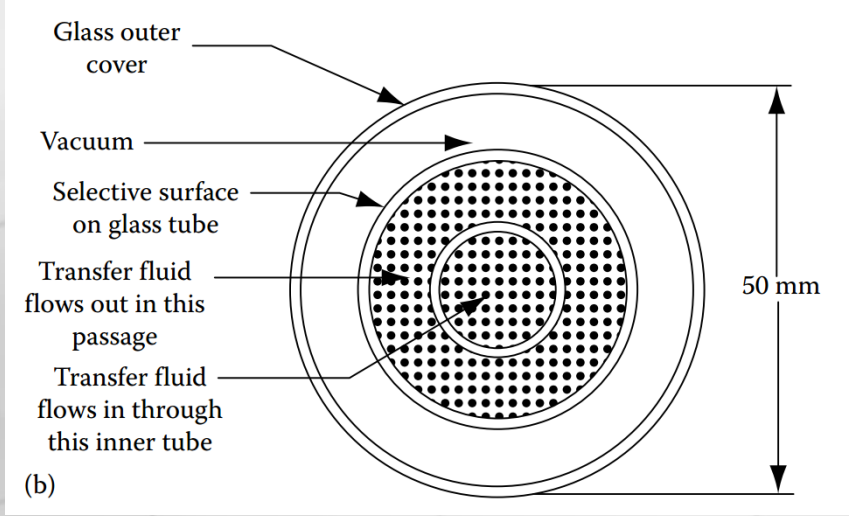
- ✓ The flat-plate collector is the most common conversion device in operation today, since it is most economical and appropriate for delivering energy at temperatures up to about 100°C.



Evacuated Tubular Collectors

- ✓ One method of obtaining temperatures between 100°C and 200°C is to use evacuated tubular collectors.
- ✓ The advantage in creating and being able to maintain a vacuum is that convection losses between glazing and absorber can be eliminated.



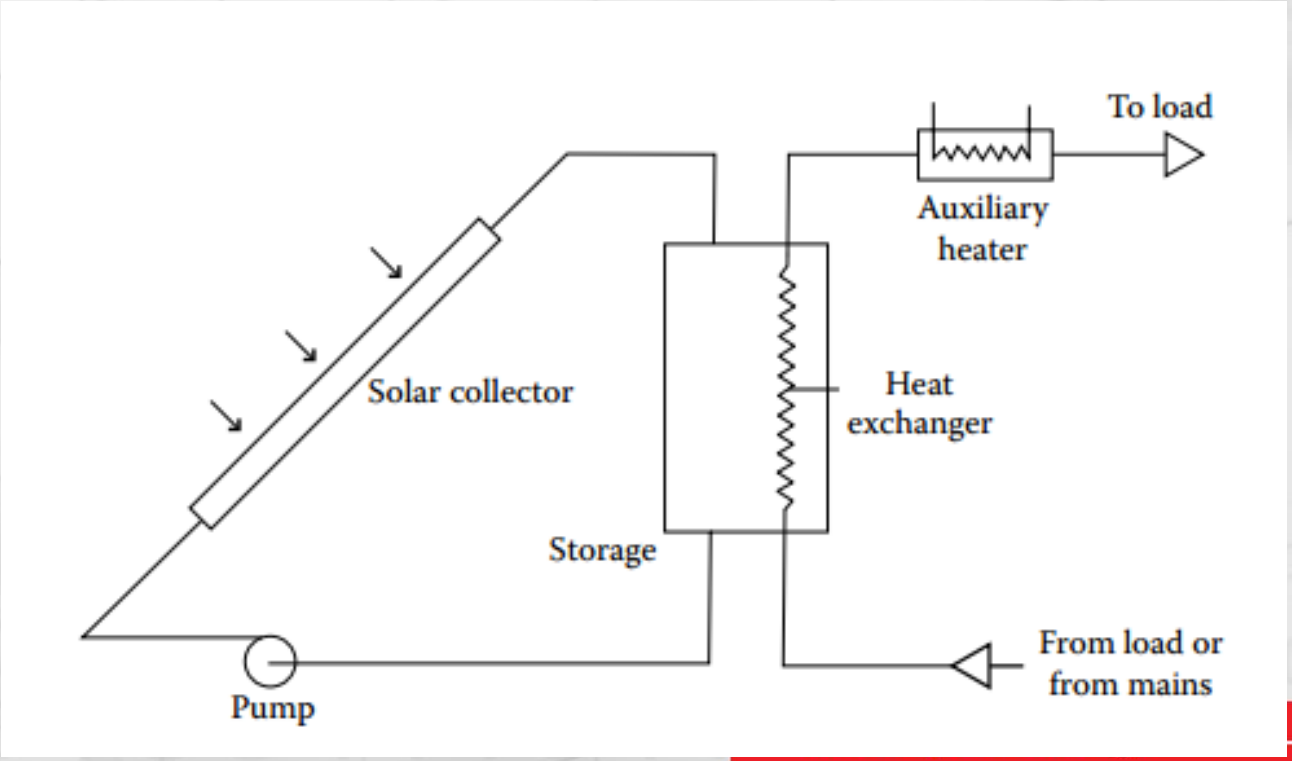


Tubular vacuum collector with absorber plate and fluid single-pass

Tubular vacuum collector with concentric tubes and fluid two-pass

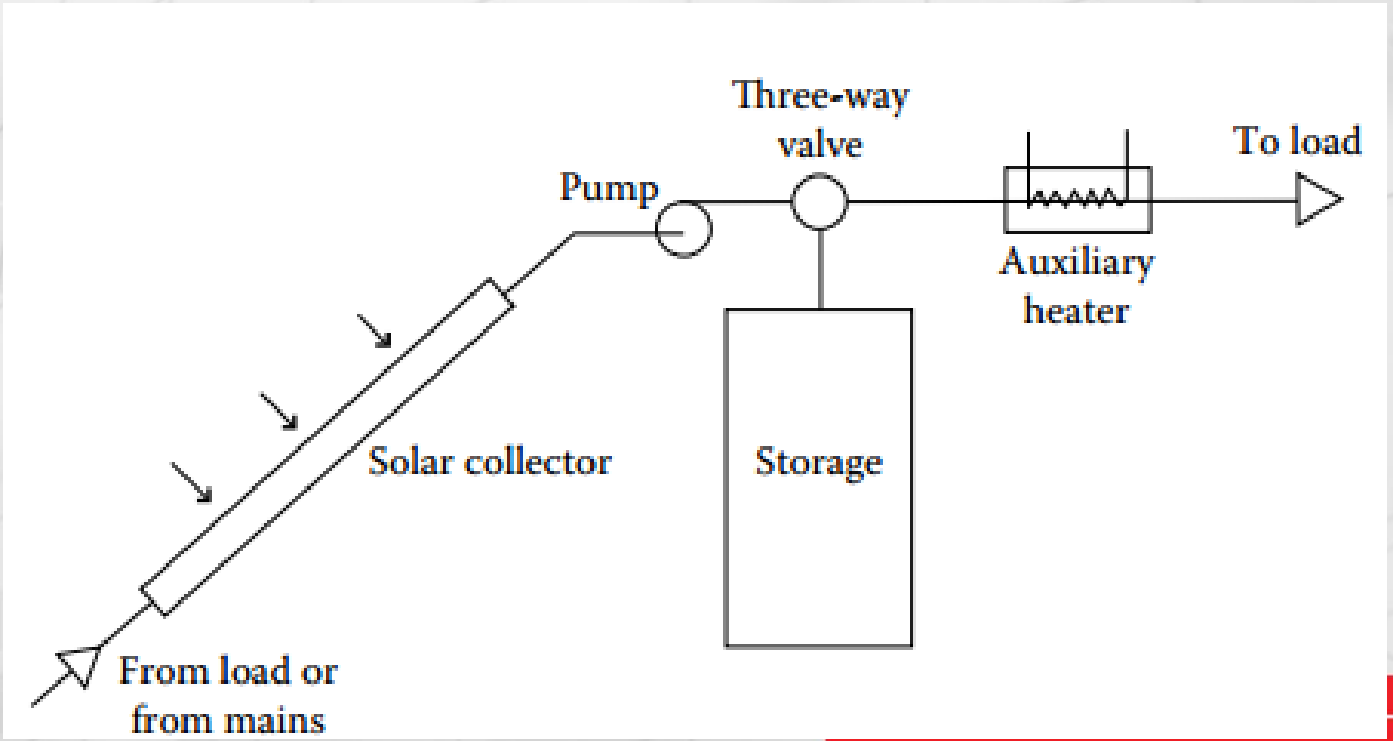
Closed-loop system

- ✓ A closed-loop system has been defined as a circuit in which the performance of the solar collector is directly dependent on the storage temperature.
- ✓ The storage volume makes about 5–10 passes through the collector during a typical sunny day, and this is why such systems are called *multipass* systems.



Open-loop system

- ✓ Open-loop systems are defined as systems in which the collector performance is independent of the storage temperature.
- ✓ The working fluid may be rejected if contaminants are picked up during its passage through the load.
- ✓ The working fluid could be directly recirculated back to the entrance of the solar collector field.



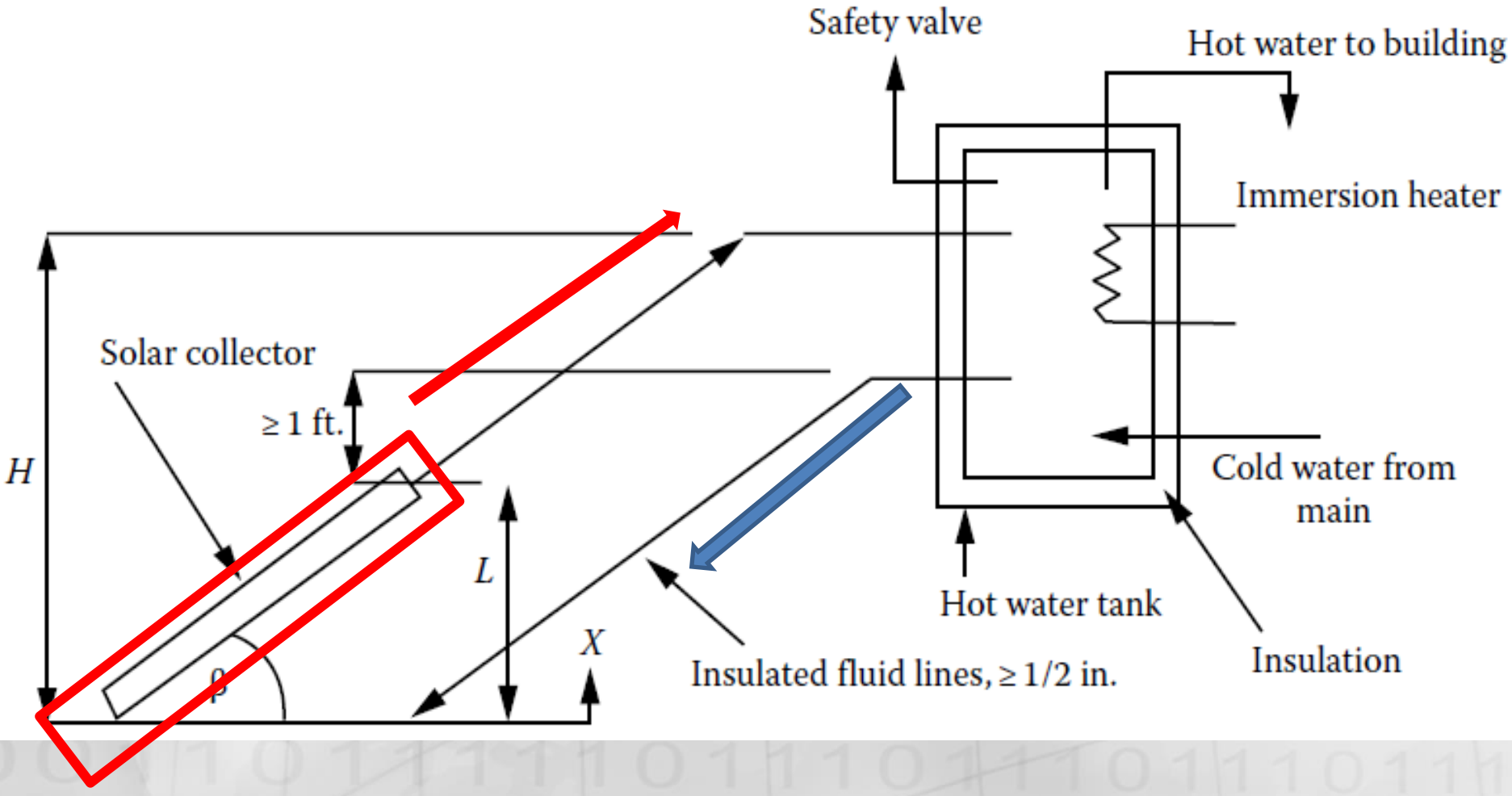
Passive solar heating

- ✓ Passive system: the *thermal energy flow is by natural means*: by conduction, radiation, and natural convection.

- ✓ A *passive heating system* is one in which the sun's radiant energy is converted to heat upon absorption by the building.

- ✓ The absorbed heat can be transferred to thermal storage by natural means or used to directly heat the building.

Solar Thermosyphon Water Heating



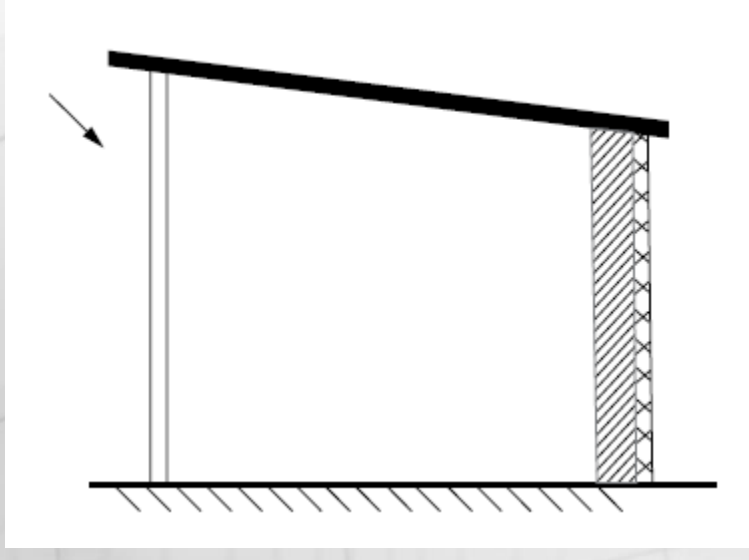
Solar Thermosyphon Water Heating

- ✓ Solar hot water heating systems are composed of a collector and a storage tank.
- ✓ When the flow between the collector and tank is by natural circulation (No pumps), these passive solar hot water systems are referred to as *thermosyphon systems*.
- ✓ The natural tendency of a less dense fluid to ***rise*** above a more dense fluid can be used in a simple solar water heater to *cause fluid motion through a collector.*
- ✓ The density difference is created within the solar collector where heat is added to increase the temperature and decrease the density of the liquid.
- ✓ This collection concept is called a *thermosyphon*

Types of Passive Heating Systems

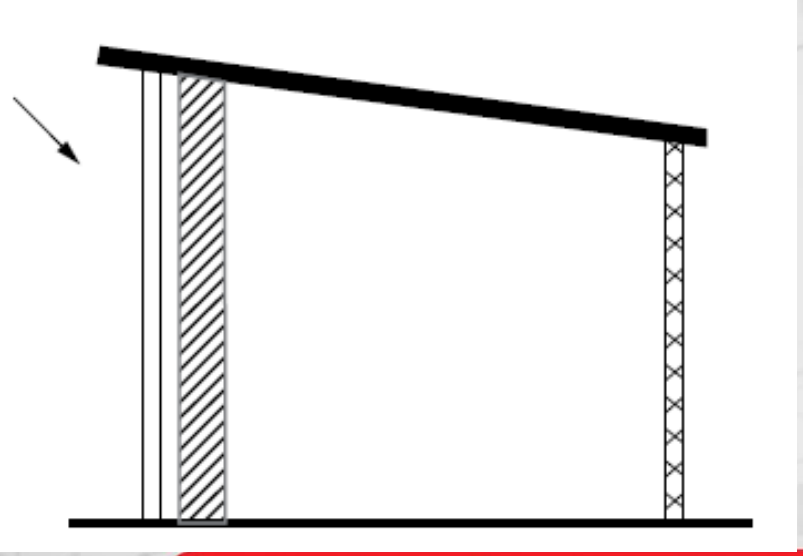
- ✓ A commonly used method of cataloging the various passive system concepts is to distinguish three general categories: **direct, indirect, and isolated gain.**

- ✓ **For direct gain,** sunlight enters the heated space and is converted to heat at absorbing surfaces.
- ✓ This heat is then distributed throughout the space and to the various enclosing surfaces and room contents.



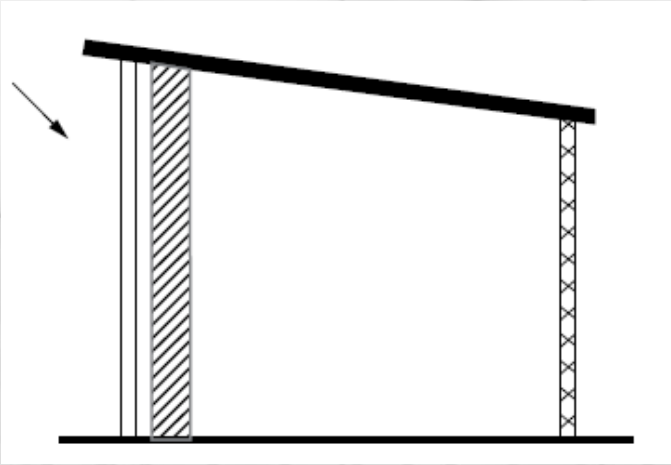
Types of Passive Heating Systems

- ✓ For **indirect gain** category systems, sunlight is **absorbed and stored** by a *mass interposed* between the *glazing* and the *conditioned space*.
- ✓ The conditioned space is partially enclosed and bounded by this thermal storage mass, so a natural thermal coupling is achieved.
- ✓ Examples of the indirect approach are the thermal storage wall, the thermal storage roof, and the northerly room of an attached sunspace.



Types of Passive Heating Systems

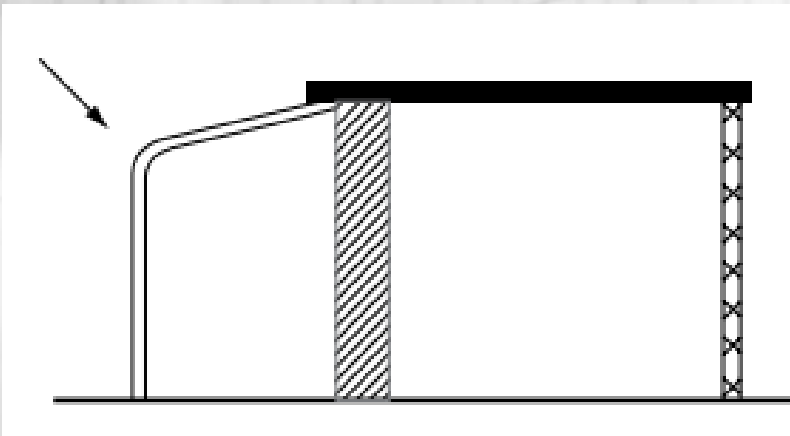
- In the thermal storage wall, sunlight penetrates the glazing and is absorbed and converted to heat at a wall surface interposed between the glazing and the heated space.
- The wall is usually masonry (Trombe wall) or containers filled with water (water wall), although it might contain phase-change material.



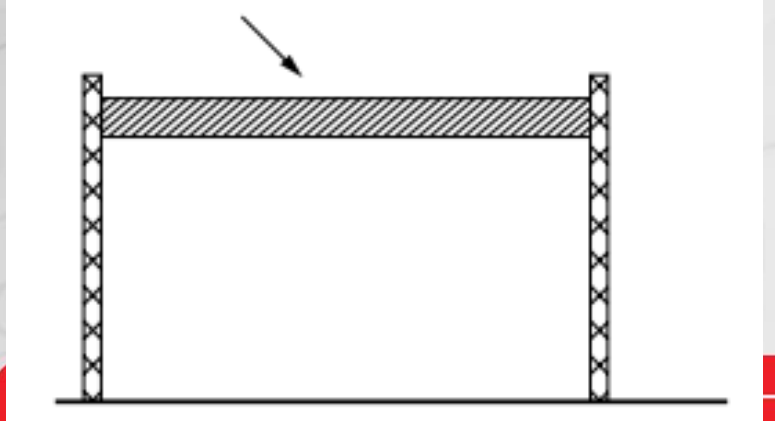
Thermal storage wall

Types of Passive Heating Systems

- ✓ The attached sunspace is actually a *two-zone combination* of direct gain and thermal storage wall.
- ✓ Sunlight enters and heats a direct gain southerly *sunspace* and also heats a mass wall separating the northerly buffered space, which is indirectly heated.
- ✓ The *sunspace* is frequently used as a greenhouse, in which case, the system is called an *attached greenhouse*.



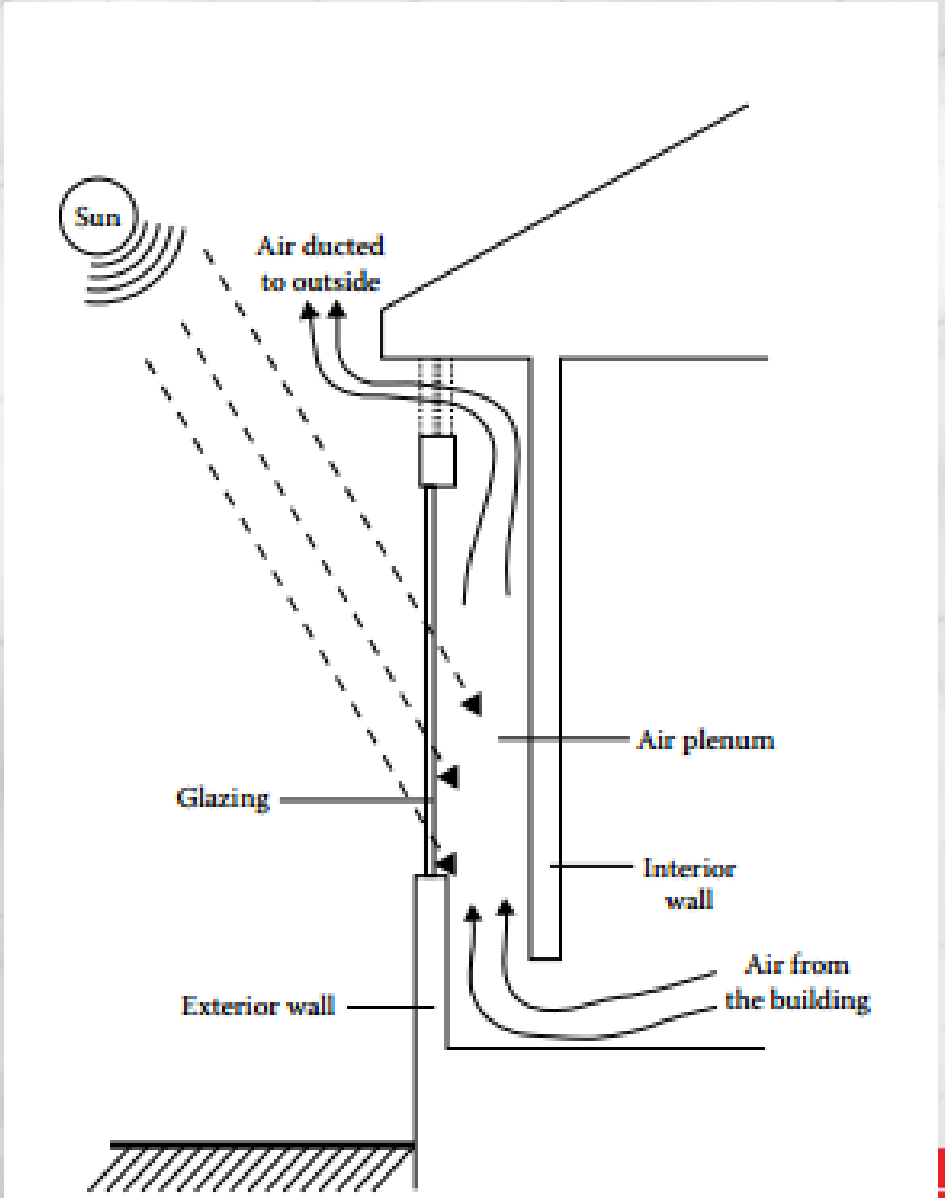
- ✓ The thermal storage roof is similar to the thermal storage wall except that the interposed thermal storage mass is located on the building roof.



Passive cooling: Chimney effect



KAUST Chimney effect



Solar concentration and Solar Thermal Plants (STP)

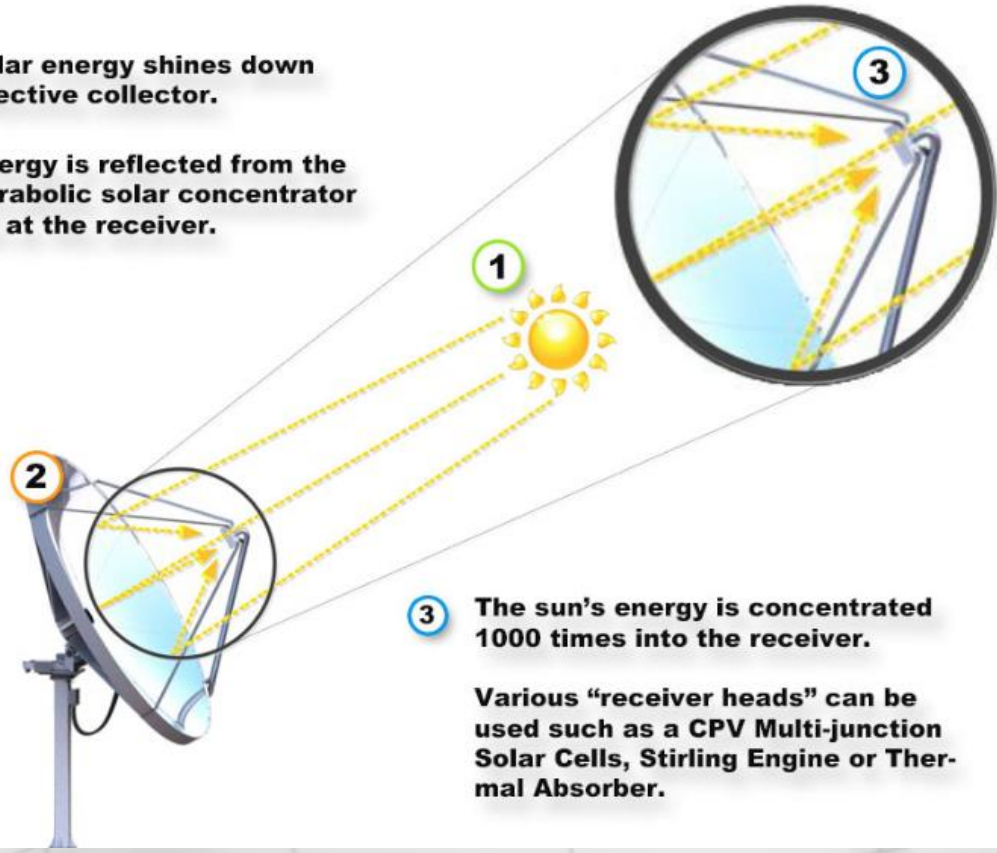
- ✓ Maximum attainable temperatures in typical fossil fuel burners are on the order of 2100°C.
- ✓ For nuclear fission, the expected maximum useful temperature in Generation-IV gas-cooled reactors is below 1000°C.

- ✓ The use of electricity for this purpose represents about a 50%–75% conversion loss.
- ✓ In contrast, solar radiation has the potential to reach temperatures close to the apparent temperature of the sun (5777 K).

- ✓ Solar concentrating systems are characterized by the use of devices, like mirrors or lenses, able to redirect the incident solar radiation received onto a particular surface, collector surface A_c, and concentrate it onto a smaller surface, absorber surface A_{abs}, or absorber.
- ✓ The quotient of these two areas is called the geometric concentration ratio, Conc = A_c/A_{abs}.

E-JUST

1. The sun's solar energy shines down onto the reflective collector.
2. The sun's energy is reflected from the reflective parabolic solar concentrator and directed at the receiver.



3. The sun's energy is concentrated 1000 times into the receiver.

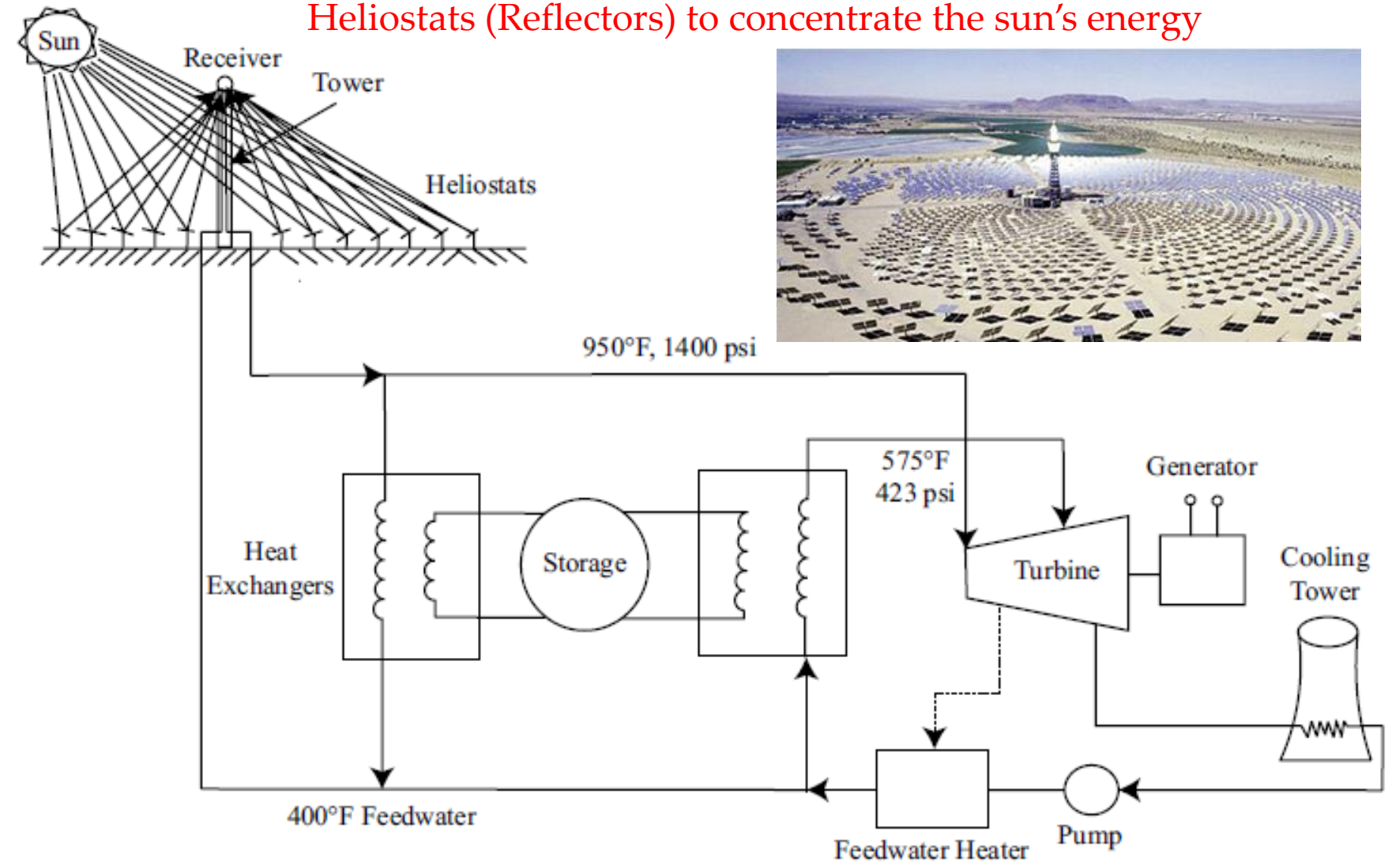
Various "receiver heads" can be used such as a CPV Multi-junction Solar Cells, Stirling Engine or Thermal Absorber.





Concentrating solar power

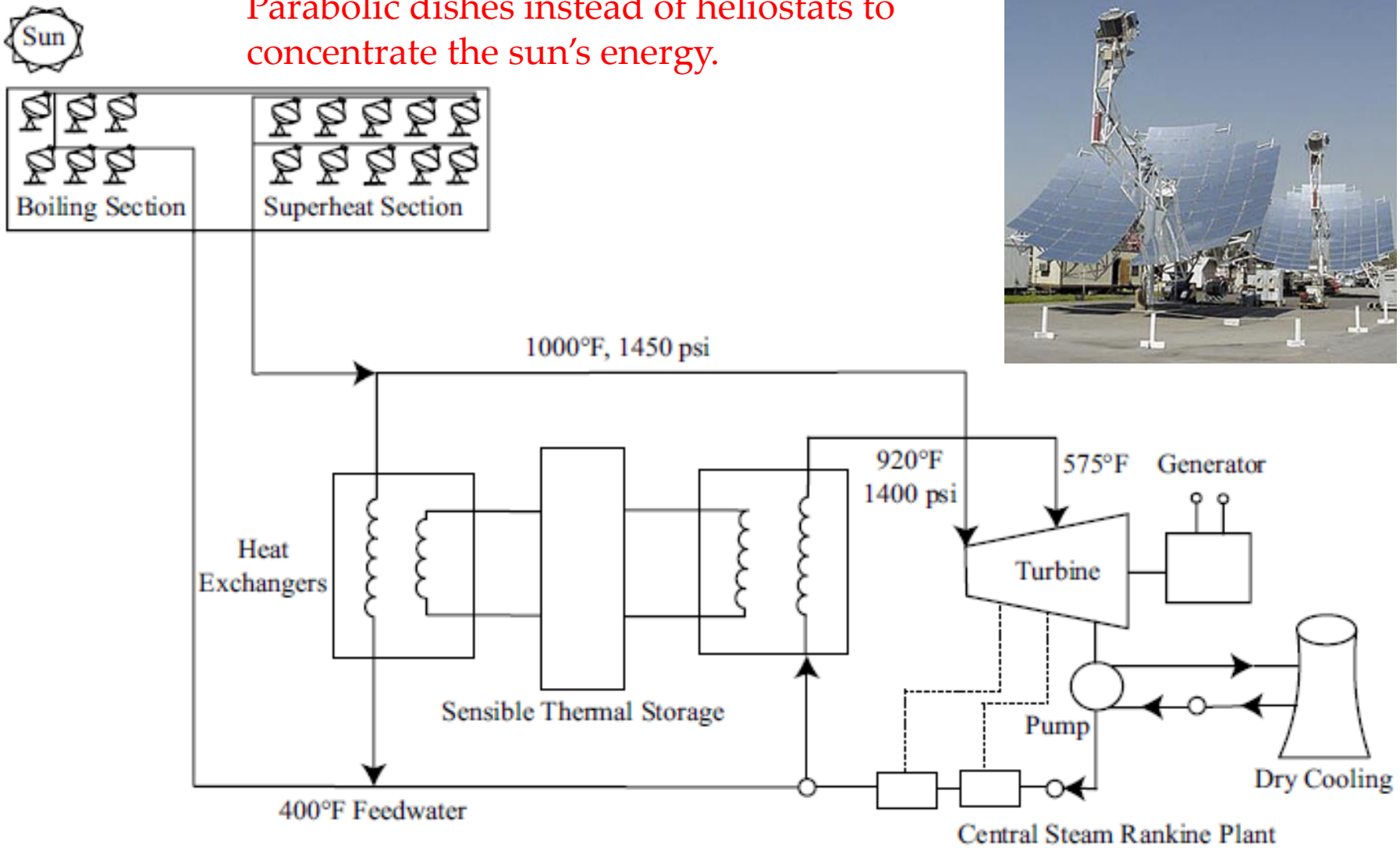
Heliostats (Reflectors) to concentrate the sun's energy



Central Receiver Solar Thermal Electric Power Plant (STEC), 10 - 200 MWe

Concentrating solar power

Parabolic dishes instead of heliostats to concentrate the sun's energy.

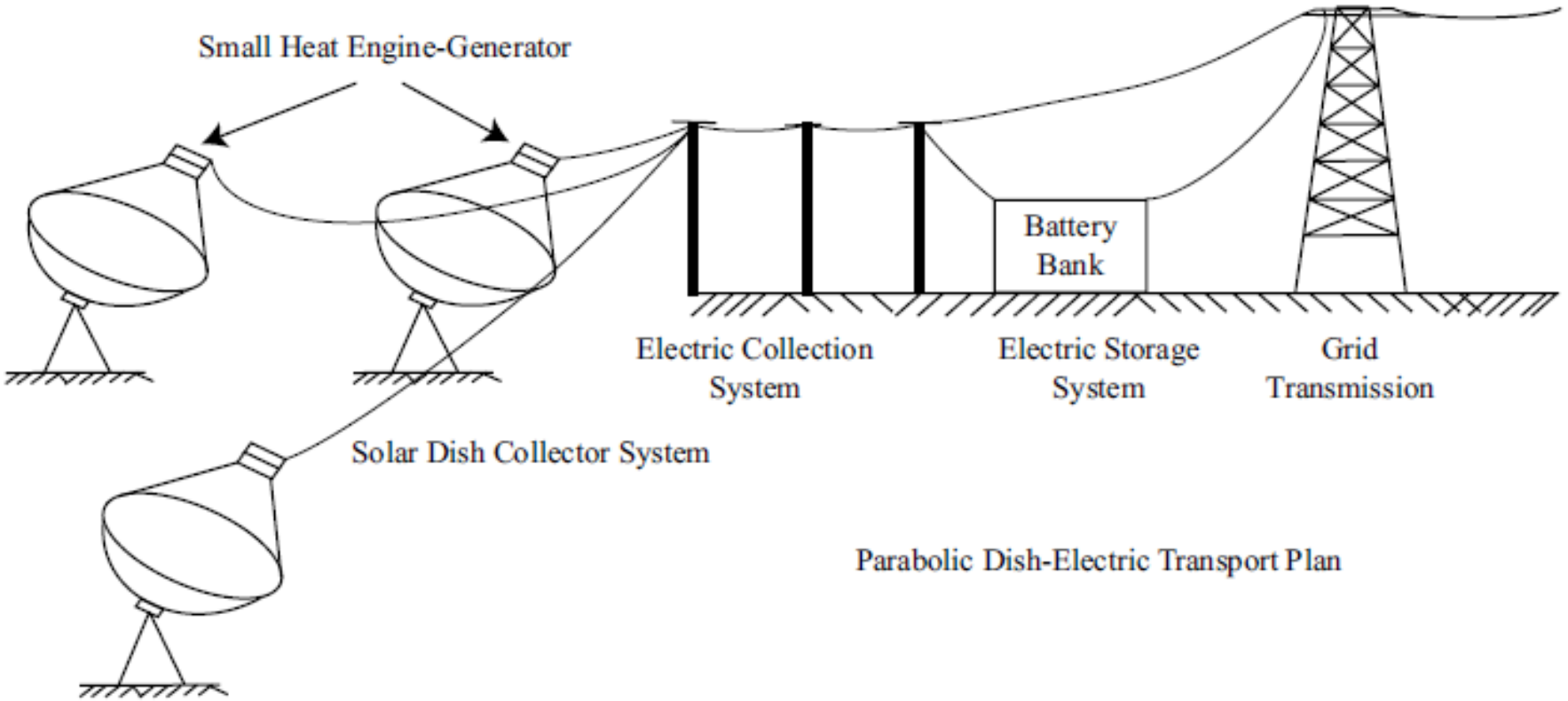


Parabolic Dish-Steam Transport and Conversion System

Concentrating solar power

The parabolic dishes are also equipped with a small heat engine generator





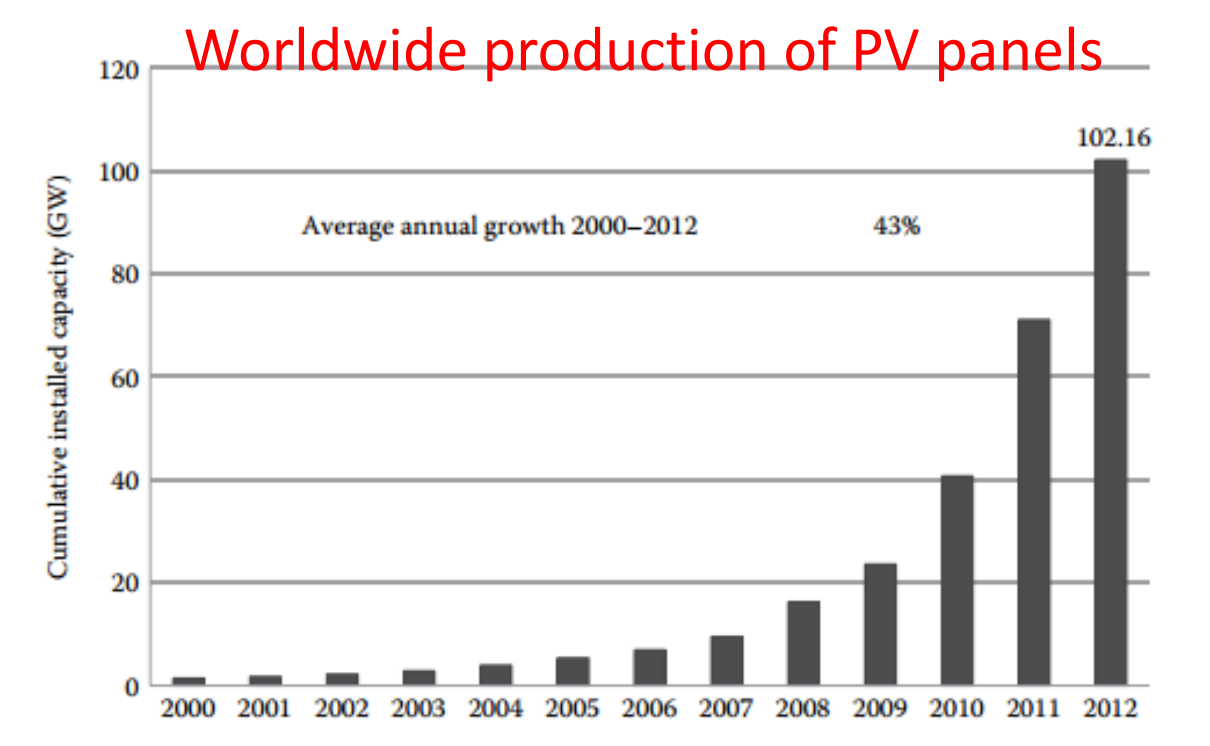
Photovoltaics



Photovoltaics

- ✓ Photovoltaic (PV) conversion is the direct conversion of sunlight into *electricity* with no intervening heat engine.
- ✓ PV devices are solid state; therefore, they are rugged and simple in design and require very little maintenance.
- ✓ Perhaps the biggest advantage of solar PV devices is that they can be constructed as stand-alone systems to give outputs from microwatts to megawatts.
- ✓ That is why they have been used as the power sources for calculators, watches, water pumping, remote buildings, communications, satellites and space vehicles, and even megawatt-scale power plants.
- ✓ PV panels can be made to form components of building skin, such as roof shingles and wall panels.

Photovoltaics



- ✓ The costs of PV panels have come down from about $30 to less than $1 per peak watt over the last three decades.
- ✓ With *panel costs* as low as $0.55/W (large quantities) to $2/W (retail), the *system costs* are in the range of about $1.50/W for megawatt-range systems to about $3–$4/W for small systems, and the PV systems have finally become cost-effective for on-grid applications.

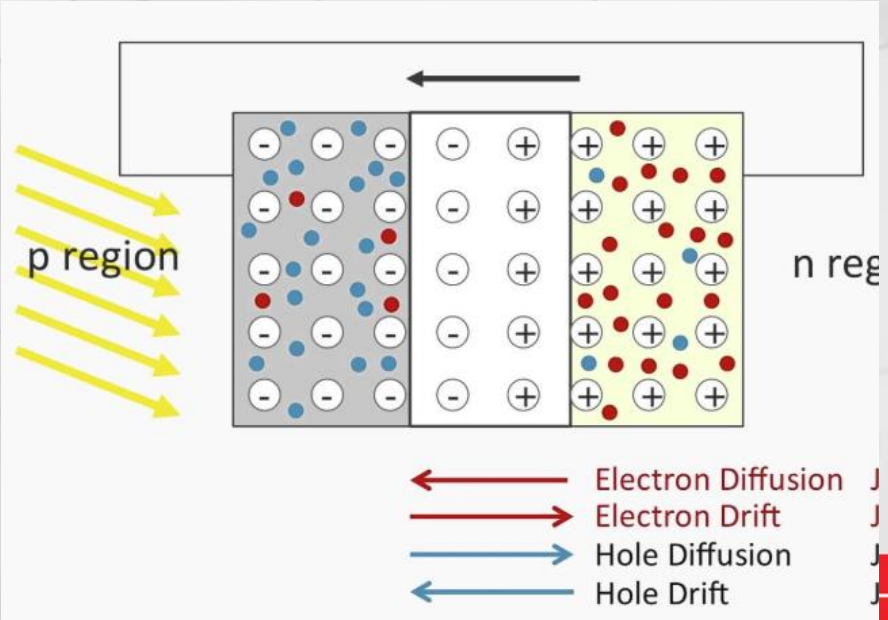
PV cell

E-


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Construction Of DC Generator In Detail || AP Tech||
DC Generator (Dynamo)

PV cell

- ✓ PV cells have been made with silicon (Si), gallium arsenide (GaAs), copper indium diselenide (CIS), cadmium telluride (CdTe), and a few other materials.
- ✓ The common denominator of PV cells is that a *p*–*n* junction
- ✓ Understanding the *p*–*n* junction is thus at the heart of understanding how a PV cell converts sunlight into electricity.
- ✓ The junction consists of a layer of *n*-type Si joined to a layer of *p*-type Si, with an uninterrupted Si crystal structure across the junction.
- ✓ The *n*-layer has an abundance of free electrons and the *p*-layer has an abundance of free holes.



Sizing of PV system

Major components of a solar PV system

a. PV panel
b. Battery
c. Controller
d. Wires
e. Load in the form of appliances, lights, or motors

- The daily load requirement (DLR) must be first calculated by adding the individual load-in watts and their corresponding hours of operation (in hours).
- Note that energy can be added but not power.
- DLR is simply the product of load-in watts and time in hours and has the units of watt-hours.
- On larger systems, the kWh load is usually the unit used.

$$DLR(kWh) = L(kW) \times t(h)$$

Sizing of PV system

E-JUST

Estimate battery size

- Battery capacity is usually rated in either watt-hours or ampere-hours.
- The latter is most common.

$$BC_{wh} = DLR \times 10$$

The constant factor that is commonly used for calculating battery capacity is 10, meaning that it will take 10 days before the battery will totally be out of power.

The battery capacity in ampere-hours $BC_{AH} = \frac{BC_{wh}}{V}$ *V:* system voltage

The PV panel watt-peak (Wp) requirement is then calculated using two factors—the climate generation factor (CGF) and the overall system efficiency (%).

$$W_p\,(Watts) = \frac{DLR}{CFG \times \xi_s}$$

The CGF varies depending upon the climate of the site location and also upon global geographic location.
For example, the CGF in Thailand is 3.43, whereas in European countries (the European Union), the value used is 2.93

Example: Sizing of PV system

A typical barn lights up its entrance from 6 p.m. to 9 p.m., the feeding stall lamp for an hour, and the feed water pump for two hours. The entrance lamp wattage is 10 watts, and the feeding stall lamp requires 10 watts. The pump consumes 1.4 amperes at 12 volts. (a) What is the energy requirement of the barn? (b) Compute the size of the battery and (c) the PV panel needed for the system in watt-peak (Wp), assuming a CGF of 3 and overall system efficiency of 85%.

Load	Wattage	Hours	Watt-Hour
Entrance Lamp 1	10	3 hrs	30
Feeding Stall Lamp 2	10	1 hr	10
Pump	1.4×12 V = 16.8	2 hrs	33.6
Total	36.8 watts		73.6 watt-hour

$$BC_{wh} = DLR \times 10 = 73.6 \times 10 = 736\ Watt - hrs$$

$$BC_{AH} = \frac{BC_{wh}}{V} = \frac{736\ Watt - hrs}{12\ Volts} = 61.3\ Ampere - hours$$

$$W_p\,(Watts) = \frac{DLR}{CFG \times \xi_s} = \frac{73.6\ Watt - hours}{3 \times 0.85} = 28.86\ Watts$$

Example: Sizing of PV system

A household owner would like to net meter his residential electrical power load using solar energy. The average solar radiation in his area is around 5 kWh/m^2/day. He uses about 10,000 kWh of electricity per year. He pays for electricity at \$0.10/kWh. (a) What is the minimum area of solar PV system his household would need if the overall conversion efficiency is 11%? (b) At a price of \$2/Wp, what is the initial investment and payback period required to recover the initial capital cost only and based on the payment for electricity alone? Assume a CGF of 3 and efficiency of 89% for the calculation of peak wattage.

a. The first step is to calculate the DLR for this household. This can simply be calculated using the yearly requirement divided by the number of days in a year, assuming 365 days/yr for a non–leap year as shown:

$$DLR\left(\frac{kWh}{day}\right)=\frac{10,000\ kWh}{yr}\times\frac{1\ yr}{365\ days}=27.4\frac{kWh}{day}$$

b. Then, using the efficiency equation, we find the amount of solar energy needed to satisfy this DLR:

$$\eta = \frac{Output}{Input}\times 100\%$$

$$0.11 = \frac{27.4\ kWh/day}{Input}\times 100\%$$

$$Input\left(\frac{kWh}{day}\right) = \frac{27.4\ kWh/day}{0.11}\times 100\% = 249\frac{kWh}{day}$$

Example: Sizing of PV system

A household owner would like to net meter his residential electrical power load using solar energy. The average solar radiation in his area is around 5 kWh/m^2/day. He uses about 10,000 kWh of electricity per year. He pays for electricity at \$0.10/kWh. (a) What is the minimum area of solar PV system his household would need if the overall conversion efficiency is 11%? (b) At a price of \$2/Wp, what is the initial investment and payback period required to recover the initial capital cost only and based on the payment for electricity alone? Assume a CGF of 3 and efficiency of 89% for the calculation of peak wattage.

c. Then the minimum area needed to satisfy this output energy is calculated as follows:

$$Area\left(m^2\right)=\frac{249\ kWh}{day}\times\frac{m^2-day}{5\ kWh}=49.8\ m^2$$

d. Next, the watt-peak calculation is found with the following equation, assuming a CGF of 3 and efficiency of 85%:

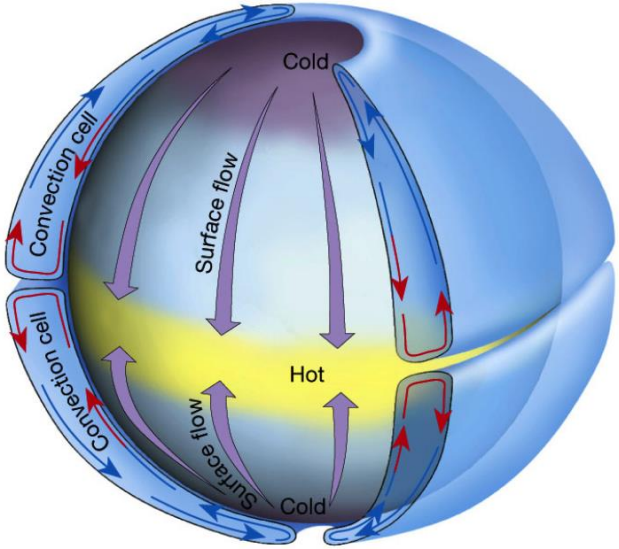
$$W_p\left(Watts\right)=\frac{DLR}{CFG\times\xi_s}=\frac{27,400\ Watt-hours}{3\times0.89}=10,262\ Watts$$

e. At \$2/Wp, the initial investment is about \$20,524. If he uses the grid power at \$0.10/kWh, he would be paying around \$1,000/yr for electricity. Consequently, recovering the initial capital investment alone and paying for net-metered electricity would require 20.52 years.

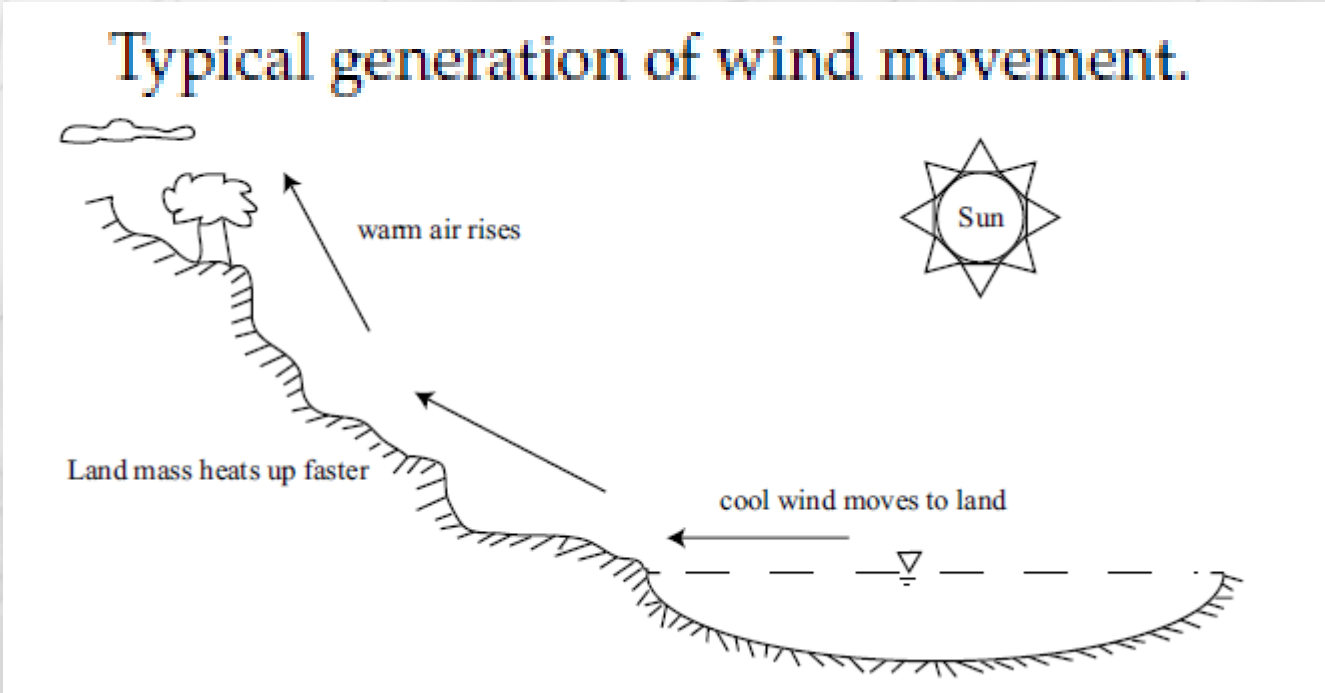
Wind energy



Wind Origins



- ✓ The primary causes of horizontal atmospheric air motion, or wind, are the uneven heating of the earth and its atmosphere by solar radiation and the earth's rotation.
- ✓ The earth moves on its own axis, a tangential component of wind movement exists, giving way to easterly or westerly wind directions, called trade winds.



Energy and Power Calculations

E-JUST

The main basis for the calculation of wind energy potential comes from the kinetic energy

$$KE = \frac{1}{2} m v^2$$

where

m = mass of wind, in kg

v = wind speed, in m/s

Example 3.1: Kinetic Energy from Wind

Estimate the kinetic energy from wind moving at a speed of 10 m/s having a mass of 100 kg. Convert this into English units assuming a conversion of 1,055 Joules per Btu.

SOLUTION:

a. The kinetic energy is simply calculated using Equation 3.1 and the Newton's Law relationship shown in Equation 3.2:

$$KE = \frac{1}{2} m v^2 = \frac{1}{2} \times 100\,kg \times \left(\frac{10\ m}{s}\right)^2 \times \frac{N - sec^2}{kg - m} \times \frac{Joules}{N - m} = 5{,}000\ Joules$$

b. In English units, the conversion is quite simple:

$$KE = 5{,}000\ Joules \times \frac{Btu}{1{,}055\ Joules} = 4.7\ Btu$$

Energy and Power Calculations

$$Wind\ Power\ (Watts) = \frac{1}{2}\dot{m}v^2$$

$$Wind\ Power\ (Watts) = \frac{1}{2}\rho \times A \times v \times v^2$$

$$Wind\ Power\ (Watts) = \frac{1}{2}\rho A v^3$$

where

wind power = watts

ρ = air density, usually 1.2 kg/m^3 at standard temperature and pressure (STP) [or a value of 0.075 lb/ft^3 in English units]

A = cross-sectional area perpendicular to flow (m^2)

v = wind speed (m/s)

Wind Power

Example 3.2: Theoretical Power Extracted from Wind

Determine the power being extracted from wind with a mass rate of flow of 100 kg/s [220 lbs/s] at a velocity of 10 m/s [32.8 ft/s]. Assume an air density of 1.2 kg/m^3 [0.075 lbs/ft^3] over a cross-sectional area of 10 m^2 [107.584 ft^2]. Express the units in the English system.

$$Wind\ Power\ (Watts) = \frac{1}{2}\rho A v^3$$

$$Power\ (W) = \frac{1}{2} \times \frac{1.2\ kg}{m^3} \times 10\ m^2 \times \left(\frac{10\ m}{s}\right)^3 \times \frac{N - s^2}{kg - m} \times \frac{J}{N - m} \times \frac{W - s}{J} = 6{,}000\ W$$

$$Power\ (hp) = 6{,}000\ W \times \frac{hp}{746\ W} = 8.04\ hp$$

Wind Power

Example 3.2: Theoretical Power Extracted from Wind

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$$Power\ (hp) = 6{,}000\ W \times \frac{hp}{746\ W} = 8.04\ hp$$

Example 3.3: Underestimation of Wind Power Using Single Average Wind Speed Data

Estimate the available theoretical power from wind in an area where the daily average wind speed was recorded as 5 m/s [16.4 ft/s] and compare this to an area where the readings are done in the morning, at noon, and in the afternoon, with readings of 4, 5, and 6 m/s, respectively [13.12, 16.4, and 19.68 ft/s]. Calculate the power at each measurement period and calculate the average power for the day assuming air density of 1.2 kg/m^3 [0.075 lbs/ft^3] and an area of 100 m^2 [1,075.84 ft^2].

$$Wind\ Power\ (W) = \frac{1}{2} \times \frac{1.2\ kg}{m^3} \times 100\ m^2 \times \left(\frac{4\ m}{s}\right)^3 = 3{,}840\ W$$

$$Wind\ Power\ (hp) = 3{,}840\ Watts \times \frac{hp}{746\ Watts} = 5.15\ hp$$

$$Wind\ Power\ (W) = \frac{1}{2} \times \frac{1.2\ kg}{m^3} \times 100\ m^2 \times \left(\frac{5\ m}{s}\right)^3 = 7{,}500\ W$$

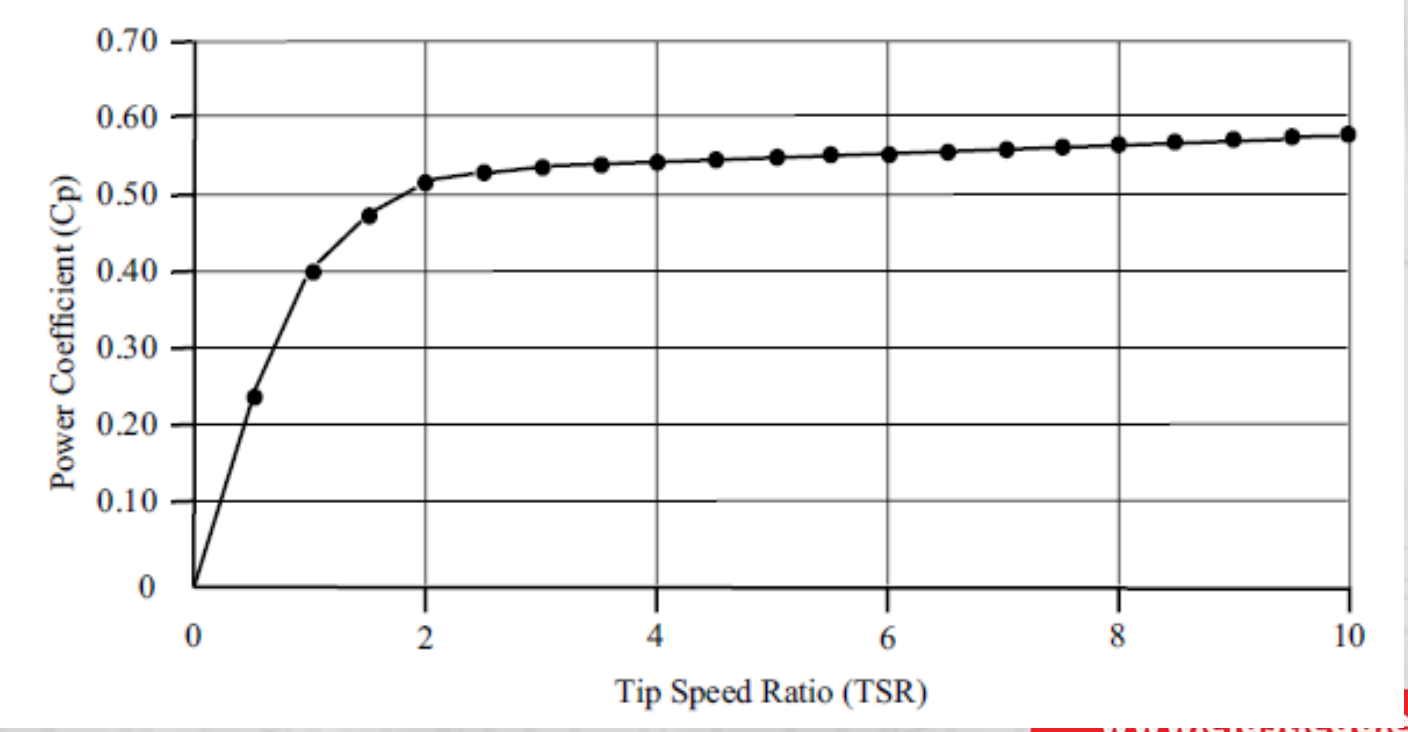
$$Wind\ Power\ (hp) = 7{,}500\ Watts \times \frac{hp}{746\ Watts} = 10.05\ hp$$

$$Wind\ Power\ (W) = \frac{1}{2} \times \frac{1.2\ kg}{m^3} \times 100\ m^2 \times \left(\frac{6\ m}{s}\right)^3 = 12{,}960\ W$$

$$Wind\ Power\ (hp) = 12{,}960\ Watts \times \frac{hp}{746\ Watts} = 17.37\ hp$$

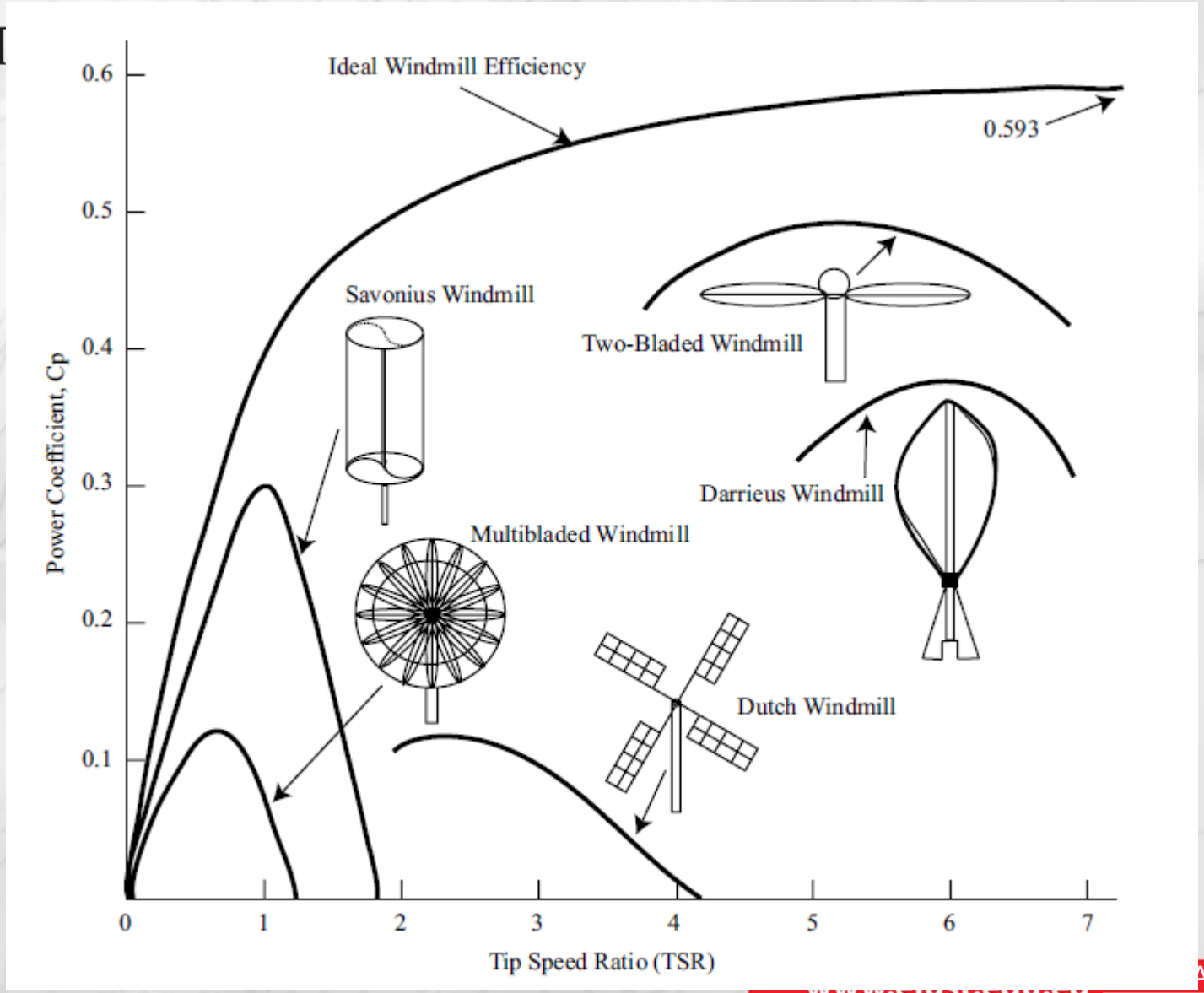
Actual Wind Power

- ✓ The maximum power or energy that could be recovered from the wind is determined by momentum theory and reported to be only 59.3% of the theoretically available power.
- ✓ This is the classical Betz coefficient.
- ✓ No wind turbine can capture more than 59.3% (or 16/27) of the kinetic energy available in the wind



The tip-speed ratio or (TSR) is the ratio of the speed of the tip of the blade divided by the wind speed.

Actual Wind Power



Actual Wind Power

$$Actual\ Wind\ Power\ (Watts) = \frac{1}{2}\rho A v^3 E_o$$

where

actual wind power = watts

ρ = air density, usually 1.2 kg/m^3 [0.075 lbs/ft^3] at STP

A = cross-sectional area perpendicular to flow (m^2)

v = wind speed (m/s)

E_o = overall efficiency (includes mechanical and electrical efficiencies)

Wind Energy

Wind turbine is a machine which converts the power in the wind into electricity.

Windmill is a machine which converts the wind into mechanical power.



General Electric 1.5MW (10000 unit till 2010)

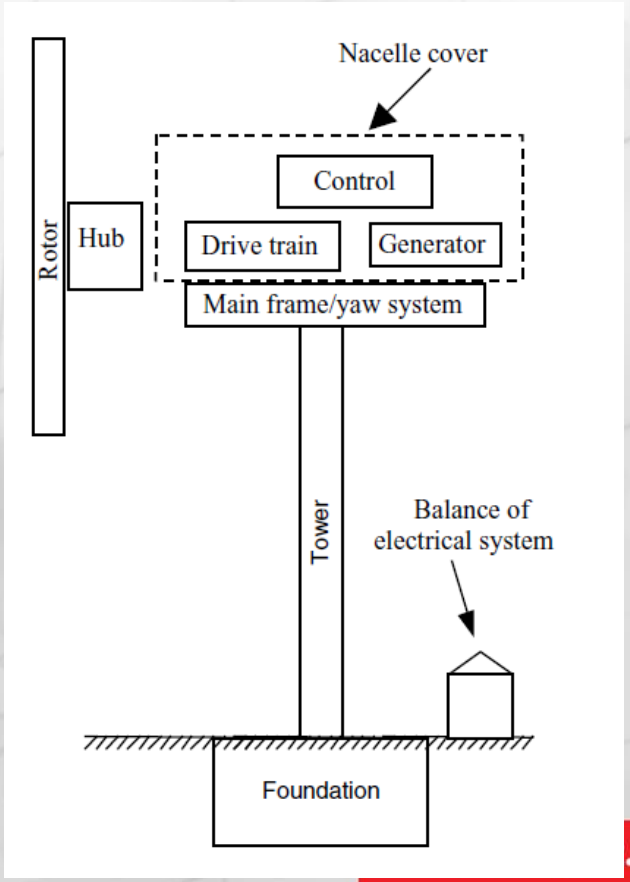
Manwell, J.F., McGowan, J.G. and Rogers, A.L., 2010. *Wind energy explained: theory, design and application*. John Wiley & Sons.

Modern Wind Turbine Design

- ✓ The most common design of wind turbine is the horizontal axis wind turbine (HAWT)
- ✓ The axis of rotation is parallel to the ground.

Main components

- ❑ROTOR
  - ➢Blades
  - ➢Hub
- ❑NACELLE
  - ➢Low speed shaft
  - ➢Brake
  - ➢Gear Box
  - ➢High speed shaft
  - ➢Generator
  - ➢Controller
  - ➢Anemometer & Wind vane
- ❑Yaw System
- ❑TOWER

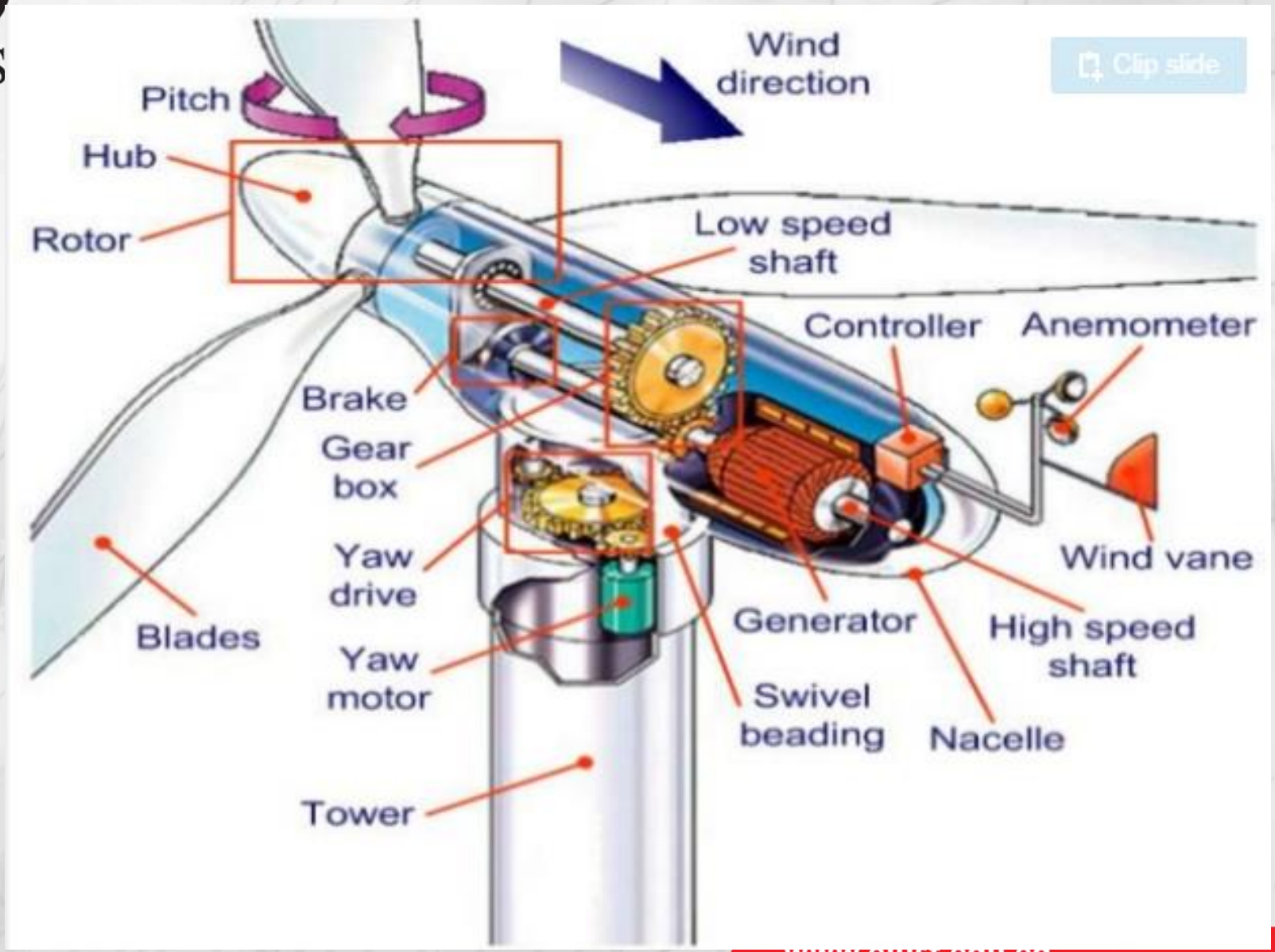




Main Components

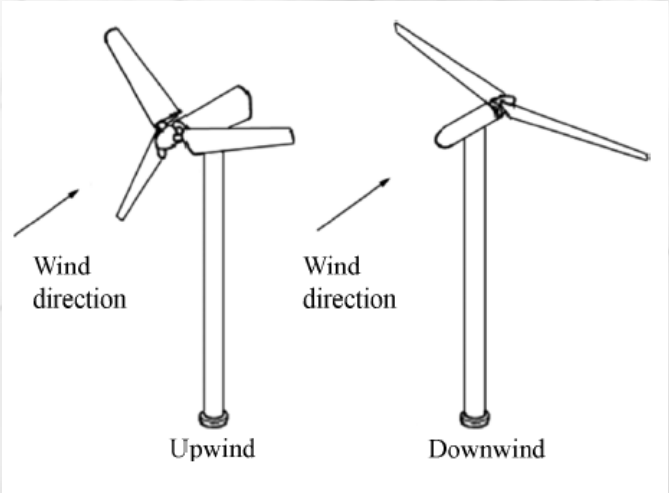


Main Components



Rotor: Blades

- ✓ The rotor consists of the hub and blades of the wind turbine.
- ✓ These are often considered to be the turbine's most important components from both a performance and overall cost standpoint.



avoids the wind shade behind the tower

- the rotor may be made more flexible
- built somewhat lighter than an upwind machine

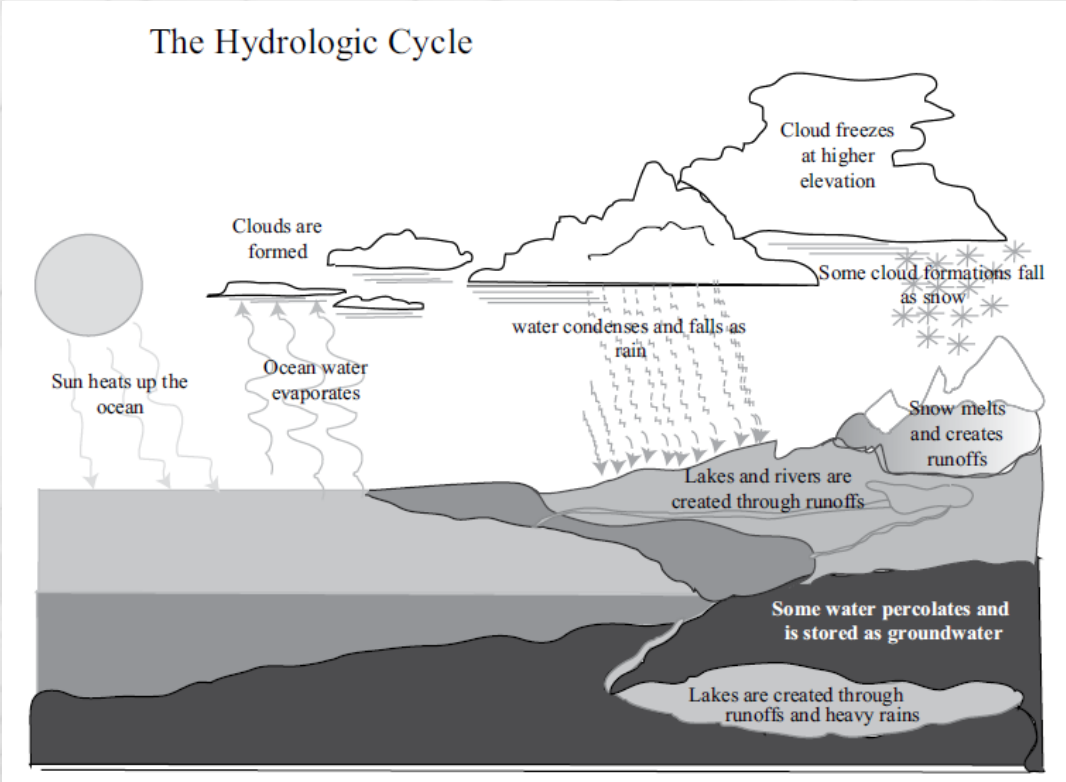
- the rotor needs to be made rather inflexible and placed at some distance from the tower
- upwind machine needs a yaw mechanism to keep the rotor facing the wind

- the fluctuation in the wind power due to the rotor passing through the wind shade of the tower.
- this may give more fatigue loads on the turbine than with an upwind design

Hydropower

Introduction

- Humankind has harnessed power and energy from water bodies over centuries.
- The hydrologic cycle is one of the earth's fundamental processes—water from the ocean evaporates, turns into clouds, and is brought back to land in the form of rain.



Hydropower

Introduction

- Hydropower supplies about 16% of the world's electricity requirement from all sources (WEC, 2017).
- Hydropower is the leading renewable energy source for electrical power worldwide and was found to make up 71% of all renewable electricity.
- China is the leading nation in hydropower generation, followed by the United States, Brazil, Canada, India, and then Russia.
- In the United States, there are over 2,000 hydropower plants with a total capacity of over 79,985 MW.

Hydropower in Egypt

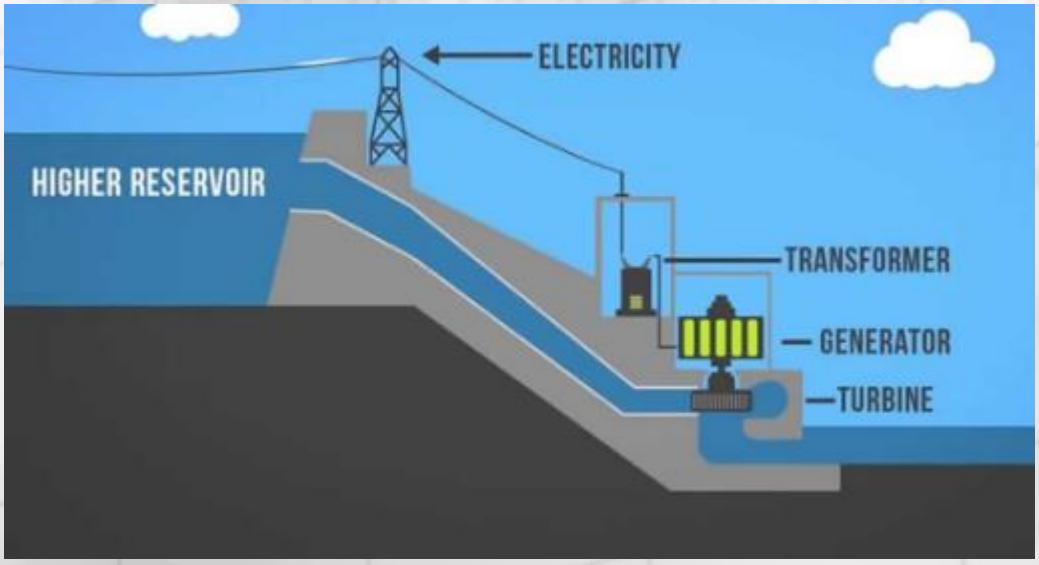


Power from Water

The theoretical power derived from water is a consequence of the potential energy of water as it is dropped from a given height.

$$Theoretical\ Power\,(Watts) = Q \times H$$

where
Q = volumetric flow rate (kg/s)
H = dynamic head (meters)



Example:
Determine the theoretical power in watts for water flowing at 200 kg/s and dropped from a dynamic head of 20 meters.

$$P_t\,(Watts) = \frac{200\ kg}{s} \times \frac{20\ m}{} \times \frac{N - s^2}{kg - m} \times \frac{9.8\ m}{s^2} \times \frac{W - s}{N - m} = 39,200\ Watts$$

$$P_t\,(hp) = 39,200\ W \times \frac{hp}{746\ W} = 52.55\ hp$$

Hydropower plants

Inefficiencies in Hydro Power Plants

The energy losses in a hydropower plant are mainly due to the following:

a. Hydraulic loses in conduits and turbines
b. Mechanical losses in bearings and the power transmission system
c. Electrical losses in the generator, station use, and transmission (for hydroelectric power)

- The overall effect reduces the theoretical power by a factor of 0.60 to 0.80.
- Large-scale hydro power plants have overall conversion efficiencies of 60% to 80%.
- These values are much higher than those of conventional internal combustion engines and coal power plants.

Inefficiencies in Hydro Power Plants

A small commercial micro–hydro power unit had the following data: the volumetric flow rate was 35 liters/s [555 gpm] and the dynamic head was 1.5 meters [4.92 ft]. This unit was reported to produce 200 watts [0.268 hp] of continuous power output at those conditions.
Determine the actual efficiency of this unit.
Also calculate the yearly output of this small device in units of kWh/year, assuming 24 hours per day and 365 days in a year.

The volumetric flow rate is converted into units of kg/s:

$$Q\left(\frac{kg}{s}\right) = \frac{35\ L}{sec} \times \frac{1\ kg}{L} = 35\frac{kg}{s}$$

Then, the theoretical power is calculated:

$$Power(W)_t = Q \times H = \frac{35\ kg}{sec} \times \frac{1.5\ m}{} \times \frac{N - s^2}{kg - m} \times \frac{9.8\ m}{s^2} \times \frac{W - s}{N - m} = 514.5$$

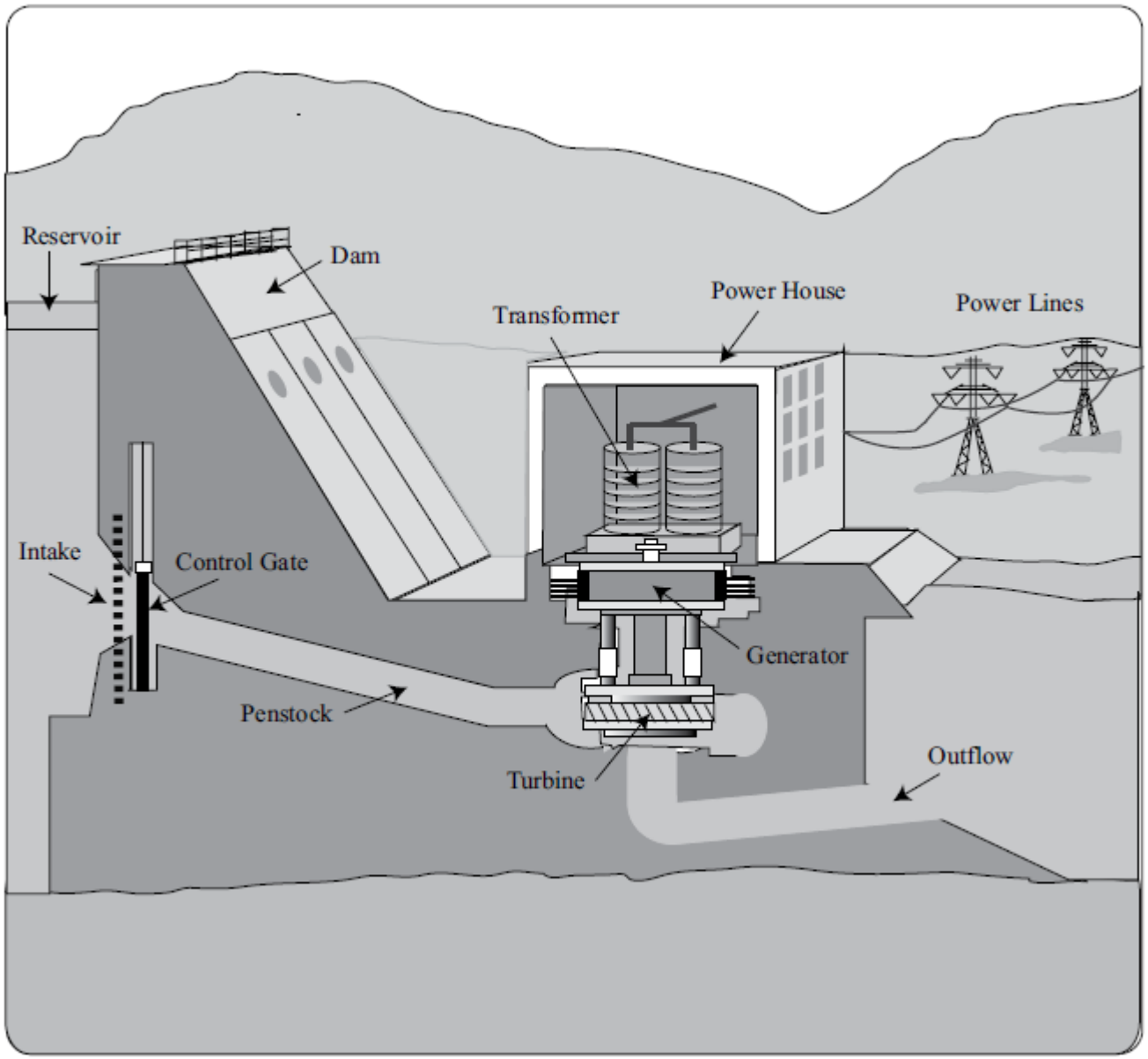
The efficiency is calculated using the ratio of actual and theoretical power as shown:

$$\eta(\%) = \frac{Actual\ Power}{Theoretical\ Power} \times 100\% = \frac{200\ W}{514.5\ W} \times 100\% = 38.9\%$$

The yearly output may then be estimated as follows:

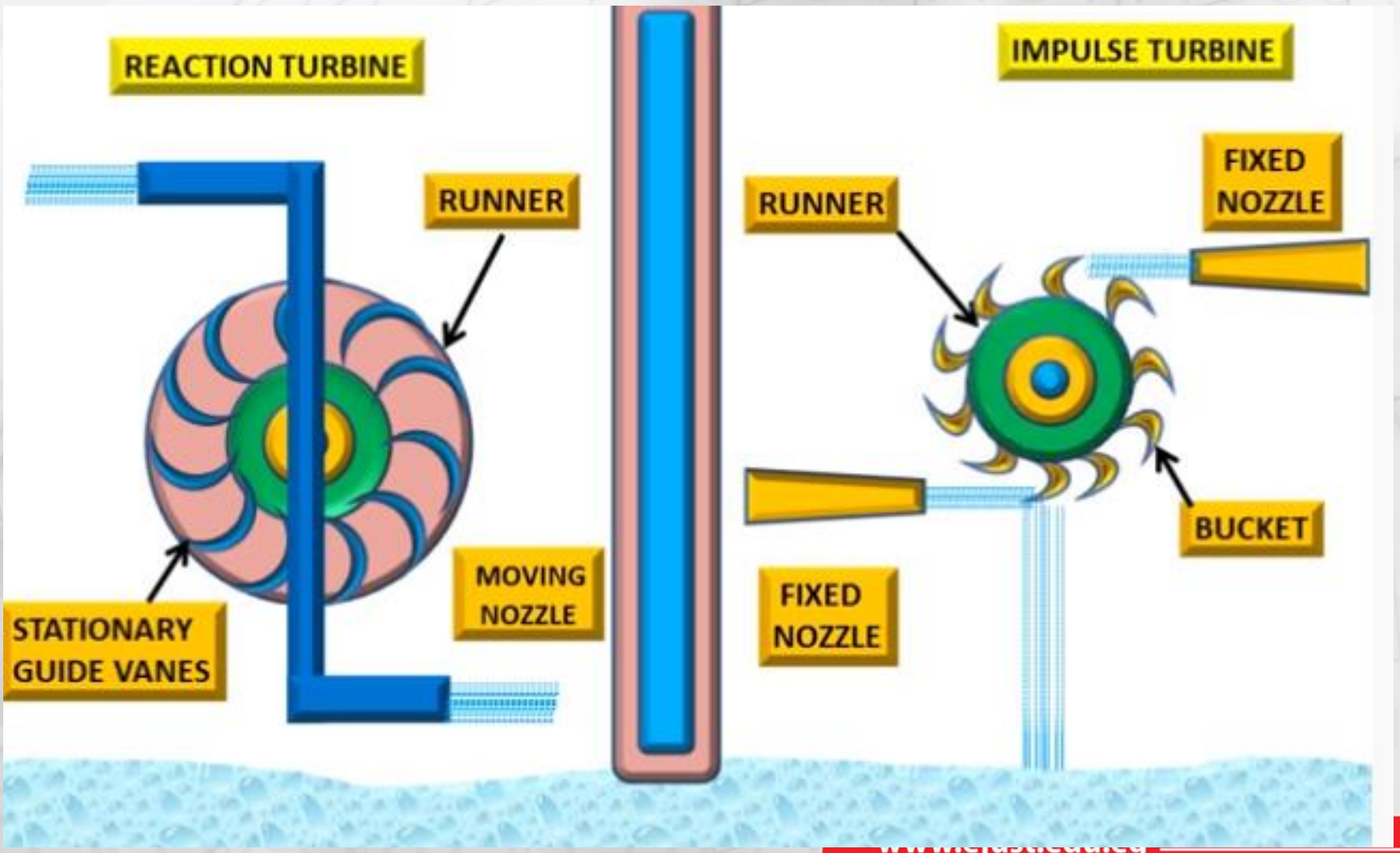
$$Yearly\ Output\left(\frac{kWh}{yr}\right) = 200\ W \times \frac{24\ hrs}{day} \times \frac{365\ days}{year} \times \frac{kW}{1{,}000\ W} = 1{,}752\frac{kWh}{yr}$$

Basic Components of a Hydro Power Plant

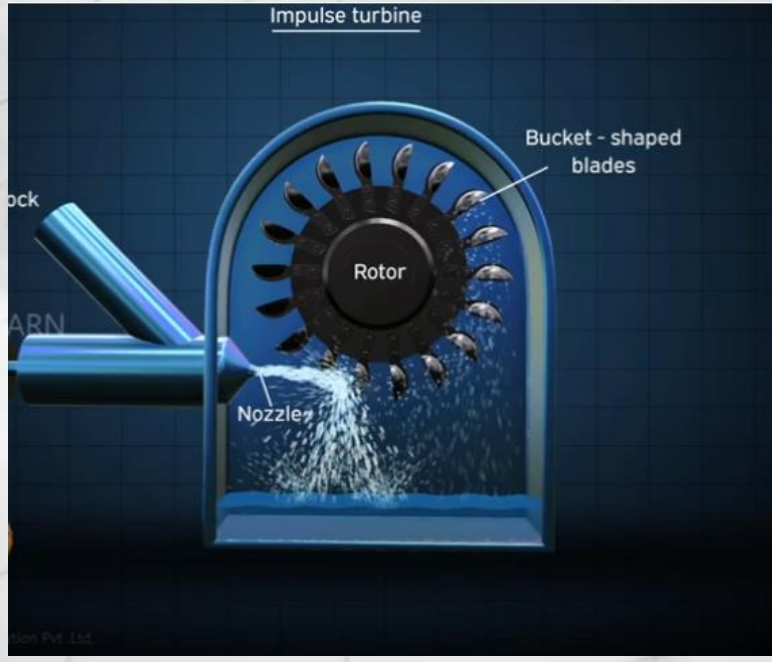


Types of Turbines

There are two types of turbines:
a. Impulse turbines b. Reaction turbines



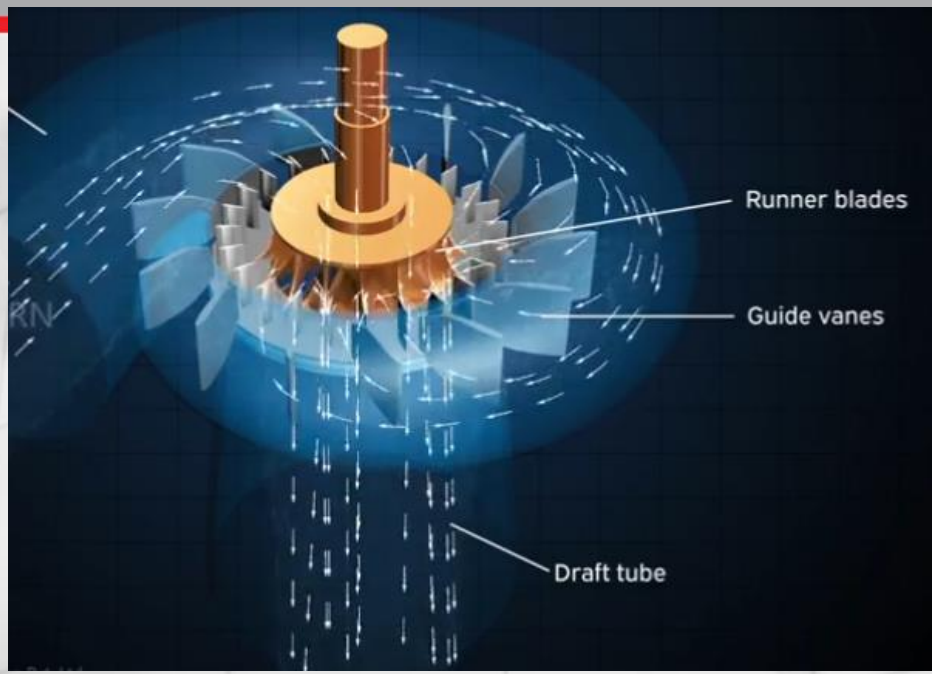
Impulse turbines



- In impulse turbines, the momentum (or impulse) of water jets is used to move the turbine blades, resulting in a rotary motion of the shafting where the blades are attached.
- Since the turbine is spinning, the force acts over a distance (work), and the diverted water flow is left with diminished energy.

- Newton's Second Law (force = mass × acceleration) describes the transfer of energy in impulse turbines.
- Impulse turbines are most often used in very high head applications.
- Prior to hitting the turbine blades, the water's pressure (its potential energy) is converted into kinetic energy by a nozzle focused on the turbine.
- No pressure change occurs in turbine blades, and the turbine actually does not require housing for this operation.

Reaction turbines



- Reaction turbines, on the other hand, are acted on by water, which changes pressure and gives up energy as it moves through the turbine.

- The turbines must be either encased to contain the water pressure (or suction) or fully submerged in the water flow.

- Newton's Third Law (law of reciprocal reactions or conservation of momentum) describes the transfer of energy for reaction turbines.
- Most reaction turbines are used in low and medium head applications.

Turbines for Power Generation

The most common types of water turbines for power generation are Francis Turbine, Kaplan Runner, and Pelton Runner



Turbines for Power Generation

a. Francis Turbine. The majority of large hydroelectric power plants use Francis turbines. Typical head ranges for this turbine type are 300 to 500 meters [984 to 1,640 ft]. Specific speeds for these turbines range from 60 to 400 rpm.

b. Kaplan Runner. The Kaplan runner is a propeller-type turbine, similar to those found in axial flow fans or rotors for pump boats. The dynamic head for Kaplan runners ranges from 2 to 70 meters [6.56 to 229.6 ft]. Its specific speeds range from 300 to 1,100 rpm.

c. Pelton Runner. This is an example of an impulse turbine with blades that look like cups. The water jet is impinged, moving the blades in the opposite direction and turning the shafting where the blades are attached. Pelton runners are very common standby emergency power platforms for hydro power plants. They are usually fed with a pipe of around a fraction of a meter (about 1 ft) and may generate about a MW or a fraction of a MW. The dynamic head for Pelton runners ranges from 50 to 1,300 meters [164 to 4,264 ft]. The specific speeds range from 4 to 70.

Specific Speeds for Turbines

- Specific speeds (*Ns*) are used to compare different impellers.

- The specific speed describes the speed of the turbine at its maximum efficiency with respect to the power and flow rate.
- The specific speed is derived to be independent of turbine size.
- If one is given the fluid flow conditions and the desired shaft output speed, the specific speed can be calculated, and an appropriate turbine design selected.

$$N_s = \frac{\omega q^{0.5}}{(gh)^{0.75}}$$

where

ω = pump shaft rotational speed (radians per second)
q = flow rate at the point of best efficiency (m^3/s)
h = total head (m)
g = acceleration due to gravity (9.8 m/s^2)

Specific Speeds for Turbines

Given a pump rotational speed of 1,760 rps, flow rate of 1,500 m_3/s, and head of 100 m, calculate the specific speed.

$$N_s = \frac{\omega q^{0.5}}{(gh)^{0.75}} = \frac{1,760 \times 1,500^{0.5}}{(9.8 \times 10)^{0.75}}$$

$$N_s = \frac{1760 * 1500^{(1/2)}}{(9.8 \times 10)^{(3/4)}} = 2188$$

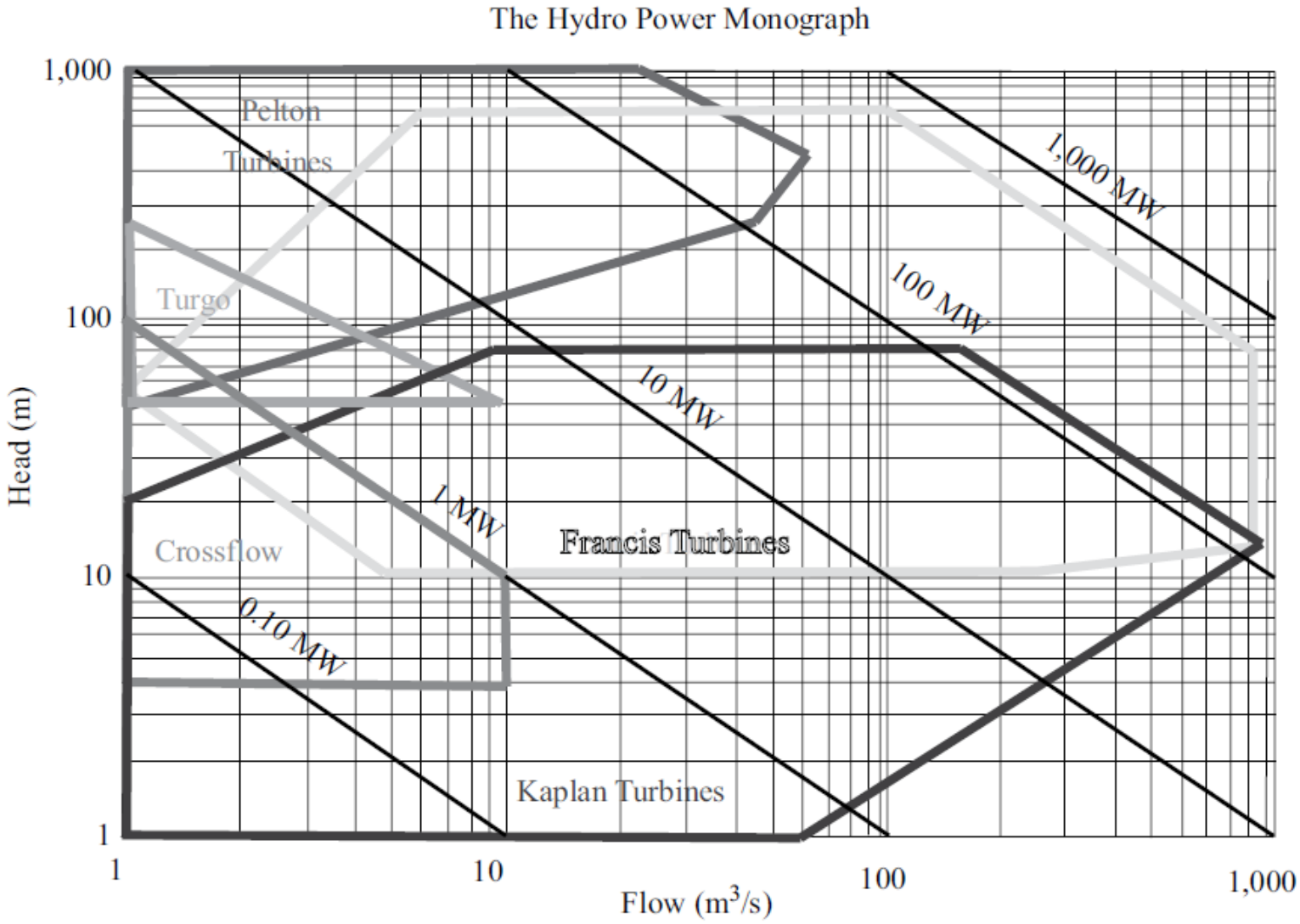
This is the specific speed of a typical centrifugal pump.

- The specific speeds for various types of pumps and impellers are as follows:
- Radial flow turbines have specific speeds ranging from 500 to 4,000,
- mixed flow turbine speeds range from 2,000 to 8,000,
- and axial flow turbines have specific speeds ranging from 7,000 to 20,000.

Turbine Selection

Turbine application chart: a monograph for turbine type selection

Francis and Pelton turbines have the highest potential power output



Types of Hydro Power Plant

1. On the Basis of Operation

- Based upon its type of operation, a hydro power plant is categorized as either a base load plant or a peak load plant.

- A base load plant operates 24 hours a day, 365 days in a year, and provides continuous power year-round.
- They are designed to operate at this base load indeterminately.
- They are neither allowed to produce power higher than the rated design nor permitted to operate way below their rated capacity.
- Engineers designing these facilities must have several years of running data of their chosen water source, including forecasts for droughts and heavy downpours.

- A hydropower plant may also be designed to operate on a peak load basis.
- It would operate for staggered periods, timed with peak loads during a given season or for a given time of day.
- Some mega-cities use too much power at the beginning of a day and require a dedicated peak load power plant to address their intermittent needs.
- Some peak loads are twice the average requirement and would take quite the toll on a power plant not designed to increase production at such a rate.

Types of Hydro Power Plant

2. Based on Plant Capacity

- Micro-capacity (<5 MW),
- Medium-capacity (5 to 100 MW),
- High-capacity (100 to 1,000), and
- Super-power plant (>1,000 MW)

3. Based on Head

- A low head power plant is said to have a head less than 15 meters [49.2 ft],
- Medium head hydroelectric power plants have dynamic heads between 15 and 50 meters [49.2 to 164 ft],
- High head plants are those with dynamic heads greater than 50 meters.

Types of Hydro Power Plant

4. Based on Hydraulic Features

Conventional

- Conventional hydroelectric power plants generate electrical power using conventional means, taking advantage of the potential and kinetic energy of falling water bodies.
- Water is neither recycled nor reused for subsequent power production and instead is allowed to take its course down the level.
- Dams are usually built through rivers and streams.
- The stored water is used to generate power continuously throughout the year.

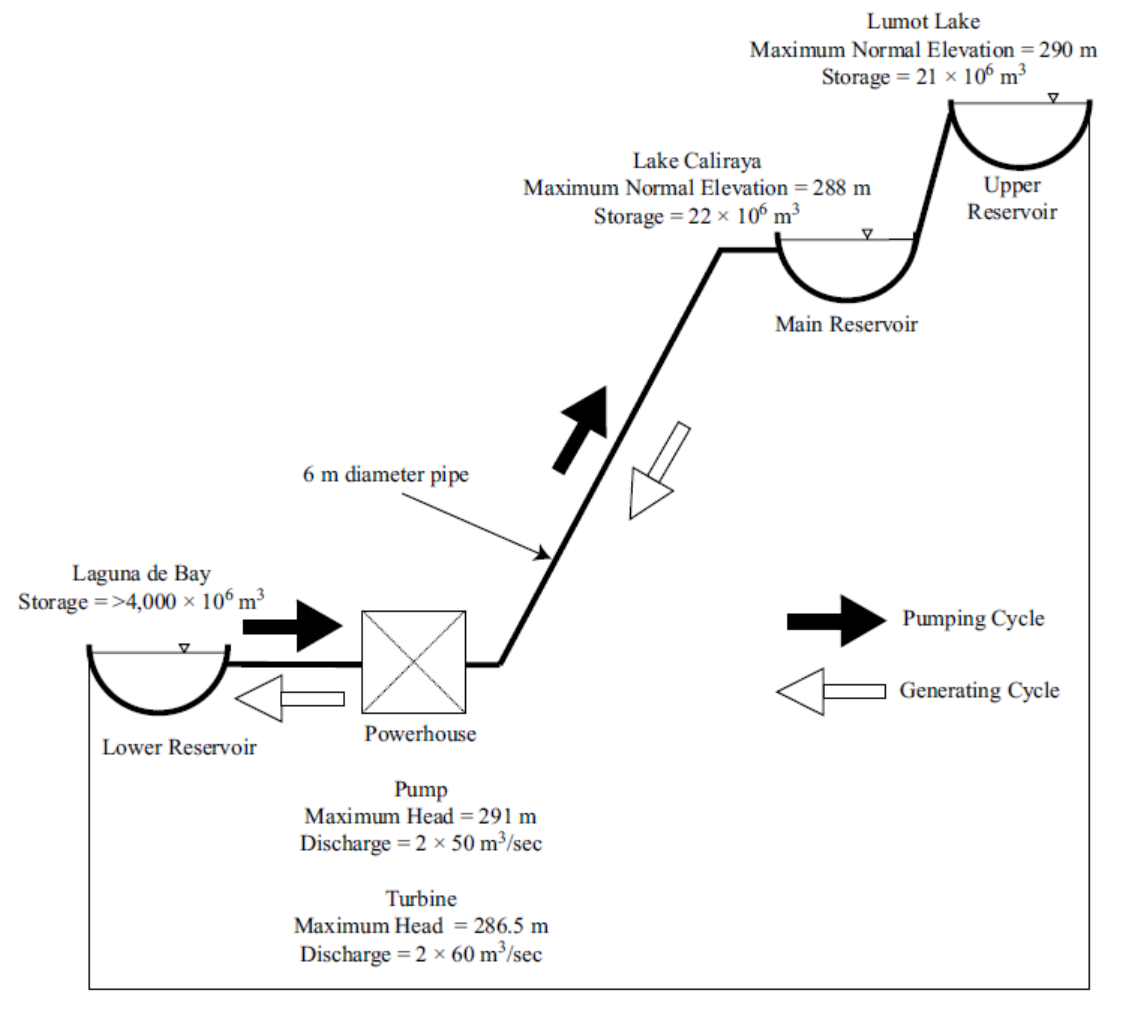
Pumped Storage Systems

- Pumped storage hydro power plants are used in electric power systems to address peak load problems.
- In mega-cities, power demands skyrocket at the beginning of the work day.
- During this time, the demand for power is at its maximum, and if there is no excess power available, brownouts and outages can occur.
- A pumped storage system is built to ensure that when peak power conditions are experienced, it can deliver the needed power.
- Once the power requirement goes down, the water released to generate power is brought back to storage in preparation for the next peak power demand.

Types of Hydro Power Plant

4. Based on Hydraulic Features

Pumped Storage Systems



Turbine Selection

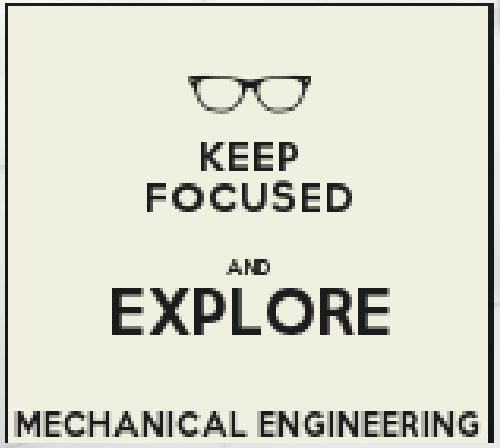
Determine the total maximum time (in hrs and days) for the water from the Caliraya Lake to be emptied when power is generated continuously at the facility. The volumetric flow rate is 280 cubic meters per second [4,434,375 gpm], and Lake Caliraya has a storage volume of 22×10^6 m^3 [5,807 million gallons].

$$Time(hrs) = \frac{Storage\ Volume(m^3)}{Volumetric\ Flow\ Rate\left(\frac{m^3}{s}\right)} \times \frac{1\ hr}{3{,}600\ s} \times \frac{22 \times 10^6\ m^2}{280\frac{m^3}{s}} \times \frac{1\ hr}{3{,}600\ s} = 21\ hrs$$

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